



User Manual



Table Of Contents

1. Read Before Operating	7
2. Warnings and Cautions	8
3. Supported G-codes&M-codes Quick Reference Table	9
4. Main Pages	13
4.1 JOG	14
4.2 Probe	17
4.3 Offsets	21
4.4 MDI	24
4.5 Mixed Wizards	28
4.6 SETTING	32
4.6.1 Six-step Setup (Must Read)	33
4.6.2 Main Page of Settings	37
4.6.3 Tool Change Settings	45
4.6.4 RTCP Configuration	56
4.6.5 Debugging Procedures for Different RTCP Structures	62
4.6.6 485 IO Extension Configuration	70
4.6.7 Modbus Communication Settings for Servomotor with Absolute Encoder using 485 Protocol	72
4.6.8 STA Settings	74
4.6.9 Screen Settings	76
4.6.10 Startup Configuration	77
4.6.11 User-Defined Input Buttons Configuration	79
4.7 RUN	81
4.8 Error Report	83
4.9 SD Run	84
4.10 List Machining	88
4.11 Drilling	90
4.12 Tapping	93

4.13 Cut Circle	97
4.14 Cut Rectangle	100
4.15 Rectangular Pocket	103
4.16 Circular Pocket	106
4.17 Cut RoundRect	109
4.18 Surfacing	112
4.19 I/O Monitor	115
4.20 Slotting	116
5. Upgrade Instructions	119
5.1 Touch Screen Firmware Upgrade Instructions	120
5.2 Controller Firmware Upgrade Instructions	121
5.3 WIFI page Firmware Upgrade Instructions	122
6. Frequently Asked Questions (Must Read)	124
7. Error And Alarm Code Reference Table (Must Read)	128
7.1 Error Code Reference Table	129
7.2 Alarm Code Reference Table	131
8. Controller Connection Instructions	132
8.1 Wiring Instructions (Must Read)	133
8.1.1 Power Supply	133
8.1.1 Power Supply	134
8.1.2 Wiring Instructions for 10-inch Screen	135
8.1.3 Driver Wiring	136
8.1.4 Spindle Wiring	138
8.1.5 Tool Setter Wiring	140
8.1.6 Limit Switch Wiring	141
8.1.7 Buttons Wiring	142
8.1.8 M6 and M7	143
8.1.9 Power Port And Touchscreen Port	144
8.1.10 Reset Wiring	145

8.1.11 MPG Handwheel Pin Wiring Instructions	146
8.1.12 Controller Appearance	147
8.2 Bluetooth Connection Instructions	149
8.3 WIFI Connection Instructions	150
9. RS485 Expansion Module Interface Terminals Description (Optional Feature) ...	153
9.1 Wiring	154
9.2 Functionality	155
10. Specific G-code Format Instructions for POCENDER CNC Control System	157
10.1 G6/G14 - Cut Rectangular/Rounded Rectangular Pocket	158
10.2 G7 - Milling Hole/Thread With Helical Interpolation (Operation Mode G90)	160
10.3 G8 - Vortex Milling Circular Pocket (Operation Mode G90)	161
10.4 G9 - Zigzag Surfacing a Rectangular Plane	162
10.5 G11 - Cut Rectangle/RoundRect. (Operation Mode G90)	163
10.6 G12 - Matrix Drilling (Operation Mode G90)	164
10.7 G13 - Index Drilling (Operation Mode G90)	165
10.8 G73/G81/G82/G83 - Drilling/Tapping	166
1) G73	167
2) G81	168
3) G82	169
4) G83	170
11. Detailed Explanation for Supported G Codes & M Codes	171
11.1 G0 - Rapid Motion Positioning	173
11.2 G1 - Linear Interpolation Motion	174
11.3 G2, G3 - Circular Interpolation Motion	175
11.4 G4 - Dwell	181
11.5 G10 L2 - Set Coordinate System	182
11.6 G10 L20 - Set Coordinate System	184
11.7 G17, G18, G19 - Plane Select	185
11.8 G20, G21 - Units	186

11.9 G28, G28.1 - Go/Set Predefined Position	187
11.10 G30, G30.1 - Go/Set Predefined Position	188
11.11 G38.n - Straight Probe	189
11.12 G40 - Cutter Compensation Cancel	190
11.13 G43 - Tool Length Offset	191
11.14 G43.1 - Dynamic Tool Length Offset	192
11.15 G43.4 - Rotary Tool Center Point (Similar to FUNAC)	193
11.16 G49 - Cancel Tool Length Offset and RTCP (Similar to FUNAC)	194
11.17 G53 - Move in Machine Coordinates	195
11.18 G54, G55, G56, G57, G58, G59 - Select Coordinate System	196
11.19 G60 - Program Loop	197
11.20 G61 - Exact Path Mode	198
11.21 G68.2, G69 - Enable/Cancel 3+2 Function	199
11.22 G73 - Drilling Cycle with Chip Breaking	200
11.23 G80 - Cancel Canned Cycle	201
11.24 G81 - Drilling Canned Cycle	204
11.25 G82 - Drilling Canned Cycle With Dwell	209
11.26 G83 - Peck Drilling Canned Cycle for Deeper Holes	210
11.27 G90, G91 - Distance Mode	211
11.28 G91.1 - Arc IJK Incremental Mode	212
11.29 G92 - Coordinate System Offset	213
11.30 G92.1 - Reset G92 Offsets	214
11.31 G93, G94 - Feed Rate Mode	215
11.32 G98, G99 - Canned Cycle Return Level	216
11.33 M0 - Program Pause	217
11.34 M2, M30 - Program End	218
11.35 M3, M5 - Spindle Control	219
11.36 M6 - Auto Tool Change	220

11.37 M7, M8, M9 - Coolant Control	221
11.38 M56 - Parking Override and Cancel	222
11.39 M62, M63 - Digital Output Control	223
11.40 M66 - Wait on Input	224
11.41 \$HX(\$HZ,\$HY...) - Homing	225

1. Read Before Operating

If you find that the interface or function is different from what you currently have, it means that there is a new firmware. Please add customer service WeChat and provide pictures of the touch screen and controller (model confirmation is required).

2. Warnings and Cautions



WARNING: Indicates circumstances or practices that can lead to personal injury as well as damage to the controller, the machine, or other equipment.



CAUTION: Indicates circumstances or practices that can lead to damage to the controller or other equipment.



INFORMATION: Indicates important information.

3. Supported G-codes&M-codes Quick Reference Table

CODE	DESCRIPTION	NOTES
G00	Rapid Motion Positioning	
G01	Linear Interpolation Motion	
G02	Circular Interpolation Motion CW	
G03	Circular Interpolation Motion CCW	
G04	Dwell	eg: G04P5 indicates to pause the program for 5 seconds. The P value can be a decimal greater than zero.
G10 L2, G10 L20	Set Work Offset Values	
G17	XY Plane Selection	
G18	XZ Plane Selection	
G19	YZ Plane Selection	
G20	Set Machine Units To Inches	
G21	Set Machine Units To Millimetres	
G28	Go Predefined Position	
G28.1	Set Predefined Position	
G30	Go Predefined Position	
G30.1	Set Predefined Position	
G38.2	Straight Probe, Positive	
G38.3	Straight Probe, Negative	
G38.4	Straight Probe, Positive with Signal	
G38.5	Straight Probe, Negative with Signal	
G40	Cutter Compensation Cancel	(G41/G42 not supported)
G43	Tool Length Offset	eg: G43H2 indicates to activate tool length

		compensation for T2.
G43.1	Dynamic Tool Length Offset	
G43.4	Enable Rotary Tool Center Point (RTCP)	eg: G43.4H1 indicates to activate tool length compensation and RTCP/coordinate system rotation function for T1.
G49	Cancel Tool Length Offset and RTCP Mode	
G53	Move in Absolute Coordinates	
G54	Select Work Coordinate System #1	
G55	Select Work Coordinate System #2	
G56	Select Work Coordinate System #3	
G57	Select Work Coordinate System #4	
G58	Select Work Coordinate System #5	
G59	Select Work Coordinate System #6	
G60	Program Loop	eg: G60P10 indicates to repeat the program 10 times.
G61	Exact Path Mode	
G68.2	Enable 3+2 Function	
G69	Cancel 3+2 Function	
G73	High-Speed Peck Drilling Canned Cycle	When the pitch value O is added after the drilling cycle command, it becomes a tapping cycle command. This applies to G73, G81, G82, and G83. eg: G81Z-10R0O0.2F100
G80	Cancel Canned Cycle	
G81	Drilling Canned Cycle	
G82	Drilling Canned Cycle With Dwell	
G83	Peck Drilling Canned Cycle for Deeper Holes	
G90	Set Distance Mode to Absolute	

G91	Set Distance Mode to Incremental	
G91.1	Arc IJK Incremental Mode	
G92	Coordinate System Offset	
G92.1	Reset G92 Offsets	
G93	Inverse Time Feed Rate Mode	
G94	Units per Minute Mode	
G98	Canned Cycle – Retract Back To The Initial Z	
G99	Canned Cycle – Retract Back To R Plane	
M0	Program Pause	
M2	Program End	
M3	Spindle ON (Clockwise)	eg: M3S1000 indicates to start spindle clockwise rotation at 1000 RPM
M5	Spindle OFF	
M6	Auto Tool Change	Both servo-driven tool changer carousel and linear tool changer are supported, with a total capacity of 20 tools. eg: T2M6 indicates to select tool 2 and perform a tool change.
M7	Turn Mist Coolant On	
M8	Turn Flood Coolant On	
M9	To Turn All Coolant Off	
M30	End The Program And Rewind	
M56, M56P0	Parking Override and Cancel	
M62P0	Turn On Tool Touch-Off Signal Output	
M63P0	Turn Off Tool Touch-Off Signal Output	

M66	Wait on Input	eg: M66 P0 L3 Q10 indicates to wait for input port 0 to be ON (High), and pause with timeout after 10 seconds. L3 waits for a high signal, and L4 waits for a low signal. (Expandable I/O supported.)
\$HX(\$HZ,\$HY...)	X home (Z home, Y home...), only one axis can be homed in a single line.	



INFORMATION: When using the tapping command, remember to set the tapping axis as axis letter A, B, or C, which correspond to the axis numbers 4, 5, and 6 respectively. Then, set the pulse equivalent of the tapping axis to the number of pulses required for the tap to rotate one degree **(for firmware versions prior to June 27, 2023)** or one revolution, and set the servo spindle to the position mode.

4. Main Pages

Read other subtopics below:

- 4.1) JOG
- 4.2) Probe
- 4.3) Offsets
- 4.4) MDI
- 4.5) Mixed Wizards
- 4.6) SETTING
- 4.7) RUN
- 4.8) Error Report
- 4.9) SD Run
- 4.10) List Machining
- 4.11) Drilling
- 4.12) Tapping
- 4.13) Cut Circle
- 4.14) Cut Rectangle
- 4.15) Rectangular Pocket
- 4.16) Circular Pocket
- 4.17) Cut RoundRect
- 4.18) Surfacing Wizard

4.1 JOG

Click on the **[JOG]** button located at the bottom left corner of the screen to access this page.



INFORMATION: Clicking on coordinates display area will cancel the step motion or positioning mode. If the status box displays an "Alarm" message, clicking on it will forcefully clear the alarm; if the status box displays a "Dwell" message, clicking on it will resume the operation. In any other status, clicking on it will cancel the action.

Control Descriptions:

NAME / LOCATION	TYPE	DESCRIPTION
RTCP	Button	Click to activate the MPG RTCP mode for the current tool. To exit the RTCP mode, click it again or click on the coordinate display area or the idle

		status box.
X, Y, Z, A, B, and C on the left	Button	Click to enter the coordinate value of the selected axis (delete the axis letter from the keyboard), and click [OK] to update the workpiece coordinate (G54-G59) value of the axis.
MPOS	Display Field	Refers to the G53 coordinate system. It is automatically cleared when the machine finds the origin. This coordinate can only be read and used, and cannot be set manually.
WPOS	Display Field	Refers to the G54-G59 coordinate systems, which can be cleared (SET0), set, and returned to zero (ZERO) at any time. The default program uses the workpiece coordinate system. When a limit switch is triggered, the font color of this coordinate turns green.
SET0	Button	Click to reset the workpiece coordinate (G54-G59) of the current axis to zero.
ZERO	Button	Click to quickly return the current axis position back to the zero point of the workpiece coordinate system (G54-G59).
X+, Y+, Z+, A+, B+, C+ X-, Y-, Z-, A-, B-, C-	Button	Short press to incrementally jog the current axis in the positive or negative direction, and hold down to continuously move the current axis in the positive or negative direction.
F	Display Field	Current feedrate.
[Ovr] to the right of [F]	Display Field	Current feedrate override value.
S	Configurable Parameter	Current spindle speed.
[Ovr] to the right of [F]	Display Field	Current spindle speed override value

Tool#	Button	Click to perform tool change to the new tool number entered on the right box. The current tool number displayed on the box can be changed by clicking and entering a new tool number.
POS	Mode Selection with Input box	The input in the box represents the distance of movement in step motion mode, and the coordinate position in positioning mode.
SPD	Mode Selection with Input box	The feedrate input for Step Motion/Positioning mode.
Gears	Mode Button	Toggle between jog speed levels by click, which are 0.01, 0.1, 1, 10, 100.
Probe	Button	Click to access the "Probe" page. The text turns green when the probing signal is triggered.
Offsets	Button	Click to access the "Offsets" page.
Jogging	Button	Click to toggle between Jogging/Step Motion/Positioning mode.
G54	Button	Current coordinate system display, click to switch current workpiece coordinate to G54-G59.
JOG	Button	Click to access the jogging page.
MDI	Button	Click to access the manual data input page.
SETTING	Button	Click to access the device parameters setting page.
RUN	Button	Click to access the run options page.

4.2 Probe

Click the **[Probe]** button on JOG page to access this page.



POS: X0Y0Z0

GOTO

	M POS	W POS
X	2.497	2.497
Y	0.978	0.978
Z	-9.331	-9.331
A	6.159	6.159
B	-6.162	-6.162
C	4.994	4.994

Probe POS 0

Prob SPD 50

Probe + 0.000

Prob TRA 20.000

Probe - 0.000

Plate THK 0.000

X Probe Y Probe Z Probe

X- Probe Y- Probe X Reset

Find X Midpoint Find Y Midpoint Y Reset

Prob POS First Prob Auto TLO

MANUAL </> CODING POCENDER SETTING RUN



INFORMATION:

- 1) Coordinate display area: Click to release lock and cancel current probing.
- 2) During probing, the page will be locked and all keys are disabled.
- 3) After tool setting failure, the system will display an ALRM error and lock until resolved. To resume operation, click to unlock and resolve any errors.

Control Descriptions:

NAME / LOCATION	TYPE	DESCRIPTION
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X0Y0Z0	Configurable Parameter	Clicking on it will prompt a keyboard input where you can enter the desired position to quickly move the axes to that coordinate location. It allows for rapid movement to the specified position.
GOTO	Button	Click to rapidly move the axis to the coordinate position entered in the left box.
The box to the right of [GOTO]	Display Field (clickable)	If an error occurs, it will be displayed here (e.g. error: 9). Click to view the error message. For example, error 9 indicates a premature probe trigger, which means the probe signal needs to be switched to either forward or reverse.
Probe	Button	Click to quickly move to the machine coordinates(NOT WORKPIECE COORDINATES) of tool setter position X and Y set on the setting page.
The box to the right of [Probe]	Display Field	Display the machine coordinates (G53) after completing the tool setting.
Probe+	Button	After clicking the X/Y letter on the left side of the page (X/Y turns green after click), click [Probe+] to manually set the current X/Y machine coordinates to those of [Probe+] .
The box to the right of [Probe+]	Display Field	Displays the machine coordinates of X/Y after a successful probing in the positive direction.
Probe-	Button	After clicking the X/Y letter on the left side of the page (X/Y turns green after click), click [Probe-] to manually set the current X/Y machine coordinates to those of [Probe-] .
The box to the right of [Probe-]	Display Field	Displays the machine coordinates of X/Y after a successful probing in the negative direction.
Probe SPD	Configurable Parameter	Probe speed.
Probe TRA	Configurable Parameter	Click to set the maximum probing distance of the probe. If the probe is not found within this range, an error will be reported

The box to the right of [Plate THK]	Configurable Parameter	Click to set the plate thickness, usually the thickness of the tool setter. If set, the position of automatic resetting will be subtracted by the plate thickness. When the plate thickness is set to 0, the coordinate will be reset when the tool touches the probe.
Plate THK	Button	Click to save the entered plate thickness value
X Probe Y Probe	Button	Click to probe towards the positive direction, and the workpiece coordinates will automatically reset to zero after tool setting is completed. After clicking, the screen will be locked and other functions cannot be operated. To unlock or cancel the tool setting operation, please click on the coordinate display area.
Z Probe X- Probe Y- Probe	Button	Click to probe towards the negative direction, and the workpiece coordinates will automatically reset to zero after tool setting is completed. After clicking, the screen will be locked and other functions cannot be operated. To unlock or cancel the tool setting operation, please click on the coordinate display area.
Find X Midpoint Find Y Midpoint	Button	Before using these two buttons, please follow these steps in order: 1. Complete tool setting in the X/Y positive direction. 2. Complete tool setting in the X/Y negative direction. 3. Raise the tool to avoid collision with the workpiece during movement. 4. Clicking Find X/Y Midpoint to move the axis to the midpoint between the positive and negative tool setting positions or the manually set midpoint between the + and - probe points. Click X/Y Reset to reset the coordinates to zero if necessary.
X Reset Y Reset	Button	Click to reset the current workpiece coordinate of the X/Y-axis to zero.

Tool Setter Pos	Button	Click to quickly move to the MPos(NOT WPos) set in the [Tool Setter X/Y] in the Setting page.
First TLM	Button	When multiple tools are required for processing, use the [Tool Setter Pos] button to move the tool to the tool setter after setting up the first tool. Then click this button to automatically measure and record the current tool length for the first tool.
Auto TLO	Button	After finishing the previous machining operation with a tool, move the tool to the tool setter by clicking [Tool Setter Pos] . Then click this button to perform automatic tool length offset operation for the Z-axis of the current coordinate system, without the need to re-measure the Z-axis tool offset. (Note: multi-tool paths need to be output in the same coordinate system.)
JOG	Button	Click to access the jogging page.
MDI	Button	Click to access the manual data input page.
SETTING	Button	Click to access the device parameters setting page.
RUN	Button	Click to access the run options page.

4.3 Offsets

Click the **[Offsets]** button on JOG page to access this page.



INFORMATION:

- 1) If necessary, click on the **[End]** button to cancel the origin search and release the current locked state.
- 2) When homing, the page will be locked, and other buttons cannot be operated.

Control Descriptions:

NAME / LOCATION	TYPE	DESCRIPTION
G54, G55, G56, G57, G58, G59, G28, G92, Probe (last machine coordinates of the tool setter) , MPos, WPos	Display Field	Display corresponding coordinates. Use the [Next Page] button to switch between them.
ZXY Home	Button	Homing sequence starts with Z-axis homing, followed by X-axis homing, and finally Y-axis homing. (Homing function needs to be enabled in the Setting page by clicking on [Home OFF] to switch to [Home ON], and corresponding travel distances should be set up, otherwise it will result in an error.) Clicking this button will lock the screen and disable other buttons until homing is completed. To cancel the homing process or cancel the lock in emergency, click the [End] button.
X Home, Y Home, Z Home, A Home, B Home, C Home	Button	Single-axis homing. (Homing function needs to be enabled in the Setting page by clicking on [Home OFF] to switch to [Home ON], and corresponding travel distances should be set up, otherwise it will result in an error.) Clicking this button will lock the screen and disable other buttons until homing is completed. To cancel the homing process or cancel the lock in emergency, click the [End] button.
G28.1, G30.1	Button	Click to set the current MPos (G53) as the return position for G28/G30.
G28, G30	Button	Clicking to quickly return to the machine coordinate position set by the G28.1/G30.1 mentioned above.
G54, G55, G56, G57, G58, G59	Button	Click to switch the current workpiece coordinate to G54-G59.

Next Page	Button	Click to switch coordinate system between G54, G55, G56, G57, G58, G59, G28, G92, Probe (last machine coordinates of the tool setter), MPos and WPos
Spindle Off	Button	Display the current spindle status. Click to turn on the spindle with a spindle speed of 1000 RPM. The button label will display "Spindle On" .
The box to the right of [Spindle Off] , which shows "1000" by default	Display Field (clickable)	Click to set the spindle speed online.
Unlock	Button	Click to unlock and resume normal operation if an "Alarm" message appears on the manual page or other pages.
Reset	Button	Click to set the current workpiece position of all axes to zero.
Zero Return	Button	Click to move all axes to the home position of the workpiece coordinate system.
CLNT Off	Button	Display the current coolant status. Click to turn on the coolant. The button label will display "CLNT On" .
The box to the left of [Zero Reset]	Display Field	Displaying current machine parameters.
JOG	Button	Click to access the jogging page.
MDI	Button	Click to access the manual data input page.
SETTING	Button	Click to access the device parameters setting page.
RUN	Button	Click to access the run options page.

4.4 MDI

Click the **[MDI]** button on the bottom of the page to access this page.



INFORMATION: Input without cursor at the end.

Note:

The G-code programming page supports: spindle control M3S1000/M5, rapid movement G0, linear interpolation G1, clockwise circular interpolation G2, counterclockwise circular interpolation G3, programming with IJKR, and delay for 5 seconds with G4P5. **The first line of G1 code must include an F value (feed rate).**

Control Descriptions:

NAME / LOCATION	TYPE	DESCRIPTION
Status	Display Field	Display the current running status: SGL Step / SGL

		BLK / Cycling / FXD Count Cycling / Idle.
Count	Display Field	Count of cycles.
Error	Display Field	Display error NO. in the right box.
Line	Display Field	Display the current line number of the program being executed. Click to display the total number of characters in the program.
Total Cycles	Configurable Parameter	Click to set the number of repetitions for [FXD Count Cycling] .
[File Name] on the top left corner	Configurable Parameter	Click to bring up the keyboard and enter a file name (Chinese characters are not supported).
The input preview box to the right of [File Name]	Input Box	Click to insert the currently entered preview content before the selected line.
ENT	Button	Means "Enter", which is used to add the just entered G-code to the programming box.
CLR	Button	1. Double-clicking the [CLR] button when a line is selected will delete the selected line. 2. Single-clicking the CLR button when there is content in the preview box will delete the content in the preview box. 3. If no line is selected and the insert line has no content, clicking CLR once will clear all content in the programming box.
Select	Button	Click to enter the "SD Run" page, select the desired file, and then click the [Select] button on the "SD Run" page. Return to the "MDI" page, and the file name will change to the selected file name.
Read	Button	Click to read the first 200 lines of the current file name. If no card is inserted or the file does not exist, an error message (error42) will be reported(Chinese file names are not supported).

Save	Button	Click to save the current program to the SD card with the current file name. If no card is inserted or the file does not exist, an error message (error42) will be reported(Chinese file names are not supported). SGL BLK / Cycle / FXD Count Cycling require the program to be saved as a file before execution.
F+	Button	Click to increase the feed rate by 10%.
F-	Button	Click to decrease the feed rate by 10%.
S+	Button	Click to increase spindle speed by 10%.
S-	Button	Click to decrease spindle speed by 10%.
SGL Step	Button	Click to run one line of code at a time. Clicking again will run the next line and so on. Once the end is reached, it will return to the first line. When activated, the screen is locked and only [Pause], [Cont.], [End], [F+], [F-], [S+] and [S-] can be used. Clicking [End] will cancel the current operation and unlock the screen.
SGL BLK	Button	Click to run the entire block of code once. When activated, the screen is locked and only [Pause], [Cont.], [End], [F+], [F-], [S+] and [S-] can be used. Clicking [End] will cancel the current operation and unlock the screen.
Cycle	Button	Click to run the entire program in an infinite loop. When activated, the screen is locked and only [Pause], [Cont.], [End], [F+], [F-], [S+] and [S-] can be used. Clicking [End] will cancel the current operation and unlock the screen.

FXD Count	Button	Click to run the entire program repeatedly for a fixed number of times. When activated, the screen is locked and only [Pause], [Cont.], [End], [F+], [F-], [S+] and [S-] can be used. Clicking [End] will cancel the current operation and unlock the screen.
Pause	Button	Click to pause running.
Cont.	Button	Click to resume running.
End	Button	Click to end the current running and release the lock.

4.5 Mixed Wizards

Click the **[Mixed Wizards]** button on MDI page or RUN page to access this page.

Circular Pocket ▾	Center X	0	Cutting Depth	3	Tool R	1.12
Conv. Milling ▾	Center Y	1	Down Step	4	First Cut R	10.1
	Z Start	2	Line Spacing	5	Machining R	10.2

G-Codes

G6X0Y1Z2I10.1J10.2K3O0P5Q4R1.12F3000;Rect. Pocket

G6X0Y1Z2I10.1J10.2K3O2P5Q4R1.12F3000;Rect. Pocket

G8X0Y1Z2I10.1J10.2K3O1P5Q4R1.12F3000;Circular Pocket

Feed 3000 RPM 6000

Add/Replace
Delete
Send
Pause
End
Copy
Back



INFORMATION:

- 1) The current page supports a maximum of 16 processes. To add more, click **[Copy]** to copy to the **MDI** page, then clear the current program and continue editing and copying.
- 2) When milling threads, make sure that the quotient obtained by dividing the depth of cut by the pitch value is an integer.

Control Descriptions:

NAME / LOCATION	TYPE	DESCRIPTION
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The first dropdown menu on the left.	Dropdown Menu	Click to open a drop-down menu for selecting a wizard: Rect Pocket, RoundRect Pocket, Circular Pocket, Zigzag Surfacing, Cut Rectangle, Cut Circle, Cut RoundRect, Slotting, Matrix Drilling, Index Drilling, Positioning Drilling and Positioning Tapping
The second dropdown menu on the left.	Dropdown Menu	Click to open a drop-down menu for selecting a machining mode: 1) Loop Inward Climb Milling, Loop Inward Conv. Milling, Loop Outward Climb Milling, Loop Outward Conv. Milling 2) Climb Milling Internal, Climb Milling External, Conv. Milling Internal, Conv. Milling External 3) Climb Milling, Conv. Milling 4) X-axis Milling, Y-axis Milling
The third dropdown menu on the left.	Dropdown Menu	Click to open a drop-down menu for selecting the machining mode: Fixed Number Distribution, Fixed Angle Increment
Center X	Input Box	The center X coordinate of the machining target in the workpiece coordinate system.
Center Y	Input Box	The center Y coordinate of the machining target in the workpiece coordinate system.
Z Start	Input Box	The initial Z coordinate of the machining target in the workpiece coordinate system.
Cutting Depth	Input Box	The total depth of the machining. A value greater than zero.
Down Step	Input Box	The depth of cut for each pass of the Z-axis during the machining process. It should be a value greater than zero.
Line Spacing	Input Box	The spacing between adjacent tool paths. When cutting a pocket or performing surfacing, the line spacing should be a positive value that is smaller than the tool diameter.
Tool R	Input Box	The radius of the tool used for machining.

Lead-in R	Input Box	Refers to the spiral lead-in radius when cutting a circular pocket. This value should be a positive one smaller than the tool radius, otherwise a ring shape will be machined instead of a complete circular shape.
Machining R	Input Box	Refers to the machining radius when cutting a circular pocket.
Length X	Input Box	Refers to the length in the X-direction for rectangle-related wizards.
Width Y	Input Box	Refers to the width in the Y-direction for rectangle-related wizards.
Retract Mode	Dropdown Menu	Click to open a drop-down menu for selecting a tool retract mode when cutting a circle: 1) Rapid Retract: a rapid motion of the tool away from the workpiece to a safe position. 2) Retract at Center: can be used for milling internal threads. 3) Retract at Outside: can be used for milling external threads.
Start Angle	Input Box	Refers to the starting angle for indexing machining, measured from the positive Y-axis.
End Angle	Input Box	Refers to the end angle for indexing machining, measured from the positive Y-axis. It should be greater than the start angle.
Number of Holes	Input Box	For " Fixed Number Distribution " mode, the number of holes drilled between the start and end angles. For " Fixed Angle Increment " mode, the number of holes drilled for each incremental angle increase from the start angle.
Increment Angle	Input Box	Refers to the incremental angle added to the starting angle for each drilling operation in " Fixed Angle Increment " drilling.
Feed	Configurable Parameter	Click to set the machining feed rate.
RPM	Configurable Parameter	Click to set the spindle speed.

Add/Replace	Button	Replace the selected item or add a new process if nothing is selected.
Delete	Button	Delete the selected item or delete all if nothing is selected.
Send	Button	Click to send the code and start execution.
Pause	Button	Click to pause running.
Cont.	Button	Click to resume running.
End	Button	Click to end running.
Copy	Button	Click to copy the code to the MDI page.
Back	Button	Click to go back to the previous page.

4.6 SETTING

Click the **[SETTING]** button at the bottom of the screen to access.

Read other subtopics below:

- 4.6.1) Six-step Setup **(Must Read)**
- 4.6.2) Main Page of Settings
- 4.6.3) Tool Change Settings
- 4.6.4) RTCP Configuration
- 4.6.4) Debugging Procedures for Different RTCP Structures
- 4.6.6) 485 IO Extension Configuration
- 4.6.7) STA Settings
- 4.6.8) Screen Settings
- 4.6.9) Startup Configuration
- 4.6.10) User-Defined Input Buttons Configuration

4.6.1 Six-step Setup (Must Read)

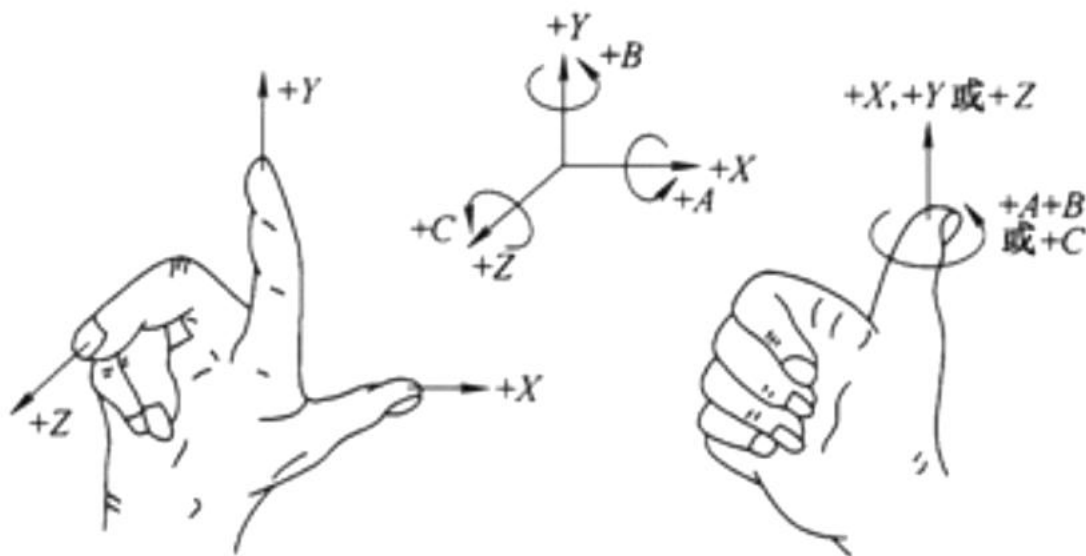
To ensure normal operation, it is necessary to set the following parameters.

1) Pulse Equivalent:

This step is to correspond the distance traveled (XYZ axis) or angle rotated (ABC axis) by the machine correspond with the distance of the coordinate movement on the screen. Refer to the “**Pulse Equivalent**” setting on the settings main page for details. After setting, click **[Apply]** twice and then **[Read]**. **Remember to power off and restart the system for the setting to take effect.**

2) Direction Setting:

This step aligns the movement direction of the machine's axis with the positive and negative coordinates on the screen, in accordance with the right-hand rule of the Cartesian coordinate system. It is important to note that this pertains to the direction of the tool's movement. For instructions on how to switch the direction, please refer to the motor direction setting in the main page of “**SETTING**”.



The right-hand rule of the Cartesian coordinate system

(referring to the direction of tool movement, not platform movement direction)

After setting the above two parameters, the machine should be able to move and operate normally.

3) Homing Setting:

A. The switch can be either two-wire normally open or three-wire, but the latter requires an NPN normally open switch. Connect the switch to the limit input port, as the limit switch will serve as the home switch during homing. Please see the “**limit switch wiring instructions**” in this manual for details on how to wire it.

B. To use the homing function, first go to the settings main page and click on **[Home OFF]** to toggle it to **[Home ON]** and enable the homing function. This will trigger an alarm indicating lost coordinates upon each power-on, and the machine's axes will be unable to move until homing is complete. After homing, the alarm will be automatically cleared.

C. Subsequently, configure the corresponding travel range and then set the retract distance. The recommended retract distance is around 2-5mm (specifically based on the distance required for the switch to reset after homing). This distance must be sufficient for the switch to reset without interfering with the home switch signal of the next axis.

D. Next, set the homing speed. We recommend a range of 1000-2000.

E. The next step is to set the homing speed, which is recommended to be around 1000-2000. **After that, set the crawling speed, which is recommended to be around 20-50. Once done, click [Apply] twice and then [Read] to save the settings.**

→Following the previous step

F. After wiring, trigger each home switch manually in the “**JOG**” page and check if the font color of the workpiece coordinate turns green. If it does, the wiring is correct and the function is normal. If it turns green without triggering the switch, go to the settings main page and switch the limit direction to the opposite state. Then return to the JOG page and manually trigger the home switch again. If the font color of the workpiece coordinate doesn't turn green, it indicates a wiring error. Please check the wiring.

G. If the limit switches are functioning properly, go to the “**Offsets**” page and sequentially click on the homing button of each axis. **(after clicking, the screen will be locked and you need to click [End] to unlock. Before returning to the origin, do not trigger any home switches prematurely.)**

H. If the homing direction is found to be inconsistent with the desired direction, the homing direction needs to be set. By default, “Homing Direction 3” represents the X-axis towards the negative direction, Y-axis towards the negative direction, and ZABC axes towards the positive direction. To switch the direction, refer to the Homing Direction setting in the main page of “**SETTING**”.

The Homing function cannot be manually triggered for simulation and must be tested on the machine. It has two triggers, and cannot be simulated by manual triggering.

4) Tool Setting Setup:

The switch can be either a two-wire or three-wire switch, and it can be normally open (NO) or normally closed (NC). For a three-wire switch, an NPN (negative-positive-negative) type is required. Please refer to the wiring instructions in this manual for proper wiring.

A. After wiring the tool setter, go to the “**JOG**” page and manually trigger the tool setter. Check if the font color of the **[Probe]** button on the “**JOG**” page turns green. If the font color turns green, the wiring is correct and the function is working properly.

B. If the font color of the **[Probe]** button turns green without triggering the tool setter, go to the main page of “**SETTING**” and switch the probe direction by clicking on **[Probe FWD]** or **[Probe REV]**. Return to the “**JOG**” page and manually trigger the tool setter again to see if the font color turns green. If it does not, there is a wiring error and you should check the connections.

C. If the function is normal, move the tool within 20mm directly above the tool setter and perform tool setting on the “**Probe**” page.

5) Spindle Speed Upper/Lower Limit Settings:

The spindle upper limit corresponds to the output of 10V on AVI when the spindle reaches maximum speed, and the spindle lower limit is set to 0, which corresponds to 0V output. After setting, remember to click **[Apply]**.

6) Limit Switch Setting:

A.Hard Limit:

It's not recommended to use the hard limit switch. It's better to use the soft limit switch instead, as triggering the hard limit switch will cause the system to immediately stop and lock and require manual intervention to move the axis away and restart.

Connect the positive and negative limit switches according to the wiring instructions in this manual. The positive and negative limit switch signals share the same limit port. Then, click on the **[HPL Off]** button in the main page of “**SETTING**” to toggle to **[HPL On]**, indicating that the hard limit switch function is activated. When the homing function is enabled, the hard limit function will automatically be disabled. Please use the soft limit instead.

B.Soft Limit:

This function is based on the homing function, so homing must be enabled and

successfully completed before soft limit can be activated. Once soft limit is enabled, each axis must home first when the machine is powered on. Failure to home and force unlock can result in incorrect movement and travel limits.

Step 1: Set the corresponding machine travel in the settings page. The machine travel is the distance from the origin to the reverse limit position in the direction of homing, regardless of positive or negative direction. **Setting the value to 0 will disable the soft limit for that axis.**

Step 2: Click on **[Home OFF]** in the main page of “**SETTING**” to enable the homing function, then click on **[SPL Off]** to switch to **[SPL On]** to enable the soft limit function. After setting the parameters, click **[Apply]** twice and then click **[Read]** to check if the settings have been applied successfully. After that, power off and restart the system.

Step 3: The soft limit function is highly intelligent. During operation, it will automatically stop before exceeding the travel limit and trigger an “ALRM2” alarm. Simply unlock without losing the coordinates. There is no need to restart and home the machine.

4.6.2 Main Page of Settings

Please note: The login passcode is 753951, which is required for accessing and modifying settings. Passcode modification is not supported.

	Pulse Equivalent	Max SPD	Acceleration	Max Travel	
X	320.000	3000.000	200.000	300.000	Motor DIR 7
Y	320.000	3000.000	200.000	300.000	Status 1
Z	320.000	3000.000	200.000	300.000	Home DIR 3
A	320.000	3000.000	800.000	0.000	Home SPD 2000.000
B	320.000	3000.000	800.000	0.000	Crawl SPD 50.000
C	320.000	3000.000	800.000	0.000	RPM UL 1000.000
Switch Debounce 0.0 Pulse REV 63 Retract Distance 2.000					RPM LL 0.000
EN FWD Limit REV Home ON CNC Mode Read Apply Screen Set					
SPL Off HPL Off MM Mode Probe REV BLUT WIFI NONE					
MANUAL		</>CODING		POCENDER	NEXT RUN



INFORMATION: The data on the settings page will only be updated for the first time after startup, and it needs to be manually refreshed and read at other times.

Instructions for Parameters Setting:

1) Pulse Equivalent

When the setting is for "1mm per motor pulse", the displayed unit on the coordinates will be in millimeters. When the setting is for "1 degree per motor pulse", the displayed unit on the coordinates will be in degrees. **After changing this parameter, click [Apply] and perform a power cycle to make the changes effective.**

Relevant formulas:**A. Lead Screw****➤ Direct drive screw**

$$\text{Pulse equivalent} = \text{Subdivision number} / \text{Lead of the screw}$$

For example, if the subdivision is set to 1600, it means that the motor will rotate one revolution for every 1600 pulses received. It is recommended to set the subdivision to 1600 and not go too high, as the single-axis frequency should not exceed 100kHz, that is, $\text{Pulse equivalent} * \text{Maximum speed} / 60 < 100000$.

● Subdivision

For some servo motors, you can set the number of pulses required for one rotation as the subdivision value, which is usually recommended to be set to 1600.

For other motors, you need to set the numerator and denominator values based on the number of encoder pulses per rotation and the required number of motor pulses per rotation. Then, the subdivision value can be calculated as the denominator value.

● Lead of the Screw

The distance traveled by the screw for one revolution. For example, if the screw is 1605, then the lead is 5. Therefore, the pulse equivalent is calculated as follows:

$$\text{pulse equivalent} = 1600 / 5 = 320$$

➤ Screw connected to motor via gearbox

$$\text{Pulse equivalent} = \text{subdivision} / \text{lead screw pitch} * \text{reduction ratio}$$

For example, if the subdivision is set to 1600, the lead screw pitch is 1605, and the gear ratio is 2 (i.e., the motor turns two revolutions while the lead screw turns one revolution), then the pulse equivalent is calculated as follows:

$$\text{pulse equivalent} = 1600 / 1605 * 2 = 640$$

NOTE: The pulse equivalent setting and screw calculation method for axes A, B, and C are the same as above, except that the lead should be replaced by 360.

B. Synchronous wheel directly & Synchronous Belt**➤ Synchronous wheel directly connected to the motor.**

Pulse equivalent = subdivision / (number of teeth on synchronous wheel * pitch)

For example, if an XL synchronous wheel with 10 teeth and a standard pitch of 5.08 is used, and the subdivision is set to 1600, then the pulse equivalent is calculated as follows:

$$\text{pulse equivalent} = 1600 / (10 * 5.08) = 31.496$$

➤ **Synchronous wheel connected to the motor via gearbox.**

Pulse equivalent = subdivision / (number of teeth on the synchronous wheel * pitch) * reduction ratio

For example, if an XL synchronous wheel with 10 teeth and a standard pitch of 5.08 is used, and the subdivision is set to 1600, and the reduction ratio is 2 (motor rotates 2 times while the synchronous wheel rotates once), then the pulse equivalent is calculated as follows:

$$\text{pulse equivalent} = 1600 / (10 * 5.08) * 2 = 62.992$$

C. Gear&Rack

➤ **Gear directly connected to the motor.**

Pulse equivalent = subdivision / (gear teeth number * pitch)

For example, if using a 3M gear with 40 teeth and a standard pitch of 9.4247 ($\pi * \text{module} = 3.14159 * 3 = 9.4247$), with a subdivision of 1600, the pulse equivalent would be calculated as follows:

$$\text{pulse equivalent} = 1600 / (40 * 9.4247) = 4.244$$

➤ **Gear connected to the motor via gearbox.**

Pulse equivalent = subdivision / (gear teeth number * pitch) * reduction ratio

For example, if using a 3M gear with 40 teeth and a standard pitch of 9.4247 ($\pi * \text{module} = 3.14159 * 3 = 9.4247$), with a subdivision of 1600 and a reduction ratio of 5 (i.e., the motor turns 5 times to rotate the gear once), the pulse equivalent is calculated as follows:

$$\text{pulse equivalent} = 1600 / (40 * 9.4247) * 5 = 21.22$$

2) SPD

Refers to the maximum running speed of the axis in mm/min. It is recommended to be set between 1000-15000, but the actual value should be determined based on the specific characteristics of the equipment.

3) Acceleration

Measured in meters per second squared (m/s^2). For stepper motors, a recommended range is 200-500 m/s^2 . For servo motors, a recommended range is 500-1000 m/s^2 .

4) Max Travel

Refers to the machine travel setting with soft limit function enabled. This value is not positive or negative (positive number only), and represents the reverse travel distance from the home direction **(setting to 0 can disable the soft limit for this axis)**.

This parameter is also used for the home travel distance. An alarm will be triggered if the origin is not found within the set travel range.

After setting the parameter, click **[Apply]** twice and then **[Read]** to check if it takes effect. **If the soft limit function is enabled, be sure to power off and restart the machine after changing this parameter.**

5) Motor DIR

Setting the direction of the motor.

6) Status

Do not alter.

7) Node Deviation

Circular interpolation accuracy setting. No need to modify.

8) Home DIR

Homing direction setting. **If this parameter is changed and the soft limit is enabled, the system must be restarted after the change to take effect.**

9) Crawl SPD

The crawl speed refers to the second stage speed during the homing operation.

In the first stage, the machine moves rapidly (**Home SPD**) to find the origin, then bounces back a certain distance after finding it.

In the second stage, the machine moves slowly (**Crawl SPD**) to re-find the origin, and then bounces back a certain distance again (**Retract Distance**).

10) Spindle UL

Setting the upper limit of spindle speed.

11) Spindle LL

Setting the lower limit of spindle speed.

12) Home SPD

Homing speed refers to the first stage speed during the homing operation.

In the first stage, the machine moves rapidly (**Home SPD**) to find the origin, then bounces back a certain distance after finding it.

In the second stage, the machine moves slowly (**Crawl SPD**) to re-find the origin, and then bounces back a certain distance again (**Retract Distance**).

13) Pull-off Distance

Pull-off distance refers to the distance that the machine moves back in the second stage of homing. This distance cannot be set to 0, as it is necessary to ensure that the current switch signal is reset.

In the first stage, the machine moves rapidly (**Home SPD**) to find the origin, then bounces back a certain distance after finding it.

In the second stage, the machine moves slowly (**Crawl SPD**) to re-find the origin, and then

bounces back a certain distance again (**Pull-off Distance**).

14) Pulse REV

After confirming correct wiring, click on the move button on “**JOG**” page. If the coordinates change but the motor doesn't move, try changing this value to 63, then click **[Apply]**, and finally, restart the system to see if it works.

15) Switch Debounce

Switch debounce refers to the process of eliminating or reducing electrical noise or spurious signals that can occur when switches or sensors are activated or deactivated. When homing, switch debounce refers to the amount of time the controller waits for the signal from the homing switch to stabilize before considering it valid. Setting this value to 0 should be sufficient in most cases.

Control Descriptions:

NAME / LOCATION	TYPE	DESCRIPTION
EN FWD EN REV	Button	These two labels respectively enable the forward and reverse directions, and can be toggled by clicking. (Only applicable to controllers with enable interface.)
Limit FWD Limit REV	Button	Refer to the limit switches for the forward and reverse directions, respectively. Click to toggle between them.
Home ON Home OFF	Button	A switch to turn on/off the homing function. When homing is enabled, an alarm will be triggered upon restarting/resetting, indicating that the machine has lost its position. The alarm can only be cleared by homing or unlocking the machine. Enabling homing will automatically disable the hard limit function. Use the soft limit function instead if limiting is needed. When homing is turned off, the soft limit function will also be disabled. To use the soft limit function, homing must be turned on and successfully find the origin.

CNC Mode Laser Mode	Button	Click to switch between CNC mode and laser mode. After switching, you need to power off and restart the system.
SPL On SPL Off	Button	Soft limit switch. Soft limit switch should be enabled after setting up the max travel, and then the system needs to be restarted. This feature is based on the homing function, so you must install the home switch and complete the homing process successfully before using this feature. Therefore, after enabling the soft limit switch, the first thing to do when starting up is to home each axis. If you move the axis without homing or force unlock, the direction and travel limit may be incorrect.
HPL On HPL Off	Button	Hard limit switch function switch. The hard limit switch function is automatically disabled when the homing function is turned on. It is recommended not to use the hard limit switch function because after encountering the limit switch, the equipment needs to be restarted before it responds. Please use the soft limit switch instead.
MM Mode Inch Mode	Button	Click to switch between millimeter and inch modes.
Probe FWD Probe REV	Button	Click to switch the probe switch signal.
BLUT	Button	Click to enable bluetooth mode. Upon clicking, the system will reboot, and the origin must be re-located.
WIFI	Button	Click to enable the WIFI AP mode, which sends out a WiFi signal for other devices to connect to. Upon clicking, the system will reboot, and the origin must be re-located.
NONE	Button	Click to disable the bluetooth and wifi modes. Upon clicking, the system will reboot, and the origin must be re-located.
Read	Button	Click to read the settings. After modifying the settings, click [Read] to check if the parameters have been

		successfully applied.
Apply	Button	After filling in the parameters, you need to click the [Apply] button twice for the settings to take effect.
Screen Set	Button	Click to enter the display settings page.

4.6.3 Tool Change Settings

Click the **[NEXT]** button on the main page of “**SETTING**” to access this this page.

Pulse Width	3	Tool Setter Z	0.000	Active Tool	2		
EN Time	255	Tool Setter A	0.000	Tool Max	20	X Spacing	0.000
Node Deviation	0.010	Tool Setter B	0.000			Y Spacing	0.000
Arc Tolerance	0.002	Tool Setter C	0.000			Drill D Val	0.000
Tool Setter X	0.000	Tool Change Mode	2			IP: 192.168.1.17	
Tool Setter Y	0.000	Turret Axis	0			Mode: STA	

WIFI/STA STA SET WIFI NONE Tooltable #1 Tooltable #2 Read Apply

MANUAL </>CODING **POCENDER** NEXT RUN

Control Descriptions:

NAME / LOCATION	TYPE	DESCRIPTION
Pulse Width	Display Field	Set the duration of each individual pulse in the pulse train, measured in microseconds. If the motor is not responding even with the correct wiring and a pulse reversal setting of 63, please attempt adjusting this value to 5.
EN Time	Configurable Parameter	This parameter refers to the duration for which the motor's enable signal is retained. It is not effective in systems without an enable output. To utilize this parameter, ensure that the motor is configured for power-up enable.

Junction Deviation		Set the value for early deceleration overshoot in cornering. The default value is 0.01 and should not be modified.
Arc Tolerance		Control the precision of arc interpolation. The default value is 0.002 and should not be modified.
Tool Setter X/Y	Configurable Parameter	Set the XY machine coordinates (non-workpiece coordinates) for tool setting on the "Probe" page.
Tool Setter Z		Not used.
Tool Setter A/B/C	Configurable Parameter	In automatic tool change mode, if the value for the corresponding axis is non-zero, the machine will first align the respective axis to the set mechanical coordinate (not workpiece coordinate) value before performing tool change and alignment actions.
Tool Change Mode	Configurable Parameter	Please refer to the tool change settings section for detailed information on tool change mode settings.
Turret Axis	Configurable Parameter	The servo axis used for the tool turret. Set 4 as the A-axis for the tool disc axis, 5 as the B-axis, and 6 as the C-axis. Any other value set will not use the tool disc axis.
Active Tool	Configurable Parameter	Click to manually set the current tool number. It will be automatically set after a successful tool change and will be saved in the event of a power outage.
Max Tool No.	Configurable Parameter	Set the number of tools in the tool table. The range is from 1 to 20.
X/Y Spacing	Configurable Parameter	In G73/81/82/83 modes, the L value controls the number of drilling cycles. If the X/Y spacing is set to 0, the drilling will be performed at the same position for L times. If it is non-zero, the drilling will be incrementally offset according to the specified spacing data.
Drill D Val.	Configurable Parameter	Represents the retract distance in G73 mode. If it is set to 0, the default value of 0.4 will be used. If a non-zero positive value is set, it will be used instead.
IP Address	Display Field	Enter this IP address in the client to access the device control and wireless file transfer page.
Mode	Display Field	Displays the wifi mode.

WIFI/STA	Button	Click to enable WIFI STA mode, which allows the controller to connect to other WiFi devices, such as a router. Upon clicking, the system will reboot, and the origin must be re-located.
STA SET	Button	Click to access the "STA Settings" page.
WIFI	Button	Click to enable the WIFI AP mode, which sends out a WiFi signal for other devices to connect to. Upon clicking, the system will reboot, and the origin must be re-located.
NONE	Button	Click to disable the bluetooth and wifi modes. Upon clicking, the system will reboot, and the origin must be re-located.
Tool Table #1/#2	Button	Tool Table #1: tool numbers 1-10 Tool Table #1: tool numbers 11-20 NOTE: Due to the large number of parameters, the loading and setting times may be longer. It is recommended to fill in all parameters before clicking on the [Apply] button.
Read	Button	Click to read the settings. After modifying the settings, click [Read] to check if the parameters have been successfully applied.
Apply	Button	After filling in the parameters, you need to click the [Apply] button twice for the settings to take effect.

Instructions for Parameters Setting:

1) Tool Length Setting

Tool #1 is the primary tool and is used for tool setting and setting the workpiece coordinates.

The lengths of all other tools should be set relative to tool #1 (this can be done by using a tool setter and the **"First TLM"** function in the **"Probe"** page to record the probe point for tool #1, then switching to other tools and using the **"First TLM"** function to record their respective probe points, and finally subtracting the difference between the probe points for each tool and the probe point for tool #1 to obtain the tool length for each tool).

After tool length is set, tool length compensation will be applied automatically when tools are switched. In auto tool alignment mode, only tool number 1 needs to be measured using the **"First TLM"** function; measurement and saving of the probe points for other tools will be done automatically.

2) Tool Change Mode (0-8)

- Set 0 to ignore tool change.
- Set 1 for automatic tool change with automatic tool length compensation (spindle moves to the corresponding tool position). Suitable for gantry-type linear tool changers/carousel tool changers.
- Set 2 to return to the safe position and wait for manual tool change. After completion, press the **[Cont.]** button to automatically perform tool length compensation before machining. Suitable for manual tool change.
- Set 3 for automatic tool change with automatic tool length compensation (spindle moves to the corresponding tool position). Suitable for column-type carouse tool changers.
- Set 4 for automatic tool change and automatic tool setting, then automatic tool length compensation (spindle moves to the corresponding tool position).

This mode requires first measuring the tool length for tool #1 in the **"Probe"** page (by clicking on the **[First TLM]** button without any other operations).

- Set 5 to return to the safe position and wait for manual tool change. After completion, press the **[Cont.]** button to automatically perform tool length compensation and tool setting before machining. Suitable for manual tool change.

This mode requires first measuring the tool length for tool #1 in the **"Probe"** page (by clicking on the **[First TLM]** button without any other operations).

- Set 6 for automatic tool change and automatic tool setting, then automatic tool length compensation (spindle moves to the corresponding tool position). Suitable for

column-type carouse tool changers.

This mode requires first measuring the tool length for tool #1 in the “**Probe**” page (by clicking on the **[First TLM]** button without any other operations).

- Set 7 for automatic tool change with automatic tool length compensation (spindle moves to the corresponding tool position). Suitable for pneumatically driven gantry-type linear tool changers/carousel tool changers.
- Set 8 for automatic tool change and automatic tool setting, then automatic tool length compensation (spindle moves to the corresponding tool position). Suitable for pneumatically driven gantry-type linear tool changers/carousel tool changers.

This mode requires first measuring the tool length for tool #1 in the “**Probe**” page (by clicking on the **[First TLM]** button without any other operations).

3) Turret Axis

The servo axis used for the tool turret. Set 4 as the A-axis for the tool disc axis, 5 as the B-axis, and 6 as the C-axis. Any other value set will not use the tool disc axis.

4) Active Tool#

Click to manually set the current tool number. It will be automatically set after a successful tool change and will be saved in the event of a power outage.

5) Tooltable #1 (1-10) / Tooltable #2 (11-20)

NOTE: Due to the large number of parameters, the loading and setting times may be longer. It is recommended to fill in all parameters before clicking on the **[Apply]** button.

T#	Safe X	Safe Y	Safe Z	ToolIndex Position	ToolChange Clearance	Safe Pos	Tool Pos X	Tool Pos Y	Tool Un- clamp Z	Tool L
1	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	1.000
3	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	2.000
4	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
5	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
6	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
7	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
8	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
9	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
10	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Apply
Read
Back

T#	Safe X	Safe Y	Safe Z	ToolIndex Position	ToolChange Clearance	Safe Pos	Tool Pos X	Tool Pos Y	Tool Un- clamp Z	Tool L
11	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
12	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
13	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
14	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
15	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
16	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
17	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
18	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
19	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
20	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Apply
Read
Back

Different tool change procedures in different modes of tool change:

- **Mode 0:**

Ignore tool change.

- **Mode 1:**

- a. Move the spindle to the safe Z coordinate (machine coordinate, not workpiece coordinate) of the current tool number. If the spindle and coolant are on, they will be automatically turned off. Then, the tool magazine door opens and waits for a delay.
- b. If a turret axis is set, it will rotate to the corresponding tool index position of the current tool number, then move to the safe XY coordinates (machine coordinate, not workpiece coordinate), and finally move to the un-clamp Z coordinate (machine coordinate, not workpiece coordinate).
- c. Move the spindle to the XY coordinates (machine coordinate, not workpiece coordinates) of the current tool position, close the M6 relay, and output the tool release signal. After a certain delay, raise the spindle to the new tool's tool change clearance Z coordinate (machine coordinate, not workpiece coordinate). If the tool turret axis is set, it will rotate to the tool index position of the new tool.
- d. Move the spindle to the tool position XY coordinate (machine coordinate, not workpiece coordinate) of the new tool number, then move the spindle to the unclamp Z coordinate (machine coordinate, not workpiece coordinate) of the new tool number. M6 relay disengaged and perform a tool clamping. After a certain delay, the spindle moves to the safe XY coordinates (machine coordinate, not workpiece coordinates) of the new tool number, then move to the safe Z coordinate of the new tool number for tool length compensation. If the tool turret axis is set, it will rotate to the safe position.
- e. Move the spindle to the XY coordinates (workpiece coordinates) before tool change, close the tool magazine door. If the spindle and coolant were on before the tool change, they will be turned back on automatically. After a certain delay, the process will be completed and resume machining.

- **Mode 2:**

- a. Move the spindle to the safe Z coordinate (machine coordinate, not workpiece coordinate) of the current tool number. If the spindle and coolant are on, they will be automatically turned off. Then, being dwell and waiting for manual tool change.
- b. After the manual tool change is completed, press the **[Cont.]** button for automatic tool length compensation. If the spindle and coolant were on before the tool change, they will be turned back on automatically. After a delay, the spindle moves to the Z coordinate (in workpiece coordinates) before tool change. Finally resume machining.

- **Mode 3:**

- a. Move the spindle to the safe Z coordinate (machine coordinate, not workpiece coordinate) of the current tool number. If the spindle and coolant are on, they will be automatically turned off. Then, the tool magazine door opens and waits for a delay.
- b. The turret axis should be set first, so that it will rotate to the tool index position of the current tool number. Then move to the un-clamp Z coordinate (machine coordinate, not workpiece coordinate).
- c. After that, execute the cylinder action to push out the tool turret, followed by a delay. Then, the M6 relay will close and output a signal to release the old tool, followed by another delay. The spindle then move up to the new tool's clearance Z height coordinate (machine coordinate, not workpiece coordinate). If the tool turret axis is set, it will rotate to the tool index position of the new tool.
- d. Move the spindle to the unclamp Z coordinate (machine coordinate, not workpiece coordinate) of the new tool number. Disengage the M6 relay to clamp the new tool, wait for a delay, and then execute the cylinder action to retract the tool turret. Wait for another delay, move the spindle to the safe Z coordinate (machine coordinate, not workpiece coordinate) of the new tool. After that, apply a tool length compensation. If the turret axis is set, it will rotate to the safe position.
- e. Move the spindle to the XY coordinates (workpiece coordinates) before tool change, close the tool magazine door. If the spindle and coolant were on before the tool change, they will be turned back on automatically. After a certain delay, the process will be completed and resume machining.

- **Mode 4 (use tool #1 for tool setting, then apply an automatic "First TLM" in the "Probe" page after the resetting following tool setting):**

- a. Move the spindle to the safe Z coordinate (machine coordinate, not workpiece coordinate) of the current tool number. If the spindle and coolant are on, they will be automatically turned off. Then, the tool magazine door opens and waits for a delay.
- b. If a turret axis is set, it will rotate to the corresponding tool index position of the current tool number, then move to the safe XY coordinates (machine coordinate, not workpiece coordinate), and finally move to the un-clamp Z coordinate (machine coordinate, not workpiece coordinate).
- c. Move the spindle to the XY coordinates (machine coordinate, not workpiece coordinates) of the current tool position, close the M6 relay, and output the tool release signal. After a certain delay, raise the spindle to the new tool's tool change clearance Z coordinate (machine coordinate, not workpiece coordinate). If the tool turret axis is set, it will rotate to the tool index position of the new tool.
- d. Move the spindle to the tool position XY coordinates (machine coordinate, not

workpiece coordinates) of the new tool number, then move the spindle to the unclamp Z coordinate (machine coordinate, not workpiece coordinate) of the new tool number. M6 relay disengaged and perform a tool clamping. After a certain delay, the spindle moves to the safe XY coordinates (machine coordinate, not workpiece coordinates) of the new tool number, then move to the safe Z coordinate of the new tool number for tool length compensation.

- e. Move to the tool setting XY coordinates (machine coordinate, not workpiece coordinates) and perform an automatic tool length measurement and compensation. After that, move the spindle to the safe Z coordinate (machine coordinate, not workpiece coordinate) of the new tool. If the turret axis is set, it will rotate to the safe position.
 - f. Move the spindle to the XY coordinates (workpiece coordinates) before tool change, close the tool magazine door. If the spindle and coolant were on before the tool change, they will be turned back on automatically. After a certain delay, the process will be completed and resume machining.
- **Mode 5** (use tool #1 for tool setting, then apply an automatic “First TLM” in the “Probe” page after the resetting following tool setting):
 - a. Move the spindle to the safe Z coordinate (machine coordinate, not workpiece coordinate) of the current tool number. If the spindle and coolant are on, they will be automatically turned off. Then, being dwell and waiting for manual tool change.
 - b. After the manual tool change is completed, press the **[Cont.]** button.
 - c. Move the spindle to the tool setting XY coordinates (the XY machine coordinate, not workpiece coordinates set in the “**SETTING**” page). Start automatic tool length measurement and compensation. Once completed, move the spindle to the safe Z-coordinate (machine coordinate, not workpiece coordinate) of the new tool. Automatic tool length compensation is performed. If the spindle and coolant were on before the tool change, they will automatically turn on again. Wait for a period of time and move the spindle to the XY-coordinate (workpiece coordinates) before the tool change. Then the process will be completed and resume machining.
 - **Mode 6:**
 - a. Move the spindle to the safe Z coordinate (machine coordinate, not workpiece coordinate) of the current tool number. If the spindle and coolant are on, they will be automatically turned off. Then, being dwell and waiting for manual tool change.
 - b. The turret axis should be set first, so that it will rotate to the tool index position of the current tool number. Then move to the un-clamp Z coordinate (machine coordinate, not workpiece coordinate).
 - c. After that, execute the cylinder action to push out the tool turret, followed by a delay.

Then, the M6 relay will close and output a signal to release the old tool, followed by another delay. The spindle then move up to the new tool's clearance Z height coordinate (machine coordinate, not workpiece coordinate). If the tool turret axis is set, it will rotate to the tool index position of the new tool.

- d. Move the spindle to the unclamp Z coordinate (machine coordinate, not workpiece coordinate) of the new tool number. Disengage the M6 relay to clamp the new tool, wait for a delay, and then execute the cylinder action to retract the tool turret. Wait for another delay, move the spindle to the safe Z coordinate (machine coordinate, not workpiece coordinate) of the new tool. If the turret axis is set, it will rotate to the safe position. After that, move the spindle to the tool setting XY-coordinate (machine coordinate, not workpiece coordinate) and apply a tool length measurement and a tool length compensation. Finally move the spindle to the safe Z coordinate (machine coordinate, not workpiece coordinate) of the new tool and perform a automatic tool length compensation.
- e. Move the spindle to the XY coordinates (workpiece coordinates) before tool change, close the tool magazine door. If the spindle and coolant were on before the tool change, they will be turned back on automatically. After a certain delay, the spindle moves to the Z coordinate (in workpiece coordinates) before tool change. Finally resume machining.

● **Mode 7:**

- a. Move the spindle to the safe Z coordinate (machine coordinate, not workpiece coordinate) of the current tool number. If the spindle and coolant are on, they will be automatically turned off. Then, the tool magazine door opens and waits for a delay.
- b. If a turret axis is set, it will rotate to the corresponding tool index position of the current tool number, then move to the safe XY coordinates (machine coordinate, not workpiece coordinate), and finally move to the un-clamp Z coordinate (machine coordinate, not workpiece coordinate).
- c. After that, execute the cylinder action to push out the tool turret, followed by a delay. Then, the M6 relay will close and output a signal to release the old tool, followed by another delay. The spindle move up to the new tool's clearance Z height coordinate (machine coordinate, not workpiece coordinate) and then move to the new tool's XY-coordinate (machine coordinate, not workpiece coordinate). If the tool turret axis is set, it will rotate to the tool index position of the new tool.
- d. Move the spindle to the unclamp Z coordinate (machine coordinate, not workpiece coordinate) of the new tool number. Disengage the M6 relay to clamp the new tool, wait for a delay, and then execute the cylinder action to retract the tool turret. Wait for another delay, move the spindle to the safe Z coordinate (machine coordinate, not workpiece coordinate) of the new tool. If the turret axis is set, it will rotate to the safe position and perform an automatic tool length compensation.

- e. Move the spindle to the XY coordinates (workpiece coordinates) before tool change, close the tool magazine door. If the spindle and coolant were on before the tool change, they will be turned back on automatically. After a certain delay, the process will be completed and resume machining.
- **Mode 8** (use tool #1 for tool setting, then apply an automatic “First TLM” in the “Probe” page after the resetting following tool setting):
 - a. Move the spindle to the safe Z coordinate (machine coordinate, not workpiece coordinate) of the current tool number. If the spindle and coolant are on, they will be automatically turned off. Then, the tool magazine door opens and waits for a delay.
 - b. If a turret axis is set, it will rotate to the corresponding tool index position of the current tool number, then move to the safe XY coordinates (machine coordinate, not workpiece coordinate), and finally move to the un-clamp Z coordinate (machine coordinate, not workpiece coordinate).
 - c. After that, execute the cylinder action to push out the tool turret, followed by a delay. Then, the M6 relay will close and output a signal to release the old tool, followed by another delay. The spindle then move to the new tool's XY-coordinate (machine coordinate, not workpiece coordinate). If the tool turret axis is set, it will rotate to the tool index position of the new tool.
 - d. Move the spindle to the unclamp Z coordinate (machine coordinate, not workpiece coordinate) of the new tool number. Disengage the M6 relay to clamp the new tool, wait for a delay, and then execute the cylinder action to retract the tool turret. Wait for another delay, move the spindle to the safe Z coordinate (machine coordinate, not workpiece coordinate) of the new tool. If the turret axis is set, it will rotate to the safe position. After that, move the spindle to the tool setting XY-coordinate (machine coordinate, not workpiece coordinate) and apply a tool length measurement and a tool length compensation. Finally move the spindle to the safe Z coordinate (machine coordinate, not workpiece coordinate) of the new tool and perform a automatic tool length compensation.
 - e. Move the spindle to the XY coordinates (workpiece coordinates) before tool change, close the tool magazine door. If the spindle and coolant were on before the tool change, they will be turned back on automatically. After a certain delay, the process will be completed and resume machining.

4.6.4 RTCP Configuration

Tapping Axis	0	C-axis Eccentricity	0.000	Z-Backlash	0.000	Tool Touch-Off	0
RTP Mode	1	X CC	0.000	A-Backlash	0.000	485 Station No.	1
RTP Digi	0.000	Y CC	0.000	B-Backlash	0.000	485 Band	9600
Pendulum L	0.000	Z CC	0.000	C-Backlash	0.000	485 Channels	18
A-axis Eccentricity	0.000	X-Backlash	0.000			IP:	192.168.1.17
B-axis Eccentricity	0.000	Y-Backlash	0.000			Mode:	STA

WIFI/AP
WIFI/STA
STA SET
NONE
Startup-Config
User-Defineds
Read
Apply

MANUAL
</>CODING
POCENDER
NEXT
RUN

Control Descriptions:

NAME / LOCATION	TYPE	DESCRIPTION
Taping Axis	Configurable Parameter	Tapping axis refers to the axis configuration for mixed wizards. Axis 4 corresponds to the A-axis, axis 5 corresponds to the B-axis, and axis 6 corresponds to the C-axis. These axis assignments are used for precise control during tapping operations.
RTP Mode	Configurable Parameter	In RTP (Rotary Tool Center Point) mode, the G43.4H1 code is used to enable tool length compensation for the specified tool (in this case, tool number 1) based on the rotary axis position. This mode allows for the adjustment of tool lengths based on the rotation of the tool, ensuring accurate machining in rotary applications. For more detailed information, please refer to the RTP mode settings section.

RTCP Digit		Unused parameter
Pendulum L	Configurable Parameter	Refers to the pendulum length setting of the A/B-axis swivel head structure. By manually measuring the pendulum length of T1 and inputting it, clicking [Apply] and then performing the first tool length measurement on the “Probe” page as a reference, automatic pendulum length measurement can be performed from now on. As long as the Z-axis origin position and the tool setter remain unchanged, the manually measured pendulum length is valid for life.
A/B/C-axis Eccentricity	Configurable Parameter	Eccentricity setting. For more detailed information, please refer to the Eccentricity Settings section.
X/Y/Z CC	Configurable Parameter	Click to set the calibration coordinates.
X/Y/Z/A/B/C-Backlash	Configurable Parameter	This parameter defines the clearance or play in the XYZABC axes during movement. It represents the amount of free movement or gap between the components of the axes. The value should be specified as a non-negative number. Setting the value to 0 disables the backlash compensation for the corresponding axis.
Tool Touch-Off	Configurable Parameter	When set to 0, the tool touch-off detection is ignored, and the delay function is used. When set to 1, the safety detection of the tool clamping cylinder is enabled. If the touch-off is not successful within 5 seconds, the system will pause. When set to 2, both the safety detection of the tool clamping cylinder and the detection of the tool turret retracting cylinder signal are enabled. If the touch-off is not successful within 5 seconds, the system will pause.
485 Station No.	Configurable Parameter	Set 1
485 Band	Configurable Parameter	Set 38400 for 16 channels and 9600 for 8 channels/4 channels.

485 Channels	Configurable Parameter	Set 4 or 18 for 4 channels, 8 or 18 for 8 channels, and 16 or 20 for 16 channels. There are multiple models available, and specific values can be obtained by contacting the seller for confirmation.
IP Address	Display Field	Enter this IP address in the client to access the device control and wireless file transfer page.
Mode	Display Field	Displays the wifi mode.
WIFI/AP	Button	Click to enable the WIFI AP mode, which sends out a WiFi signal for other devices to connect to. Upon clicking, the system will reboot, and the origin must be re-located.
WIFI/STA	Button	Click to enable WIFI STA mode, which allows the controller to connect to other WiFi devices, such as a router. Upon clicking, the system will reboot, and the origin must be re-located.
STA SET	Button	Click to access the "STA Settings" page.
NONE	Button	Click to disable the bluetooth and wifi modes. Upon clicking, the system will reboot, and the origin must be re-located.
Startup-Config	Button	Click to access the "Startup Configuration" page.
User-Defineds	Button	Click to access the "User-Defined Input Buttons Configuration" page.
Read	Button	Click to read the settings. After modifying the settings, click [Read] to check if the parameters have been successfully applied.
Apply	Button	After filling in the parameters, you need to click the [Apply] button twice for the settings to take effect.

Instructions for Parameters Setting:

1) Taping Axis: Tapping axis refers to the axis configuration for mixed wizards. Axis 4 corresponds to the A-axis, axis 5 corresponds to the B-axis, and axis 6 corresponds to the C-axis.

These axis assignments are used for precise control during tapping operations.

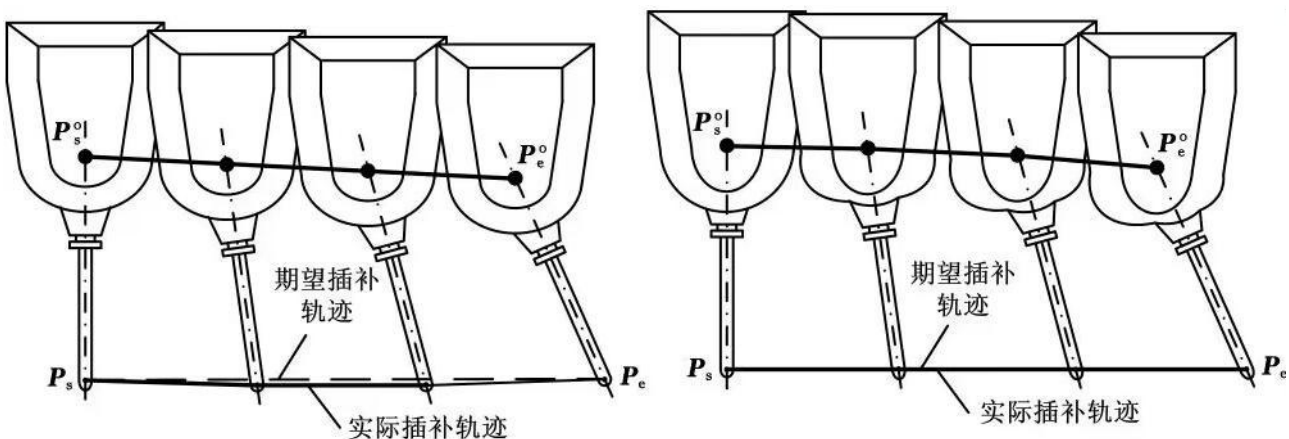
2) X/Y/Z/A/B/C-Backlash: This parameter defines the clearance or play in the XYZABC axes during movement. It represents the amount of free movement or gap between the components of the axes. The value should be specified as a non-negative number. Setting the value to 0 disables the backlash compensation for the corresponding axis.

3) RTP Mode (4-19)

Use G43.4H1 command to activate tool length compensation and RTCP function for T1. The following settings can be applied:

- **Set 4:** Rotate XY plane around X-axis by an angle corresponding to the A-axis angle.
- **Set 5:** Rotate XY plane around Y-axis by an angle corresponding to the B-axis angle.
- **Set 6:** Rotate XY plane around Z-axis by an angle corresponding to the C-axis angle.
- **Set 7:** A-axis swivel head RTCP mode.

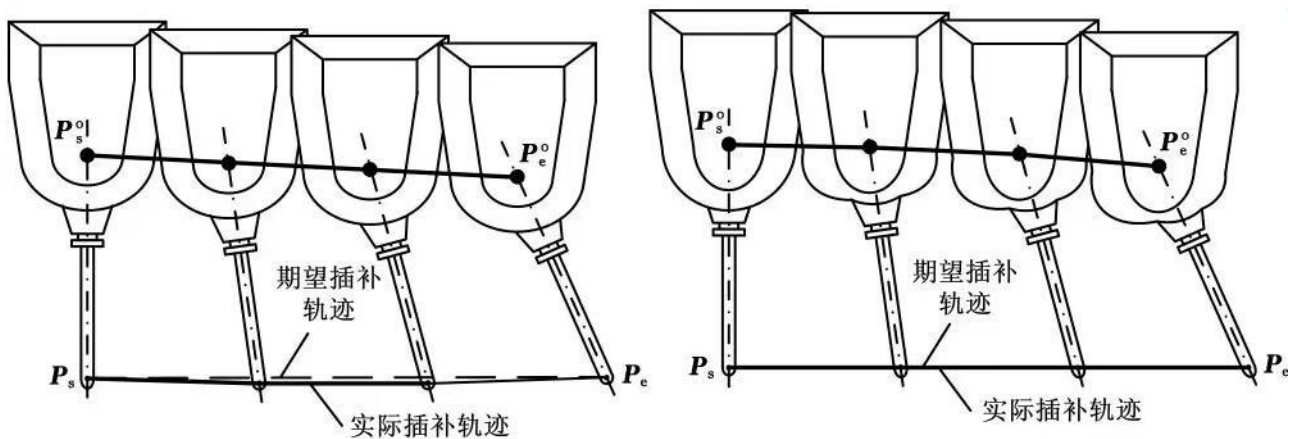
In this mode, the system automatically measures the pendulum length and compensates for the tool tip position to make it the pivot point. This eliminates the need to activate the RTCP function when programming the tool path and the pendulum length and tool length can be set to zero. Additionally, the automatic measurement of pendulum length and compensation of tool tip position can be configured to perform after a tool change. The algorithm in this mode compensates for the left structure in the lower picture and matches it to the right structure.



- **Set 8:** B-axis swivel head RTCP mode.

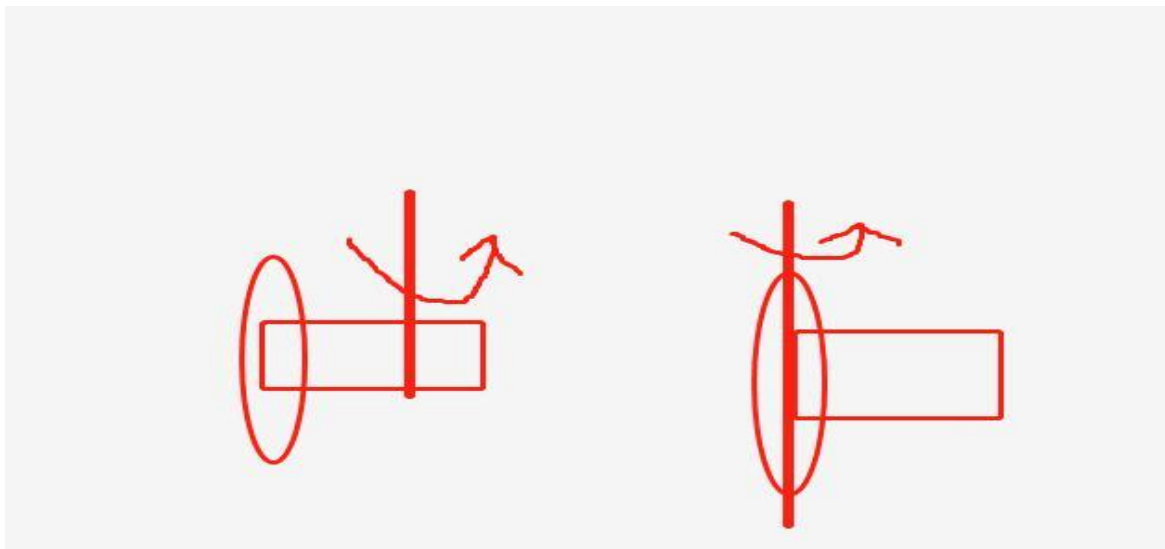
In this mode, the system automatically measures the pendulum length and compensates for the tool tip position to make it the pivot point. This eliminates the need to activate the RTCP function when programming the tool path and the pendulum length and tool length can be set to zero. Additionally, the automatic measurement of pendulum length and compensation of tool tip position can be configured to perform after a tool change. The algorithm in this mode compensates for the left structure in the lower picture and

matches it to the right structure.



- **Set 9:** C-axis swivel head RTCP mode.

This function can be used for 4-axis bridge saw machines. By setting the C-axis offset (i.e., the distance from the saw blade to the rotation center), the system can automatically compensate for the saw blade position and align its center line with the rotation center. The algorithm in this mode compensates for the left structure in the lower picture and matches it to the right structure.



- **Set 10:** C-axis rotary table RTCP mode.
- **Set 11:** B-axis swivel head and C-axis rotary table RTCP mode. Pendulum length can be measured automatically.
- **Set 12:** A-axis rotary table RTCP mode.
- **Set 13:** B-axis swivel head and A-axis rotary table RTCP mode. Pendulum length can be measured automatically.

- **Set 14:** B-axis rotary table RTCP mode.
- **Set 15:** A-C axis tilting table RTCP mode.
- **Set 16:** B-C axis tilting table RTCP mode.
- **Set 17:** A-axis swivel head and C-axis rotary table RTCP mode.
- **Set 18:** C&B axes dual swivel heads RTCP mode.
- **Set 19:** C&A axes dual swivel heads RTCP mode.

4) RTCP Digit: Unused parameter

5) Pendulum L

Refers to the pendulum length setting of the A/B-axis swivel head structure. By manually measuring the pendulum length of T1 and inputting it, clicking **[Apply]** and then performing the first tool length measurement on the “**Probe**” page as a reference, automatic pendulum length measurement can be performed from now on. As long as the Z-axis origin position and the tool setter remain unchanged, the manually measured pendulum length is valid for life.

6) Eccentricity Settings

- **A-axis Eccentricity:** The horizontal distance between the pivot point of the A-axis swivel head and the tool tip.
- **B-axis Eccentricity:** The horizontal distance between the pivot point of the B-axis swivel head and the tool tip.
- **C-axis Eccentricity:** The horizontal distance between the pivot point of the C-axis swivel head and the tool tip.

7) A/B/C-axis PL

Refers to the pendulum length of the A/B/C axis. This parameter is a preview parameter and is read-only. Clicking **[RTCP]** in the “**JOG**” page will quickly enable the RTCP mode for T1, and the pendulum length will be previewed in these boxes. To cancel RTCP mode, you can send the G49 code or click the **[End]** button in the “**MDI**” page.

8) X/Y/Z CC: Click to set the calibration coordinate of X/Y/Z axis.

4.6.5 Debugging Procedures for Different RTCP Structures



CAUTION: Coordinate zeroing should not be performed when RTCP is enabled.



INFORMATION: In manual RTCP mode, testing can only be done using the MPG instead of the moving buttons on the touch screen .

1) B-axis swivel head and C-axis rotary table



Steps:

- a. Home the machine and ensure the current tool number is 1. On the **Probe** page, swing the B-axis to the vertical position and perform a tool length measurement for T1 (**First TLM**) as the reference. Then accurately measure the distance from the B-axis pivot point to the tool tip. Enter this measurement as the swing length in the **Settings** page and click **[Apply]**. Once applied, subsequent tool changes will automatically measure the swing length. As long as the Z-axis origin position and the tool setter remain unchanged, the manual measurement is valid indefinitely.
- b. When the center axis of the tool pivot point does not intersect with the B-axis pivot point, input the eccentricity distance into the “**B-axis eccentricity**” field on the “**SETTING**” page. If they intersect, set it to 0.
- c. Align the B-axis to the vertical direction, move the tool to the C-axis rotation center, and input the current X/Y machine coordinates (not workpiece coordinates) into the corresponding X/Y calibration coordinate field in the “**SETTING**” page.
- d. Set the RTP mode to 11 and click **[Apply]**.
- e. Go to the “**JOG**” page. When not in RTCP mode, swing the B-axis to the vertical direction, then move to the position where the tool tip needs to follow, and reset the XYZBC coordinates (B-axis can be cleared only when it is in the vertical direction). Click **[RTCP]** to activate the function.

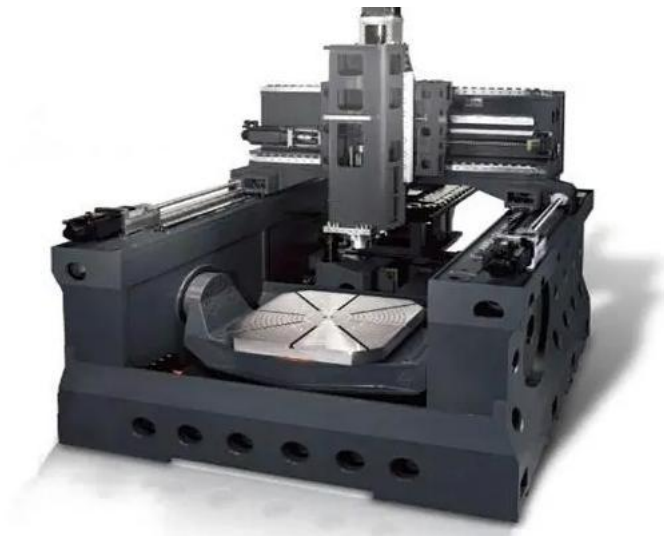
2) B-axis swivel head and A-axis rotary table



Steps:

- a. Home the machine and ensure the current tool number is 1. On the **Probe** page, swing the B-axis to the vertical position and perform a tool length measurement for T1 (**First TLM**) as the reference. Then accurately measure the distance from the B-axis pivot point to the tool tip. Enter this measurement as the swing length in the Settings page and click **[Apply]**. Once applied, subsequent tool changes will automatically measure the swing length. As long as the Z-axis origin position and the tool setter remain unchanged, the manual measurement is valid indefinitely.
- b. When the center axis of the tool pivot point does not intersect with the B-axis pivot point, input the eccentricity distance into the “**B-axis eccentricity**” field on the “**SETTING**” page. If they intersect, set it to 0.
- c. Align the B-axis to the vertical direction, move the tool to the A-axis rotation center, and input the current Y/Z machine coordinates (not workpiece coordinates) into the corresponding Y/Z calibration coordinate field in the “**SETTING**” page.
- d. Set the RTP mode to 13 and click **[Apply]**.
- e. Go to the “**JOG**” page. When not in RTCP mode, swing the B-axis to the vertical direction, then move to the position where the tool tip needs to follow, and reset the XYZBC coordinates (B-axis can be cleared only when it is in the vertical direction). Click **[RTCP]** to activate the function.

3) AC-axis tilting table



Steps:

- a. Home the machine and ensure the current tool number is 1. Align the A-axis with the horizontal plane, move the tool to the C-axis rotation center, and set the current X/Y machine coordinates (not workpiece coordinates) into the corresponding X/Y calibration coordinate field in the “**SETTING**” page.
- b. Go to the **Probe** page and perform a tool length measurement for T1 (**First TLM**) as the reference. Move the Z-axis to bring the tool tip to the height of the A-axis rotation center (not the table), and input the current Z-axis machine coordinate (not workpiece coordinate) into the Z calibration coordinate field on the “**SETTING**” page. Once applied, subsequent tool changes will automatically measure the Z-axis calibration position. As long as the Z-axis origin position and the tool setter remain unchanged, the manual measured Z-axis CC is valid indefinitely.
- c. Set the RTP mode to 15 and click [**Apply**].
- d. Go to the “**JOG**” page. When not in RTCP mode, align the A-axis with the horizontal plane, then move to the position where the tool tip needs to follow, and reset the XYZBC coordinates (A-axis can be cleared only when it is horizontally). Click [**RTCP**] to activate the function.

4) BC-axis tilting table



Steps:

- a. Home the machine and ensure the current tool number is 1. Align the B-axis with the horizontal plane, move the tool to the C-axis rotation center, and set the current X/Y machine coordinates (not workpiece coordinates) into the corresponding X/Y calibration coordinate field in the “**SETTING**” page.
- b. Go to the **Probe** page and perform a tool length measurement for T1 (**First TLM**) as the reference. Move the Z-axis to bring the tool tip to the height of the B-axis rotation center (not the table), and input the current Z-axis machine coordinate (not workpiece coordinate) into the Z calibration coordinate field on the “**SETTING**” page. Once applied, subsequent tool changes will automatically measure the Z-axis calibration position. As long as the Z-axis origin position and the tool setter remain unchanged, the manual measured Z-axis CC is valid indefinitely.
- c. Set the RTP mode to 16 and click [**Apply**].
- d. Go to the “**JOG**” page. When not in RTCP mode, align the B-axis with the horizontal plane, then move to the position where the tool tip needs to follow, and reset the XYZBC coordinates (B-axis can be cleared only when it is horizontally). Click [**RTCP**] to activate the function.

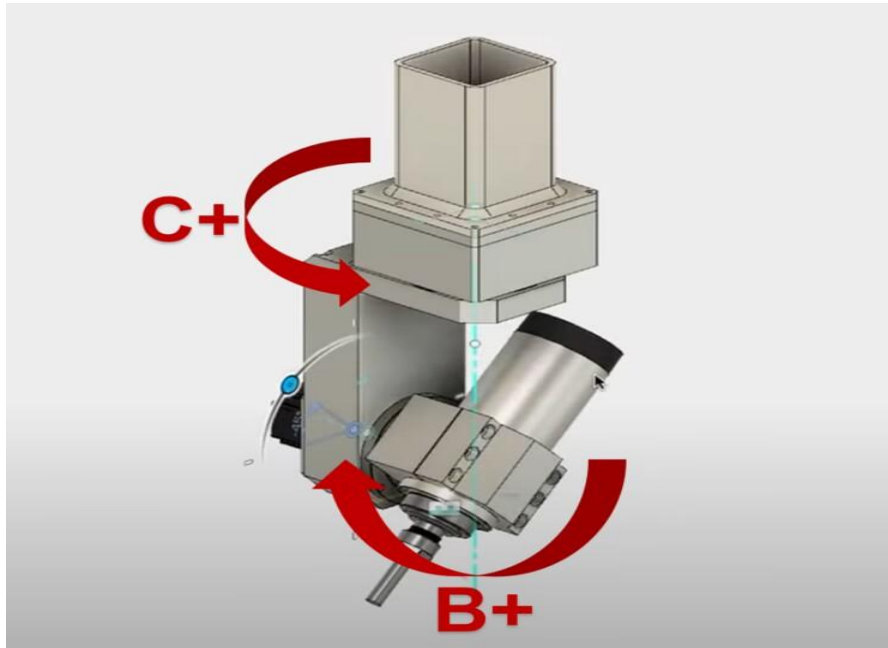
5) A-axis swivel head and C-axis rotary table



Steps:

- a. Home the machine and ensure the current tool number is 1. On the **Probe** page, swing the A-axis to the vertical position and perform a tool length measurement for T1 (**First TLM**) as the reference. Then accurately measure the distance from the A-axis pivot point to the tool tip. Enter this measurement as the swing length in the Settings page and click **[Apply]**. Once applied, subsequent tool changes will automatically measure the swing length. As long as the Z-axis origin position and the tool setter remain unchanged, the manual measurement is valid indefinitely.
- b. When the center axis of the tool pivot point does not intersect with the A-axis pivot point, input the eccentricity distance into the “**A-axis eccentricity**” field on the “**SETTING**” page. If they intersect, set it to 0.
- c. Align the A-axis to the vertical direction, move the tool to the C-axis rotation center, and input the current X/Y machine coordinates (not workpiece coordinates) into the corresponding X/Y calibration coordinate field in the “**SETTING**” page.
- d. Set the RTP mode to 17 and click **[Apply]**.
- e. Go to the “**JOG**” page. When not in RTCP mode, swing the A-axis to the vertical direction, then move to the position where the tool tip needs to follow, and reset the XYZAC coordinates (A-axis can be cleared only when it is in the vertical direction). Click **[RTCP]** to activate the function.

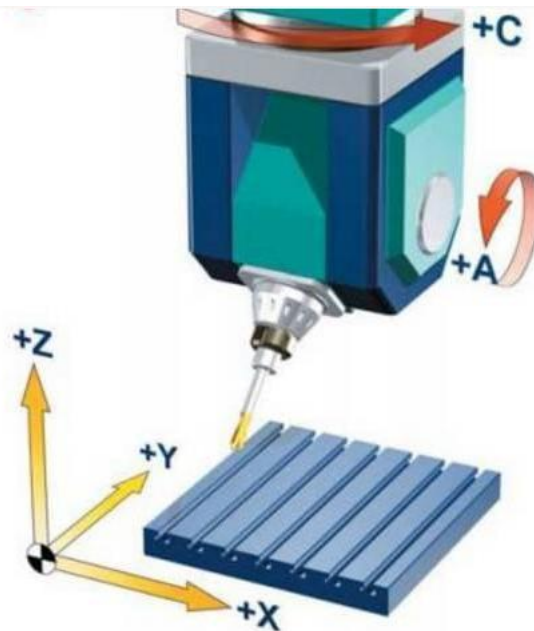
6) C&B axes dual swivel heads



Steps:

- Home the machine and ensure the current tool number is 1. On the **Probe** page, swing the B-axis to the vertical position and perform a tool length measurement for T1 (**First TLM**) as the reference. Then accurately measure the distance from the B-axis pivot point to the tool tip. Enter this measurement as the swing length in the Settings page and click **[Apply]**. Once applied, subsequent tool changes will automatically measure the swing length. As long as the Z-axis origin position and the tool setter remain unchanged, the manual measurement is valid indefinitely.
- When the center axis of the tool pivot point does not intersect with the B-axis pivot point, input the eccentricity distance into the “**B-axis eccentricity**” field on the “**SETTING**” page. If they intersect, set it to 0.
- Set the RTP mode to 18 and click **[Apply]**.
- Go to the “**JOG**” page. When not in RTCP mode, swing the B-axis to the vertical direction and align C-axis to make the rotation axis of B-axis is parallel to the Y-axis. Then move XYZ axes to the position where the tool tip needs to follow, and reset the XYZBC coordinates (B-axis and C-axis can be cleared only when B-axis is in the vertical direction and its rotation center axis is parallel to the Y-axis by aligning C-axis). Click **[RTCP]** to activate the function.

7) C&A axes dual swivel heads



Steps:

- Home the machine and ensure the current tool number is 1. On the Probe page, swing the B-axis to the vertical position and perform a tool length measurement for T1 (First TLM) as the reference. Then accurately measure the distance from the B-axis pivot point to the tool tip. Enter this measurement as the swing length in the Settings page and click [Apply]. Once applied, subsequent tool changes will automatically measure the swing length. As long as the Z-axis origin position and the tool setter remain unchanged, the manual measurement is valid indefinitely.
- When the center axis of the tool pivot point does not intersect with the A-axis pivot point, input the eccentricity distance into the “**A-axis eccentricity**” field on the “**SETTING**” page. If they intersect, set it to 0.
- Set the RTP mode to 19 and click [Apply].
- Go to the “**JOG**” page. When not in RTCP mode, swing the A-axis to the vertical direction and align C-axis to make the rotation axis of A-axis is parallel to the X-axis. Then move XYZ axes to the position where the tool tip needs to follow, and reset the XYZAC coordinates (A-axis and C-axis can be cleared only when A-axis is in the vertical direction and its center axis is parallel to the X-axis by aligning C-axis). Click [RTCP] to activate the function.

4.6.6 485 IO Extension Configuration

Tapping Axis	0	C-axis Eccentricity	0.000	Z-Backlash	0.000	Tool Touch-Off	0
RTP Mode	1	X CC	0.000	A-Backlash	0.000	485 Station No.	1
RTP Digi	0.000	Y CC	0.000	B-Backlash	0.000	485 Band	9600
Pendulum L	0.000	Z CC	0.000	C-Backlash	0.000	485 Channels	18
A-axis Eccentricity	0.000	X-Backlash	0.000			IP:	192.168.1.17
B-axis Eccentricity	0.000	Y-Backlash	0.000			Mode:	STA

WIFI/AP
WIFI/STA
STA SET
NONE
Startup-Config
User-Defineds
Read
Apply

MANUAL
</>CODING
POCENDER
NEXT
RUN

Instructions for Parameters Setting:

1) Tool Release/Clamp Detection:

- Setting it to 0 disables the detection and activates the delay function.
- Setting it to 1 enables the safety check for the tool clamping cylinder. If the check fails within 5 seconds, the system will halt.
- Setting it to 2 enables the safety check for tool clamping cylinder and signal check for the tool turret pop out cylinder. If the check fails within 5 seconds, the system will halt.

2) 485 Station No.: Set 1

3) 485 Band: Set 38400 for 16 channels and 9600 for 18 channels/8 channels/4 channels.

4) 485 Channels: Set 4 for 4 channels, 8 for 8 channels, 16 for 16 channels, and 18 for 18 channels. Various models are available, please consult the seller for specific values and details.

**INFORMATION:**

- 1) The settings are applied by clicking on **[Apply]** and then restarting the system.
- 2) For detailed usage instructions, please refer to Chapter 9 of this Manual titled **"RS485 Expansion Module Interface Terminals Description"**.

4.6.7 Modbus Communication Settings for Servomotor with Absolute Encoder using 485 Protocol

X-POS		X-Station	0	X-Register	0	Function Code	3
Y-POS		Y-Station	0	Y-Register	0	Registers Number	2
Z-POS		Z-Station	0	Z-Register	0	H/L Change	1
A-POS		A-Station	0	A-Register	0		
B-POS		B-Station	0	B-Register	0	IP:	192.168.1.23
C-POS		C-Station	0	C-Register	0	:	STA

WIFI/AP
WIFI/STA
STA SET
NONE
Startup-Config
User-Defineds
Read
Apply

MANUAL
</>CODING
POCENDER
NEXT
RUN

X-Pul/Rev	1600	X-Corder Rev	10000				
Y-Pul/Rev	1600	Y-Corder Rev	10000				
Z-Pul/Rev	1600	Z-Corder Rev	10000				
A-Pul/Rev	1600	A-Corder Rev	10000				
B-Pul/Rev	1600	B-Corder Rev	10000			IP:	192.168.1.23
C-Pul/Rev	1600	C-Corder Rev	10000			Mode:	STA

WIFI/AP
WIFI/STA
STA SET
NONE
Startup-Config
User-Defineds
Read
Apply

MANUAL
</>CODING
POCENDER
SETTING
RUN

Setting Instructions:

- 1) **X/Y/Z/A/B/C-POS:** After clicking **[Read]**, the absolute positions of the servos will be displayed in this box.
- 2) **X/Y/Z/A/B/C-Station:** Set a unique value between 2 to 254 for each servo axis. Each axis should have a different station number.
- 3) **X/Y/Z/A/B/C-Register:** The register address for accessing the servo's absolute position (input as decimal value).
- 4) **Function Code:** The function code used to read the register containing the servo's absolute position. Usually set to 3 or 4.
- 5) **Registers Number:** The number of registers to read. For 32-bit position values, set to 2; for 64-bit position values, set to 4.
- 6) **H/L Change:** Specify the byte order of the data. Use 0 for "high byte first, low byte second" or 1 for "low byte first, high byte second."
- 7) **X/Y/Z/A/B/C-Pul/Rev:** The number of subdivisions for each servo axis. For example, if the motor receives 1600 pulses per revolution, set this value to 1600.
- 8) **X/Y/Z/A/B/C-Coder Rev:** Specify the absolute value encoding received from the servo feedback after one complete revolution. The difference read after one revolution represents the circle encoding value.

4.6.8 STA Settings

Click the **[STA Set]** button on “**SETTING**” page to access.




INFORMATION: The WiFi network name should consist of English letters and numbers. Please note that the device does not support 5G WiFi networks.

Control Descriptions:

NAME / LOCATION	TYPE	DESCRIPTION
Active Wireless SSID	Display Field	Display the current bound WiFi name.
Current Connected IP	Display Field	Display the current IP address obtained, which is the website address that should be filled in the browser of the mobile phone/computer.

Current Mode	Display Field	Display the current working mode: AP/STA/BLUE/NONE.
New Connection Line No.	Display Field (clickable)	Click to set the sequence number in front of the WiFi name that needs to be connected.
New Connection Password	Display Field (clickable)	Click to set the password of the WiFi that needs to be connected.
Refresh	Button	Click to refresh the current bound WiFi name, working mode, and IP address.
Disconnect	Button	Click to disconnect from the current connected WiFi and automatically switch to AP mode.
Search	Button	Click to search for available WiFi networks to connect to.
Connect	Button	After setting the line number and password for the new connection, click this button to connect to the WiFi. The system will reboot, and the coordinate will be lost, requiring rehomings.

4.6.9 Screen Settings

Click the **[Screen Set]** button on “**SETTING**” page to access.



Control Descriptions:

NAME / LOCATION	TYPE	DESCRIPTION
Screen Brightness	Slider Button	Drag the orange circular slider to adjust the screen brightness.
115200	Configurable Parameter	Click to toggle the baud rate. After switching, click [Apply] and reset the screen for the changes to take effect.
Touch Calibration	Button	Perform touch screen calibration. Usually, no need to adjust as it comes pre-calibrated from the factory. Incorrect calibration may require sending the unit back to the manufacturer for repair at your own expense.
Reset	Button	Reboot the screen, restoring it to its initial state.

4.6.10 Startup Configuration

Click the **[Startup-Config]** button on the “**SETTING**” page to access.

Startups		Spindle On-Delay	0.000	S
1		Spindle Off-Delay	0.000	S
2				
3				
4				
5				
6				
7				
8				



INFORMATION:

- 1) Ensure the machine is idle before clicking the **[Apply]** button. This will immediately run all startup items. If any item fails to run, none of them will be saved and take effect. After the application is executed, click on **[Read]** to check if the settings have taken effect.
- 2) If homing function is enabled, the machine will start in an alarm state. Only the homing code **\$HZ** or the force unlock code **\$X** will be executed at this time. After these codes are executed successfully and the machine is unlocked, the subsequent G codes can be executed.
- 3) All subsequent startup items will be ignored if the machine is not successfully unlocked. Only one action or one G code can be included in each line.

Setting Instructions:

1) Startup item examples

CODE	DESCRIPTION
\$HX	Automatically execute X-axis homing at startup, and unlock after successful homing.
\$HY	Automatically execute Y-axis homing at startup, and unlock after successful homing.
\$HZ	Automatically execute Z-axis homing at startup, and unlock after successful homing.
\$HA	Automatically execute A-axis homing at startup, and unlock after successful homing.
\$HB	Automatically execute B-axis homing at startup, and unlock after successful homing.
\$HC	Automatically execute C-axis homing at startup, and unlock after successful homing.
\$X	Unlock forcibly after startup.
G53G0X10	Upon startup, move the X axis to machine coordinate 10 (not workpiece coordinate).
G54G0X10	Upon startup, move the X-axis to the G54 coordinate position of 10.

2) Spindle On-Delay

After executing the M3 code, the subsequent G-code will be automatically delayed by the specified time value before running.

Note: After enabling spindle delay, please do not operate any other buttons/MPG during the delay period after the spindle has been turned on.

3) Spindle Off-Delay

After executing the M5 code, the subsequent G-code will be automatically delayed by the specified time value before running.

Note: After enabling spindle delay, please do not operate any other buttons/MPG during the delay period after the spindle has been turned off.

4.6.11 User-Defined Input Buttons Configuration

Click the **[User-Defined]** button on the “**SETTING**” page to access.

User-Defineds		Spindle On-Delay	0.000	S
1		Spindle Off-Delay	0.000	S
2				
3				
4				
5				
6				
7				
8				



INFORMATION:

- 1) User-defined items configuration is an exclusive function to the IO extension module. Numbers 1-8 corresponds to the 3rd input pin to the 10th input pin of the IO input module, totaling 8 pins.
- 2) When the machine is in alarm state, it will ignore the execution of G/M codes.
- 3) This feature allows for 8 items, and each item can have multiple lines of input separated by the "&" symbol. These lines can contain G codes, M codes, and other relevant commands. The total character limit for each item is 100 characters.

Setting Instructions:

1) Startup item examples

CODE	DESCRIPTION
G90G0X30&Y20	This command instructs the machine to move in absolute positioning mode to the X30 position, followed by the Y20 position. The "&" character serves as a separator between the two lines of code. So this command refers to two separate lines of code: G90G0X30 G0Y20
[ESP220]/JZ1.NC	This command runs the program named "JZ1.NC" located in the root directory of the TF card.
[ESP700]/JZ1.NC	This command runs the program named "JZ1.NC" located in the root directory of the flash memory.
\$x	Clear the alarm.
\$HZ&\$HX&\$HY	Move Z, X, and Y axes to their respective home positions.

4.7 RUN

Click the **[RUN]** button on the bottom right corner of the screen to access this page.



Control Descriptions:

NAME	TYPE	DESCRIPTION
SD Run	Button	Click to run the file in the controller's SD card. For best performance and stability, it is recommended to use the this feature. It can easily handle NC files smaller than 3GB.
Drilling	Button	Click to access the drilling page.
Tapping	Button	Click to access the tapping page.
Cut Circle	Button	Click to access the "Cut Circle" page to create spiral-shaped toolpaths for milling circular holes.
Cut Rectangle	Button	Click to access the "Cut Rectangle" page to create spiral-shaped toolpaths for milling rectangular holes.

USB Run	Button	Click to run the file in the USB drive.
Rect Pocket	Button	Click to access the "Rect Pocket" page to cut rectangular pockets.
Circular Pocket	Button	Click to access the "Circular Pocket" page to cut circular pockets.
Cut RoundRect	Button	Click to access the "Cut RoundRect" page to create spiral-shaped toolpaths for milling round-cornered rectangular holes.
Surfacing	Button	Click to access the "Surfacing" page for layered surface cleaning.
IO Monitor	Button	Click to access the "IO Monitor" page.
Slotting	Button	Click to access the "Slotting" page for milling slots.
Mixed Wizards	Button	Click to access the "Mixed Wizards" page for multi-wizard combined processing.
JOG	Button	Click to access the jogging page.
MDI	Button	Click to access the manual data input page.
SETTING	Button	Click to access the device parameters setting page.
RUN	Button	Click to access the run options page.

4.8 Error Report

Click on the error code display area to access this page.

Error

- 1.G-code words consist of a letter and a value. Letter was not found.
- 2.Missing the expected value or numeric value format is not valid.
- 3.System command was not recognized or supported.
- 4.Negative value received for an expected positive value.
- 5.Homing is not enabled via settings. Please navigate to the settings page to activate it.
- 6.Minimum step pulse time must be greater than 3usec.
- 7.An EEPROM read failed. Auto-restoring affected EEPROM to default values.
- 8.System command cannot be used unless POCENDER is IDLE.
- 9.Being locked out during alarm state. Please unlock and operate.
- 10.Soft limits cannot be enabled without HOMING also enabled.

ERR PD
ERR PU
ALM PD
ALM PU
Back

Control Descriptions:

NAME	TYPE	DESCRIPTION
Back	Button	Click to go back to previous page.
ERR PD	Button	Click to go to the next page.
ERR PU	Button	Click to go to the previous page.
ALM PD	Button	Click to go to the next page.
ALM PU	Button	Click to go to the previous page.

4.9 SD Run

Click the [SD Run] button on the “RUN” page to access.



Page 1



Page 2



INFORMATION: If the status bar displays an "Alarm" message, clicking on it will forcefully clear the alarm; if any other status, clicking on it will cancel the action. During program execution, the page will be locked, and some keys may not function. To unlock, please press **[End]**.

Control Descriptions:

NAME / LOCATION	TYPE	DESCRIPTION
The big display box on the upper left of the screen.	Display field	Displays the files in the controller's SD card. Please name the files in English. You can scroll to view all files. Click on the name of the program to be run to select it.
The box to the left of [Run] / [End]	Display field	Displays the current feedrate override.
Run	Button	Click to run the selected file. After clicking, the screen will switch to page 2 (the screen will be locked), and clicking [End] will stop the operation and unlock the screen.
Delete	Button	Click to delete the selected file.
Spindle Off Spindle On	Buton	Displays the spindle status and click to switch.
Pause Cont.	Button	Switch button for [Pause] and [Cont] . When [Pause] is clicked, please do not perform any operation other than [Cont.] and [End]
The box to the right of [Delete]	Display field (clickable)	Click to set the spindle speed.
CLNT Off CLNT On	Buton	Displays the spindle status and click to switch.
Refresh	Buton	Click to refresh the controller's SD card files. This is necessary when inserting or removing the SD card. The system will also retrieve the line number that was last executed in the previous run of the program, which is useful for power loss recovery (the line number will be a few lines before the last executed line).

Set Next Line Cancel Next Line	Buton	 Method of Use: If you need to reconfigure, click [Cancel Next Line] first, then reconfigure. 1. Steps: 1) In case of a power failure, return to the home position successfully and switch to the corresponding workpiece coordinate system for the desired program. 2) Enter the line number of the breakpoint, then click [Set Next Line], and select the preferred starting method for resuming machining. 3) Choose the program and click [Run] to resume machining. 2. Resuming Machining After Power Failure based on Axis Configuration: 1) 3/4 Axis - Fixed Line: Resume machining from the specified line number in the program. 2) 3/4/5 Axis -Find X: Resume machining by manually moving the X-axis to an approximate position near the desired X point. Then locate the closest X value in the program and initiate machining. 3) 3/4/5 Axis -Find Y: Resume machining by manually moving the X-axis to an approximate position near the desired X point. Then locate the closest Y value in the program and initiate machining. 4) AC/BC Tilting Table & 5 Axis B Table A Head: The system will first perform a quick search to find the closest
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		position for the rotational axes. Then, it will move the XY axes to the breakpoint position, and finally, the Z-axis will precisely position itself to resume machining. 5) 5 Axis B Table C Head: The system will initially find the closest position for the rotational axis. Subsequently, it will move the XZ axes to the breakpoint location, and finally, the Y-axis will precisely position itself to resume machining.
Tool Change	Buton	Click to change to the new tool specified in the left input box. The current tool number is displayed in the left box and can be changed by clicking.
Error display bar below the big display box	Display field	Displays the current error. "NONE" indicates no error.
Status display field on the right of the error display bar	Display field	Displays the current machine status.
Progress	Display field	Displays the current progress of the running task.
Ongoing	Display field	Displays the file currently being executed.
F100, F+10, F+1, F-1, F-10	Button	Click to adjust the feed rate in real time.
S100, S+1, S-1, S+10, S-10	Button	Click to adjust the spindle speed in real time.
G0/1	Button	Click to set rapid positioning feedrate to 100%.
G0/2	Button	Click to set rapid positioning feedrate to 50%.
G0/4	Button	Click to set rapid positioning feedrate to 25%.

Step-by-step Procedure for Manual Guide with MPG Handwheel:

- 1) After clicking **[Run]**, press the **[Pause]** button on the touch screen.
- 2) Set the handwheel axis to neutral mode.
- 3) Rotate the handwheel clockwise to enter manual guided forward machining mode.
- 4) To exit the manual guiding mode, click **[Cont.]** on the touch screen.

4.10 List Machining

Click the **[SGL-point Milling]** button on the wizard pages will switch to **[Multi-point Milling]** mode and enter the this page.

Line	Center X	Center	Z Start	Z End	Circle R / Length X	Width Y	Fillet / A
1	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3	0.000	0.000	0.000	0.000	0.000	0.000	0.000
4	0.000	0.000	0.000	0.000	0.000	0.000	0.000
5	0.000	0.000	0.000	0.000	0.000	0.000	0.000
6	0.000	0.000	0.000	0.000	0.000	0.000	0.000
7	0.000	0.000	0.000	0.000	0.000	0.000	0.000
8	0.000	0.000	0.000	0.000	0.000	0.000	0.000
9	0.000	0.000	0.000	0.000	0.000	0.000	0.000
10	0.000	0.000	0.000	0.000	0.000	0.000	0.000
11	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Save
Read
Back



INFORMATION: The names of the **[SGL-point Milling]** button and **[Multi-point Milling]** button may vary depending on the specific wizard page. Here are some other possible button names:

- 1) **[SGL-point Drilling]**, **[SGL-point Tapping]** and **[SGL-pos Milling]**
- 2) **[List Drilling]**, **[Multi-point Tapping]** and **[Multi-pos Milling]**

Configuration Process:

- a. Click **[SGL-point Milling]** on the wizard page to switch to **[Multi-point Milling]**.
- b. After filling in parameters or reading the previously saved parameters on the “**List Machining**” page, return to the wizard page.
- c. Enter the start line number from which the parameters need to be obtained from the “**List Machining**” page.

- d. Enter the end line number from which the parameters need to be obtained from the “**List Machining**” page.
- e. Set all the basic machining parameters highlighted in green..
- f. Select and click on the machining mode button, and the parameters will be automatically obtained from the corresponding start line in the “**List Machining**” page and run until the end line is completed.

4.11 Drilling

Click the **[Drilling]** button on the “**RUN**” page to access.

Drill		Idle		X-coord:	2.497	Y-coord:	0.978
Safe Z	5	Feedrate	2000	A-Axis Index	0.000	Z-coord:	-9.331
Z Start	0.200	Center X	0.000	Number of Copies in X	1	Spindle RPM +10%	
Z End	0.000	Center Y	0.000	Number of Copies in Y	1	Spindle RPM -10%	
Down Step	5.000			X Spacing	1.000	SGL-point Drilling	
Spindle RPM	2000			Y Spacing	1.000	Centering	
Spindle Delay	5			Feedrate +10%		Deep-Hole Drilling	
Start Line	1	Save		Feedrate -10%		Peck Drilling	
End Line	1	Read		Retract to Start Plane		Pause	

MANUAL
</>CODING
POCENDER
SETTING
→RUN

Parameter Descriptions:

- 1) **Safe Z:** Refers to the Z-coordinate of the tool at a safe distance above the workpiece after completing a machining operation. It should be set higher than the start and end depths to ensure safe tool movement during the entire process.
- 2) **Z Start:** The starting depth of Z-axis workpiece coordinate for drilling operation.
- 3) **Z End:** The ending depth of Z-axis workpiece coordinate for drilling operation.(End depth should be less than start depth.)
- 4) **Down Step:** The depth at which the drill goes into the material at each pass (must be greater than 0) in a drilling process.
- 5) **Spindle RPM:** Setting for the spindle speed.
- 6) **Spindle Delay:** The time delay setting for spindle to start after activation, measured in seconds.
- 7) **Start Line:** The line number from which to start reading parameters for list processing.

- 8) **End Line:** The line number at which to end reading parameters for list processing.
- 9) **Feedrate:** Setting for the drilling feed rate in units of mm/min.
- 10) **Center X/Y:** Setting for the X/Y coordinates of the drilling center in a regular drilling operation, or the X/Y coordinates of the first reference drilling center in a matrix drilling operation.
- 11) **A-Axis Index:** Setting for the A-axis coordinate position for drilling operations.
- 12) **Number of Copies in X:** Setting for the number of holes to be drilled in the X-axis direction during matrix drilling, must be greater than zero.
- 13) **Number of Copies in Y:** Setting for the number of holes to be drilled in the Y-axis direction during matrix drilling, must be greater than zero.
- 14) **X Spacing:** Setting for the distance between two adjacent holes on the X-axis direction during matrix drilling, must be greater than zero.
- 15) **Y Spacing:** Setting for the distance between two adjacent holes on the Y-axis direction during matrix drilling, must be greater than zero.

Control Descriptions:

NAME	TYPE	DESCRIPTION
Save	Button	Click to save the parameters in red font on the current page to the system memory.
Read	Button	Click to save the red font parameters on the current page to the system memory.
Feedrate +10%	Button	Increase feed rate by 10% in real time.
Feedrate -10%	Button	Decrease feed rate by 10% in real time.
Spindle RPM +10%	Button	Increase spindle speed by 10% in real time.
Spindle RPM -10%	Button	Decrease spindle speed by 10% in real time.
Retract to Start Plane	Button	Display the return plane for peck drilling. Click to switch to [Retract D Distance] .

SGL-point Drilling Matrix Drilling List Drilling	Button	A button to switch between [SGL-point Drilling], [Matrix Drilling], and [List Drilling]. SGL-point Drilling: Machining one hole with the parameters set on the current page. Matrix Drilling: Machining multiple holes in a matrix form with the parameters set on the current page. List Drilling: Automatically loading the start and end lines of the "List Machining" page and machining multiple holes with different parameters.
Centering	Button	Click to move the axis rapidly to the center position above the workpiece coordinate point for precise positioning.
Deep-Hole Drilling	Button	This function allows for drilling a hole in one pass, with the tool going all the way to the bottom of the hole. After completion, the tool automatically retracts to a safe height. To activate, click the [Deep-Hole Drilling] button and the screen will lock. To unlock the screen and emergency stop, click the [End] button.
Peck Drilling	Button	This function allows for drilling with a retracting cycle to remove chips and prevent tool breakage. The drill bit moves up and down by the layer thickness set on "Down Step". The operation ends with a final retraction to the safe height. To activate, click the [Peck Drilling] button and the screen will lock. To unlock the screen and emergency stop, click the [End] button.
Pause Cont.	Button	Click to pause/resume operation.

4.12 Tapping

Click the [Tapping] button on the “RUN” page to access.

Tapping		Idle	X-coord: 2.497	Y-coord: 0.978			
Safe Z	5	Pitch	0.200	Center X	0.000	Z-coord:	-9.331
Z Start	0.200	Feedrate	2000	Center Y	0.000	Centering	
Z End	0.000		A-Axis Index	0.000	Query		
Down Step	0.200		Servo Spindle	B	Retract to Start Plan		
Start Line	1		Feedrate +10%	CW	Straight Tapping		
End Line	1		Feedrate -10%		Back And Forth		
SGL-point Tapping		Save	Spindle RPM +10%	CCW	Straight Tapping		
Pause		Read	Spindle RPM -10%		Back And Forth		

 MANUAL
  CODING
 POCENDER
  SETTING
  RUN

INFORMATION:



This function requires the use of a servo spindle (position mode) connected to the B/C axis (switchable).

The pulse equivalent of the axis should be set to the number of pulses required for the motor to rotate one degree of the tap (for machines purchased before June 27, 2023, with firmware not updated, the pulse equivalent corresponds to the number of pulses required for the spindle to rotate one revolution of the tap).

When not used as the tapping spindle, it can be used as a regular drilling/milling spindle. To do so, switch the servo mode to speed mode and adjust the speed with the AVI signal.

Parameter Descriptions:

- 1) **Safe Z:** Refers to the Z-coordinate of the tool at a safe distance above the workpiece after completing a machining operation. It should be set higher than the start and end depths to ensure safe tool movement during the entire process.
- 2) **Z Start:** The starting depth of Z-axis workpiece coordinate for tapping operation.
- 3) **Z End:** The ending depth of Z-axis workpiece coordinate for tapping operation.(End depth should be less than start depth.)
- 4) **Down Step:** Refers to the depth of each thread cutting up and down, and should be greater than 0.
- 5) **Start Line:** The line number from which to start reading parameters for list processing.
- 6) **End Line:** The line number at which to end reading parameters for list processing.
- 7) **Pitch:** Setting the pitch of the thread.
- 8) **Feedrate:** Setting for the tapping feed rate in units of mm/min.
- 9) **Center X/Y:** Setting for the XY coordinates of the tapping position.
- 10) **A-Axis Index:** Setting for the A-axis coordinate position for tapping operations.
- 11) **Servo Spindle:** Setting for switching the servo spindle between B/C axes for tapping.

Control Descriptions:

NAME	TYPE	DESCRIPTION
Save	Button	Click to save the parameters in red font on the current page to the system memory.
Read	Button	Click to save the red font parameters on the current page to the system memory.
Feedrate +10%	Button	Increase feed rate by 10% in real time.
Feedrate -10%	Button	Decrease feed rate by 10% in real time.
Spindle RPM +10%	Button	Increase spindle speed by 10% in real time.
Spindle RPM -10%	Button	Decrease spindle speed by 10% in real time.

SGL-point Tapping Multi-point Tapping	Button	A button to switch between [SGL-point Tapping] and [Multi-point Tapping]. SGL-point Tapping: Machining one hole with the parameters set on the current page. Multi-point Tapping: Automatically loading the start and end lines of the "List Machining" page and machining multiple holes with different parameters.																																				
Centering	Button	Click to move the axis rapidly to the center position above the workpiece coordinate point for precise positioning.																																				
Query	Button	 <table> <thead> <tr> <th>Metric</th><th>Coarse Thread</th><th>Fine Thread</th></tr> </thead> <tbody> <tr><td>M1</td><td>0.25</td><td>0.2</td></tr> <tr><td>M1.1</td><td>0.25</td><td>0.2</td></tr> <tr><td>M1.6</td><td>0.35</td><td>0.2</td></tr> <tr><td>M2</td><td>0.4</td><td>0.25</td></tr> <tr><td>M2.5</td><td>0.45</td><td>0.35</td></tr> <tr><td>M3</td><td>0.5</td><td>0.35</td></tr> <tr><td>M4</td><td>0.7</td><td>0.5</td></tr> <tr><td>M5</td><td>0.8</td><td>0.5</td></tr> <tr><td>M6</td><td>1</td><td>0.75,(0.5)</td></tr> <tr><td>M8</td><td>1.25</td><td>1,0.75,(0.5)</td></tr> <tr><td>M10</td><td>1.5</td><td>1.25,1,0.75,(0.5)</td></tr> </tbody> </table> Click to enter the thread pitch standard inquiry page.	Metric	Coarse Thread	Fine Thread	M1	0.25	0.2	M1.1	0.25	0.2	M1.6	0.35	0.2	M2	0.4	0.25	M2.5	0.45	0.35	M3	0.5	0.35	M4	0.7	0.5	M5	0.8	0.5	M6	1	0.75,(0.5)	M8	1.25	1,0.75,(0.5)	M10	1.5	1.25,1,0.75,(0.5)
Metric	Coarse Thread	Fine Thread																																				
M1	0.25	0.2																																				
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M1.6	0.35	0.2																																				
M2	0.4	0.25																																				
M2.5	0.45	0.35																																				
M3	0.5	0.35																																				
M4	0.7	0.5																																				
M5	0.8	0.5																																				
M6	1	0.75,(0.5)																																				
M8	1.25	1,0.75,(0.5)																																				
M10	1.5	1.25,1,0.75,(0.5)																																				

Straight Tapping	Button	This function allows for tapping to the bottom in one pass. After completion, the tool automatically retracts to a safe height. To activate, click the [Straight Tapping] button and the screen will lock. To unlock the screen and emergency stop, click the [End] button.
Back and Forth	Button	This function allows for tapping with a retracting cycle to remove chips and prevent tool breakage. The tap moves up and down by the layer thickness set on "Down Step". The operation ends with a final retraction to the safe height. To activate, click the [Back and Forth] button and the screen will lock. To unlock the screen and emergency stop, click the [End] button.
Pause Cont.	Button	Click to pause/resume operation.

4.13 Cut Circle

Click the **[Cut Circle]** button on the “**RUN**” page to access.



Cut Circle		Idle	X-coord: 2.497	Y-coord: 0.978			
Safe Z	5	Feedrate	2000	Spindle RPM	2000	Z-coord:	-9.331
Z Start	0.200		Spindle Delay	5	Spindle RPM +10%		
Z End	0.000		Center X	0.000	Spindle RPM -10%		
Down Step	0.200		Center Y	0.000	Centering		
Tool R	0.000		Feedrate +10%		Climb	Spiral External Milling	
Machining R	0.000		Feedrate -10%			Spiral Internal Milling	
Start Line	1	SGL-point Milling	Pause		Conv.	Spiral External Milling	
End Line	1	Save	Read			Spiral Internal Milling	

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Parameter Descriptions:

- Cut Circle:** Click to switch to the threading function. **Note that the threading function requires the value of (ending depth - starting depth) to be an integer multiple of the layer height.**
- Safe Z:** Refers to the Z-coordinate of the tool at a safe distance above the workpiece after completing a machining operation. It should be set higher than the start and end depths to ensure safe tool movement during the entire process.
- Z Start:** The starting depth of Z-axis workpiece coordinate for cutting a circle in a spiral tool path.
- Z End:** The ending depth of Z-axis workpiece coordinate for cutting a circle in a spiral tool path.(End depth should be less than start depth.)
- Down Step:** The depth of each pass in the spiral cut, which should be greater than 0.
- Tool R:** Sets the radius of the tool and automatically applies tool compensation. The resulting hole size will be the diameter specified as the machining radius.

- 7) **Machining R:** Setting for the radius of the circle to be cut. For spiral internal milling, the radius should be larger than the tool radius.
- 8) **Start Line:** The line number from which to start reading parameters for list processing.
- 9) **End Line:** The line number at which to end reading parameters for list processing.
- 10) **Feedrate:** Setting for the milling feed rate in units of mm/min.
- 11) **Spindle RPM:** Setting for the spindle speed.
- 12) **Spindle Delay:** The time delay setting for spindle to start after activation, measured in seconds.
- 13) **Center X/Y:** Setting for the XY coordinates of the circle to be cut.

Control Descriptions:

NAME	TYPE	DESCRIPTION
Save	Button	Click to save the parameters in red font on the current page to the system memory.
Read	Button	Click to save the red font parameters on the current page to the system memory.
Feedrate +10%	Button	Increase feed rate by 10% in real time.
Feedrate -10%	Button	Decrease feed rate by 10% in real time.
Spindle RPM +10%	Button	Increase spindle speed by 10% in real time.
Spindle RPM -10%	Button	Decrease spindle speed by 10% in real time.
SGL-point Milling Multi-point Milling	Button	A button to switch between [SGL-point Milling] and [Multi-point Milling]. SGL-point Milling: Machining one hole with the parameters set on the current page. Multi-point Milling: Automatically loading the start and end lines of the "List Machining" page and machining multiple holes with different parameters.
Centering	Button	Click to move the axis rapidly to the center position above the workpiece coordinate point for precise positioning.

Spiral External Milling	Button	This function allows for external circular contour millining. To activate, click the [Spiral External Milling] button and the screen will lock. To unlock the screen and emergency stop, click the [End] button.
Spiral Internal Milling	Button	This function allows for internal circular contour milling and automatically applies tool length compensation. The resulting hole size will be the diameter specified as the machining radius. To activate, click the [Spiral Internal Milling] button and the screen will lock. To unlock the screen and emergency stop, click the [End] button.
Pause Cont.	Button	Click to pause/resume operation.

4.14 Cut Rectangle

Click the **[Cut Rectangle]** button on the “**RUN**” page to access.

Cut Rect.		Idle		X-coord:	2.497	Y-coord:	0.978	
Safe Z	5	Spindle RPM	2000	Spindle Delay	5	Z-coord:	-9.331	
Z Start	0.200	Feedrate	2000	Center X	0.000	Spindle RPM +10%		
Z End	0.000	Tool R	0.000	Center Y	0.000	Spindle RPM -10%		
Down Step	0.200			Length X	0.000	Centering		
Start Line	1			Width Y	0.000	Climb	Spiral External Milling	
End Line	1			Feedrate +10%			Spiral Internal Milling	
SGL-point Milling				Feedrate -10%		Conv.	Spiral External Milling	
Pause		Save		Spiral Internal Milling				
Read								

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SETTING
→RUN

Parameter Descriptions:

- 1) **Safe Z:** Refers to the Z-coordinate of the tool at a safe distance above the workpiece after completing a machining operation. It should be set higher than the start and end depths to ensure safe tool movement during the entire process.
- 2) **Z Start:** The starting depth of Z-axis workpiece coordinate for a cutting rectangle in a spiral tool path.
- 3) **Z End:** The ending depth of Z-axis workpiece coordinate for cutting a rectangle in a spiral tool path.(End depth should be less than start depth.)
- 4) **Down Step:** The depth of each pass in the spiral cut, which should be greater than 0.
- 5) **Start Line:** The line number from which to start reading parameters for list processing.
- 6) **End Line:** The line number at which to end reading parameters for list processing.
- 7) **Spindle RPM:** Setting for the spindle speed.
- 8) **Feedrate:** Setting for the milling feed rate in units of mm/min.

- 9) **Tool R:** Sets the radius of the tool and automatically applies tool compensation.
- 10) **Spindle Delay:** The time delay setting for spindle to start after activation, measured in seconds.
- 11) **Center X/Y:** Setting for the XY coordinates of the rectangle to be cut.
- 12) **Length X:** Setting for the length of the rectangle to be cut. For spiral internal milling, the length should be larger than the tool diameter.
- 13) **Width Y:** Setting for the width of the rectangle to be cut. For spiral internal milling, the width should be larger than the tool diameter.

Control Descriptions:

NAME	TYPE	DESCRIPTION
Save	Button	Click to save the parameters in red font on the current page to the system memory.
Read	Button	Click to save the red font parameters on the current page to the system memory.
Feedrate +10%	Button	Increase feed rate by 10% in real time.
Feedrate -10%	Button	Decrease feed rate by 10% in real time.
Spindle RPM +10%	Button	Increase spindle speed by 10% in real time.
Spindle RPM -10%	Button	Decrease spindle speed by 10% in real time.
SGL-point Milling Multi-point Milling	Button	A button to switch between [SGL-point Milling] and [Multi-point Milling]. SGL-point Milling: Machining one hole with the parameters set on the current page. Multi-point Milling: Automatically loading the start and end lines of the "List Machining" page and machining multiple holes with different parameters.
Centering	Button	Click to move the axis rapidly to the center position above the workpiece coordinate point for precise positioning.

Spiral External Milling	Button	This function allows for external rectangular contour millining. To activate, click the [Spiral External Milling] button and the screen will lock. To unlock the screen and emergency stop, click the [End] button.
Spiral Internal Milling	Button	This function allows for internal rectangular contour milling and automatically applies tool length compensation. The resulting hole size will be the diameter specified as the machining radius. To activate, click the [Spiral Internal Milling] button and the screen will lock. To unlock the screen and emergency stop, click the [End] button.
Pause Cont.	Button	Click to pause/resume operation.

4.15 Rectangular Pocket

Click the **[Rect. Pocket]** button on the “**RUN**” page to access.

Rect. Pocket		Idle		X-coord:	2.497	Y-coord:	0.978
Safe Z	5	Spindle RPM	2000	Spindle Delay	5	Z-coord:	-9.331
Z Start	0.200	Feedrate	2000	Center X	0.000	Spindle RPM +10%	
Z End	0.000	Tool R	0.000	Center Y	0.000	Spindle RPM -10%	
Down Step	0.200			Length X	0.000	SGL-point Milling	
Line Spacing	0.200			Width Y	0.000	Centering	
Start Line	1			Feedrate +10%		Inward Climb Milling	
End Line	1			Feedrate -10%		Inward Conv. Milling	
Conner R	0.000	Save		Read		Pause	

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POCENDER
SETTING
→RUN

Parameter Descriptions:

- 1) **Safe Z:** Refers to the Z-coordinate of the tool at a safe distance above the workpiece after completing a machining operation. It should be set higher than the start and end depths to ensure safe tool movement during the entire process.
- 2) **Z Start:** The starting depth of Z-axis workpiece coordinate for cutting a rectangular pocket.
- 3) **Z End:** The ending depth of Z-axis workpiece coordinate for cutting a rectangular pocket.(End depth should be less than start depth.)
- 4) **Down Step:** The depth of each pass, which should be greater than 0.
- 5) **Line Spacing:** The distance between two adjacent cuts made by the tool. It should be greater than 0 (Spacing < Tool Diameter).
- 6) **Start Line:** The line number from which to start reading parameters for list processing.
- 7) **End Line:** The line number at which to end reading parameters for list processing.

- 8) **Corner R:** Setting for the corner radius.
- 9) **Spindle RPM:** Setting for the spindle speed.
- 10) **Feedrate:** Setting for the milling feed rate in units of mm/min.
- 11) **Tool R:** Sets the radius of the tool and automatically applies tool compensation.
- 12) **Spindle Delay:** The time delay setting for spindle to start after activation, measured in seconds.
- 13) **Center X/Y:** Setting for the XY coordinates of the rectangular pocket to be cut.
- 14) **Length X:** Setting for the length of the rectangular pocket to be cut, ensuring it is larger than the tool diameter.
- 15) **Width Y:** Setting for the width of the rectangular pocket to be cut, ensuring it is larger than the tool diameter.

Control Descriptions:

NAME	TYPE	DESCRIPTION
Save	Button	Click to save the parameters in red font on the current page to the system memory.
Read	Button	Click to save the red font parameters on the current page to the system memory.
Feedrate +10%	Button	Increase feed rate by 10% in real time.
Feedrate -10%	Button	Decrease feed rate by 10% in real time.
Spindle RPM +10%	Button	Increase spindle speed by 10% in real time.
Spindle RPM -10%	Button	Decrease spindle speed by 10% in real time.
SGL-pos Milling Multi-pos Milling	Button	A button to switch between [SGL-pos Milling] and [Multi-pos Milling]. SGL-pos Milling: Machining one hole with the parameters set on the current page. Multi-pos Milling: Automatically loading the start and end lines of the "List Machining" page and machining multiple holes with different parameters.

Centering	Button	Click to move the axis rapidly to the center position above the workpiece coordinate point for precise positioning.
Inward Climb Milling	Button	Click to start inward climb milling a cuboid pocket. Upon clicking, the screen will be locked. To unlock the screen and emergency stop, click the [End] button.
Inward Conv. Milling	Button	Click to start inward conventional milling a cuboid pocket. Upon clicking, the screen will be locked. To unlock the screen and emergency stop, click the [End] button.
Pause Cont.	Button	Click to pause/resume operation.

4.16 Circular Pocket

Click the **[Circular Pocket]** button on the “**RUN**” page to access.

Circular Pocket	Idle		X-coord: 2.497	Y-coord: 0.978
Safe Z: 5	Tool R: 0.000		Feedrate: 2000	Z-coord: -9.331
Z Start: 0.200	Machining R: 0.000		Spindle RPM: 2000	Spindle RPM +10%
Z End: 0.000			Spindle Delay: 5	Spindle RPM -10%
Down Step: 0.200			Center X: 0.000	SGL-pos Milling
First Cut R: 1.000			Center Y: 0.000	Centering
Line Spacing: 0.000			Feedrate +10%	Climb Milling
Start Line: 1			Feedrate -10%	Conv. Milling
End Line: 1				
	Read	Save	Pause	
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Parameter Descriptions:

- 1) **Safe Z:** Refers to the Z-coordinate of the tool at a safe distance above the workpiece after completing a machining operation. It should be set higher than the start and end depths to ensure safe tool movement during the entire process.
- 2) **Z Start:** The starting depth of Z-axis workpiece coordinate for cutting a circular pocket.
- 3) **Z End:** The ending depth of Z-axis workpiece coordinate for cutting a circular pocket.(End depth should be less than start depth.)
- 4) **Down Step:** The depth of each pass, which should be greater than 0.
- 5) **Lead-in R:** Setting for the entry radius of the tool's initial approach. It should be cautiously set to avoid excessive enlargement, which could lead to the formation of circular paths. Ensure that the sum of the entry radius and the tool radius does not surpass the machining radius to prevent overcutting.
- 6) **Line Spacing:** The distance between two adjacent cuts made by the tool. It should be greater than 0 (Spacing < Tool Diameter).

- 7) **Start Line:** The line number from which to start reading parameters for list processing.
- 8) **End Line:** The line number at which to end reading parameters for list processing.
- 9) **Tool R:** Sets the radius of the tool and automatically applies tool compensation.
- 10) **Machining R:** Setting for the radius of the circular pocket to be cut, ensuring it is larger than the tool radius.
- 11) **Feedrate:** Setting for the milling feed rate in units of mm/min.
- 12) **Spindle RPM:** Setting for the spindle speed.
- 13) **Spindle Delay:** The time delay setting for spindle to start after activation, measured in seconds.
- 14) **Center X/Y:** Setting for the XY coordinates of the circular pocket to be cut.

Control Descriptions:

NAME	TYPE	DESCRIPTION
Save	Button	Click to save the parameters in red font on the current page to the system memory.
Read	Button	Click to save the red font parameters on the current page to the system memory.
Feedrate +10%	Button	Increase feed rate by 10% in real time.
Feedrate -10%	Button	Decrease feed rate by 10% in real time.
Spindle RPM +10%	Button	Increase spindle speed by 10% in real time.
Spindle RPM -10%	Button	Decrease spindle speed by 10% in real time.
SGL-pos Milling Multi-pos Milling	Button	A button to switch between [SGL-pos Milling] and [Multi-pos Milling]. SGL-pos Milling: Machining one hole with the parameters set on the current page. Multi-pos Milling: Automatically loading the start and end lines of the "List Machining" page and machining multiple holes with different parameters.
Centering	Button	Click to move the axis rapidly to the center position above the workpiece coordinate point for precise

		positioning.
Climb Milling	Button	Click to start climb milling. The screen will be locked upon clicking. To unlock the screen and emergency stop, click the [End] button.
Conv. Milling	Button	Click to start conventional milling. The screen will be locked upon clicking. To unlock the screen and emergency stop, click the [End] button.
Pause Cont.	Button	Click to pause/resume operation.

4.17 Cut RoundRect

Click the [Cut RoundRect] button on the “RUN” page to access.

Cut RoundRect		Idle		X-coord:	0.000	Y-coord:	1.000	
Safe Z	5	Conner R	0.000	Spindle RPM	2000	Z-coord:	7.000	
Z Start	0.200	Tool R	0.000	Spindle Delay	5	F+10%	F-10%	
Z End	0.000	Feedrate	2000	Center X	0.000	S+10%	S-10%	
Down Step	0.200			Center Y	0.000	Centering		
Start Line	1			Length X	0.000	Climb	Spiral External Milling	
End Line	1			Width Y	0.000		Spiral Internal Milling	
		SGL-point Milling		Pause		Conv.	Spiral External Milling	
		Save		Read			Spiral Internal Milling	

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SETTING
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Parameter Descriptions:

- 1) **Safe Z:** Refers to the Z-coordinate of the tool at a safe distance above the workpiece after completing a machining operation. It should be set higher than the start and end depths to ensure safe tool movement during the entire process.
- 2) **Z Start:** The starting depth of Z-axis workpiece coordinate for cutting a round rectangle.
- 3) **Z End:** The ending depth of Z-axis workpiece coordinate for cutting a round rectangle.(End depth should be less than start depth.)
- 4) **Down Step:** The depth of each pass, which should be greater than 0.
- 5) **Start Line:** The line number from which to start reading parameters for list processing.
- 6) **End Line:** The line number at which to end reading parameters for list processing.
- 7) **Corner R:** The corner radius of the round rectangle to be cut, ensuring it is larger than the tool diameter when spiral inter milling.
- 8) **Tool R:** Sets the radius of the tool and automatically applies tool compensation.

- 9) **Feedrate:** Setting for the milling feed rate in units of mm/min.
- 10) **Spindle RPM:** Setting for the spindle speed.
- 11) **Spindle Delay:** The time delay setting for spindle to start after activation, measured in seconds.
- 12) **Center X/Y:** Setting for the XY coordinates of the round rectangle to be cut.
- 13) **Length X:** Setting for the length of the round rectangle to be cut, ensuring it is larger than the tool diameter when spiral inter milling.
- 14) **Width Y:** Setting for the width of the round rectangle to be cut, ensuring it is larger than the tool diameter when spiral inter milling.

Control Descriptions:

NAME	TYPE	DESCRIPTION
Save	Button	Click to save the parameters in red font on the current page to the system memory.
Read	Button	Click to save the red font parameters on the current page to the system memory.
Feedrate +10%	Button	Increase feed rate by 10% in real time.
Feedrate -10%	Button	Decrease feed rate by 10% in real time.
Spindle RPM +10%	Button	Increase spindle speed by 10% in real time.
Spindle RPM -10%	Button	Decrease spindle speed by 10% in real time.
SGL-point Milling Multi-point Milling	Button	A button to switch between [SGL-point Milling] and [Multi-point Milling]. SGL-point Milling: Machining one hole with the parameters set on the current page. Multi-point Milling: Automatically loading the start and end lines of the "List Machining" page and machining multiple holes with different parameters.
Centering	Button	Click to move the axis rapidly to the center position above the workpiece coordinate point for precise positioning.

Spiral External Milling	Button	This function allows for external contour millining. To activate, click the [Spiral External Milling] button and the screen will lock. To unlock the screen and emergency stop, click the [End] button.
Spiral Internal Milling	Button	This function allows for internal contour milling and automatically applies tool length compensation. The final hole will be shaped according to the parameters set for length (X) and width (Y). To activate, click the [Spiral Internal Milling] button and the screen will lock. To unlock the screen and emergency stop, click the [End] button.
Pause Cont.	Button	Click to pause/resume operation.

4.18 Surfacing

Click the **[Surfacing]** button on the “**RUN**” page to access.

Surfacing		Idle		X-coord:	2.497	Y-coord:	0.978
Safe Z	5	Spindle RPM	2000	Spindle Delay	5	Z-coord:	-9.331
Z Start	0.200	Feedrate	2000	Center X	0.000	Spindle RPM +10%	
Z End	0.000	Tool R	0.000	Center Y	0.000	Spindle RPM -10%	
Down Step	0.200			Length X	0.000	SGL-pos Milling	
Line Spacing	0.200			Width Y	0.000	X-axis Milling	
Start Line	1			Feedrate +10%		Centering	
End Line	1			Feedrate -10%		Start Run	
		Save		Ramp Down		Pause	
		Read					

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POCENDER
SETTING
→RUN

Parameter Descriptions:

- 1) **Safe Z:** Refers to the Z-coordinate of the tool at a safe distance above the workpiece after completing a machining operation. It should be set higher than the start and end depths to ensure safe tool movement during the entire process.
- 2) **Z Start:** The starting depth of Z-axis workpiece coordinate for surfacing.
- 3) **Z End:** The ending depth of Z-axis workpiece coordinate for surfacing.(End depth should be less than start depth.)
- 4) **Down Step:** The depth of each pass, which should be greater than 0.
- 5) **Line Spacing:** The distance between two adjacent cuts made by the tool. It should be greater than 0 (Spacing < Tool Diameter).
- 6) **Start Line:** The line number from which to start reading parameters for list processing.
- 7) **End Line:** The line number at which to end reading parameters for list processing.
- 8) **Spindle RPM:** Setting for the spindle speed.

- 9) **Feedrate:** Setting for the milling feed rate in units of mm/min.
- 10) **Tool R:** Sets the radius of the tool and automatically applies tool compensation.
- 11) **Spindle Delay:** The time delay setting for spindle to start after activation, measured in seconds.
- 12) **Center X/Y:** Setting for the XY coordinates of the rectangle center.
- 13) **Length X:** Setting for the length of the face to be milled, ensuring it is larger than the tool diameter.
- 14) **Width Y:** Setting for the width of the face to be milled, ensuring it is larger than the tool diameter.

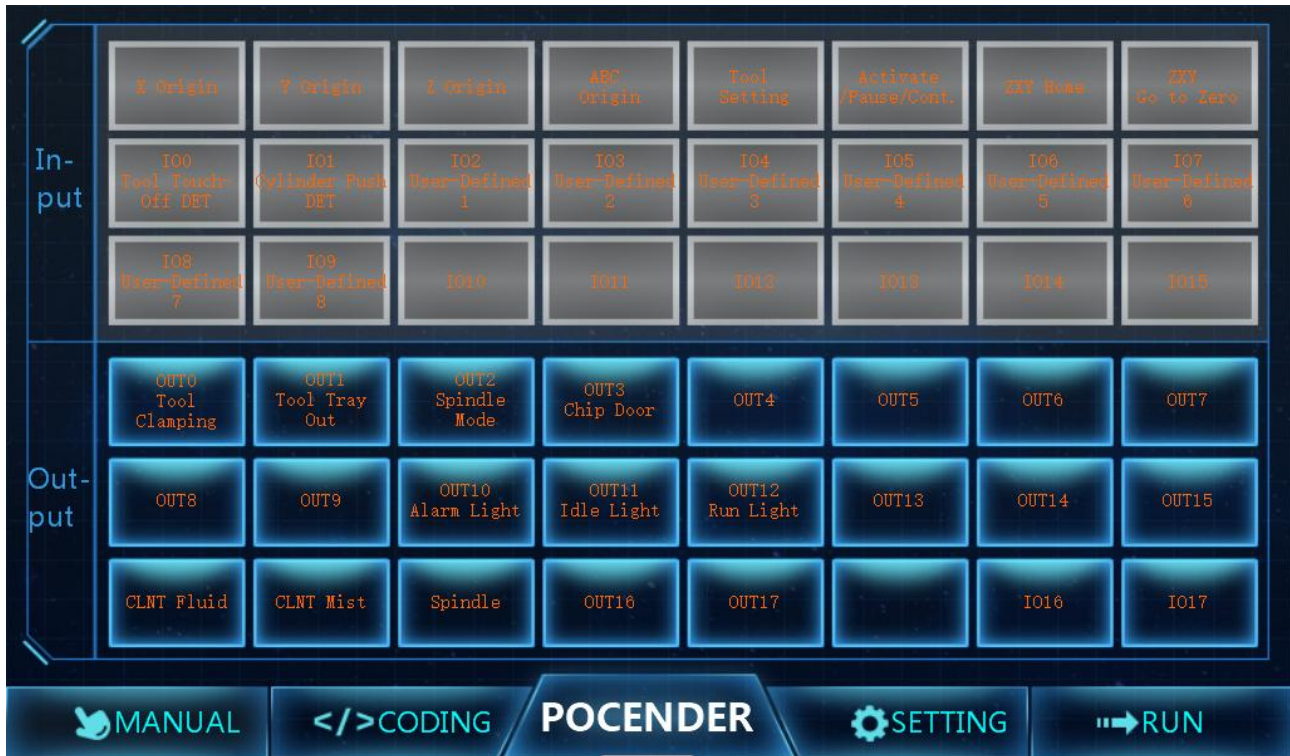
Control Descriptions:

NAME	TYPE	DESCRIPTION
Save	Button	Click to save the parameters in red font on the current page to the system memory.
Read	Button	Click to save the red font parameters on the current page to the system memory.
Feedrate +10%	Button	Increase feed rate by 10% in real time.
Feedrate -10%	Button	Decrease feed rate by 10% in real time.
Spindle RPM +10%	Button	Increase spindle speed by 10% in real time.
Ramp Down Straight Down	Button	Click to switch tool plunge way between [Ramp Down] and [Straight Down] .
Spindle RPM -10%	Button	Decrease spindle speed by 10% in real time.
SGL-pos Milling Multi-pos Milling	Button	A button to switch between [SGL-pos Milling] and [Multi-pos Milling]. SGL-pos Milling: Machining one hole with the parameters set on the current page. Multi-pos Milling: Automatically loading the start and end lines of the "List Machining" page and machining multiple holes with different parameters.
Centering	Button	Click to move the axis rapidly to the center position above the workpiece coordinate point for precise

		positioning.
X-axis Milling Y-axis Milling	Button	Click to switch between [X-axis Milling] and [Y-axis Milling] .
Start Run	Button	Click to start running. The screen will be locked upon clicking. To unlock the screen and emergency stop, click the [End] button.
Pause Cont.	Button	Click to pause/resume operation.

4.19 I/O Monitor

Click the **[I/O Monitor]** button on the “**RUN**” page to access.



Input/Output Monitoring:

- 1) When the current I/O is in an input or output state, it will be displayed in green.
- 2) Clicking the corresponding output button will trigger the corresponding signal output.

4.20 Slotting

Click the **[Slotting]** button on the “**RUN**” page to access.

Slotting		Idle		X-coord:		Y-coord:	
Safe Z	5	Spindle RPM	2000	Spindle Delay	5	Z-coord:	
Z Start	0.200	Feedrate	2000	Center X	0.000	Spindle RPM +10	
Z End	0.000	Tool R	0.000	Center Y	0.000	Spindle RPM -10	
Down Step	0.200			Slot Length	0.000	SGL-point Milling	
Start Line	1					X-axis Milling	
End Line	1			Feedrate +10%		Centering	
		Save		Feedrate -10%		Start Run	
		Read		Ramp Down		Pause	

 MANUAL
  CODING
 POCENDER
  SETTING
  RUN

Parameter Descriptions:

- 1) **Safe Z:** Refers to the Z-coordinate of the tool at a safe distance above the workpiece after completing a machining operation. It should be set higher than the start and end depths to ensure safe tool movement during the entire process.
- 2) **Z Start:** The starting depth of Z-axis workpiece coordinate for slotting.
- 3) **Z End:** The ending depth of Z-axis workpiece coordinate for slotting.(End depth should be less than start depth.)
- 4) **Down Step:** The depth of each pass, which should be greater than 0.
- 5) **Start Line:** The line number from which to start reading parameters for list processing.
- 6) **End Line:** The line number at which to end reading parameters for list processing.
- 7) **Spindle RPM:** Setting for the spindle speed.
- 8) **Feedrate:** Setting for the milling feed rate in units of mm/min.
- 9) **Tool R:** Sets the radius of the tool and automatically applies tool compensation.

10) Spindle Delay: The time delay setting for spindle to start after activation, measured in seconds.

11) Center X/Y: Setting for the XY coordinates of the slot center.

12) Slot Length: Setting for the length of the slot to be milled.

Control Descriptions:

NAME	TYPE	DESCRIPTION
Save	Button	Click to save the parameters in red font on the current page to the system memory.
Read	Button	Click to save the red font parameters on the current page to the system memory.
Feedrate +10%	Button	Increase feed rate by 10% in real time.
Feedrate -10%	Button	Decrease feed rate by 10% in real time.
Spindle RPM +10%	Button	Increase spindle speed by 10% in real time.
Spindle RPM -10%	Button	Decrease spindle speed by 10% in real time.
Ramp Down Straight Down	Button	Click to switch tool plunge way between [Ramp Down] and [Straight Down] .
SGL-pos Milling Multi-pos Milling	Button	A button to switch between [SGL-pos Milling] and [Multi-pos Milling]. SGL-pos Milling: Machining one hole with the parameters set on the current page. Multi-pos Milling: Automatically loading the start and end lines of the "List Machining" page and machining multiple holes with different parameters.
Centering	Button	Click to move the axis rapidly to the center position above the workpiece coordinate point for precise positioning.
X-axis Milling Y-axis Milling	Button	Click to switch between [X-axis Milling] and [Y-axis Milling] .
Start Run	Button	Click to start running. The screen will be locked upon clicking. To unlock the

		screen and emergency stop, click the [End] button.
Pause Cont.	Button	Click to pause/resume operation.

5. Upgrade Instructions

Read other subtopics below:

5.1 Touch Screen Firmware Upgrade Instructions

5.2 Controller Firmware Upgrade Instructions

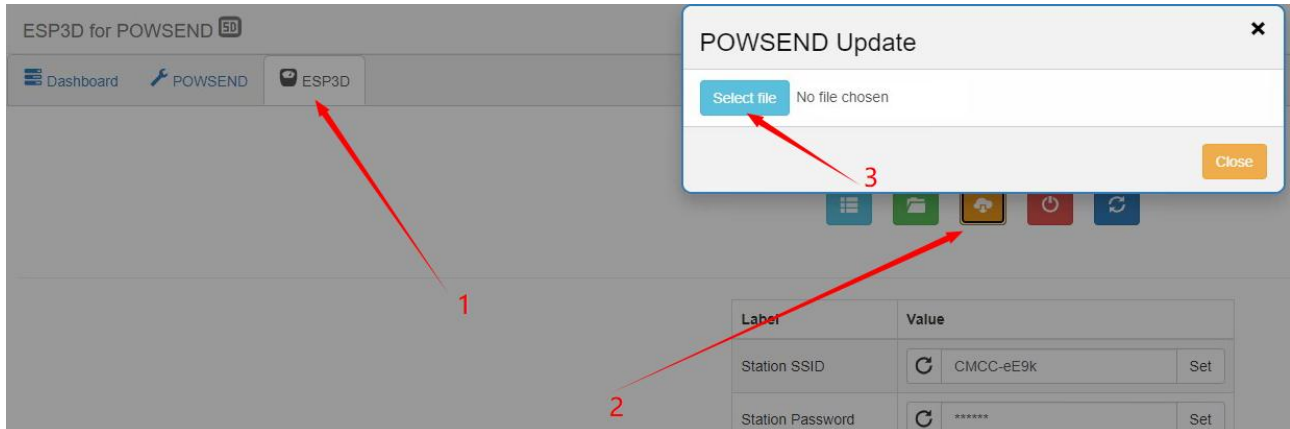
5.3 WIFI page Firmware Upgrade Instructions

5.1 Touch Screen Firmware Upgrade Instructions

Please note that if the controller firmware also needs to be upgraded, it must be updated first. Then, copy the provided touch screen update file to an SD card and insert it into the screen, and it will update automatically after power-off and restart. The previously set parameters will not be cleared. After the update is completed, power off and restart!

5.2 Controller Firmware Upgrade Instructions

First, follow the instructions in Chapter 8 of this manual to connect to WiFi and open the web-based upper computer.



Select the update provided file. If it's a compressed file, extract it first. After selecting the file, click "Update" and wait until it reaches 100%. If it displays "Restarting" afterward, the update is successful. Otherwise, repeat the process.

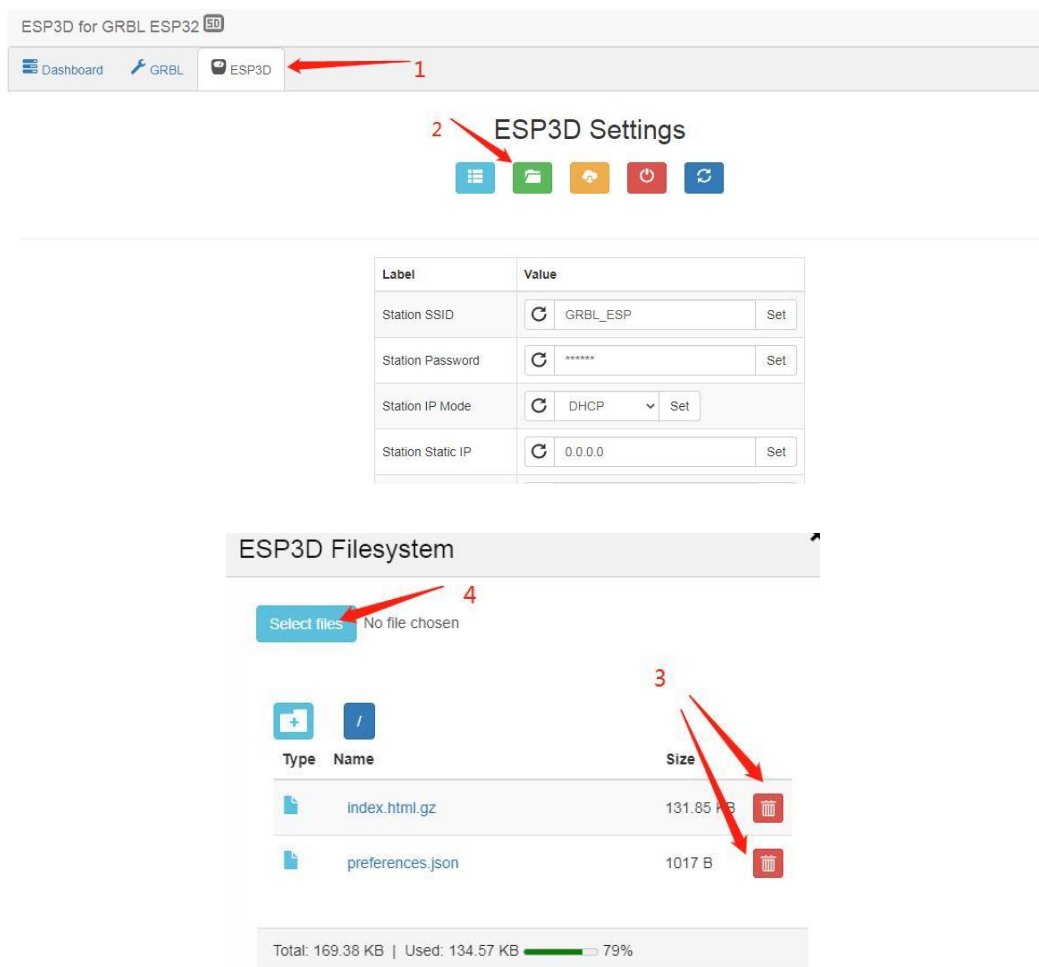
5.3 WIFI page Firmware Upgrade Instructions



INFORMATION: This is not a controller firmware upgrade or a touch screen firmware upgrade.

Steps:

- 1) Connect to the WiFi network according to the instructions in the **WIFI Function Description** and open the web-based upper computer.
- 2) Download the file named "index.html.gz" from the official website and do not unzip or rename it.
- 3) Follow the guidance numbers shown in the pictures below, and perform orderly.



The screenshots illustrate the steps for firmware upgrade on the ESP3D web interface:

- Step 1:** Navigate to the **ESP3D** tab in the top navigation bar.
- Step 2:** Click on the **ESP3D Settings** icon (a green folder icon) in the sub-menu.
- Step 3:** In the **ESP3D Settings** page, click on the **Station SSID** field to open the settings menu.
- Step 4:** In the **ESP3D Filesystem** page, click on the **Select files** button to upload files.

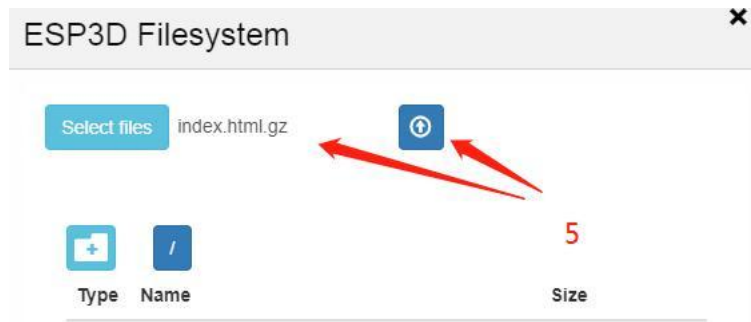
The **ESP3D Settings** page shows the following configuration options:

Label	Value	Action
Station SSID	GRBL_ESP	Set
Station Password	*****	Set
Station IP Mode	DHCP	Set
Station Static IP	0.0.0.0	Set

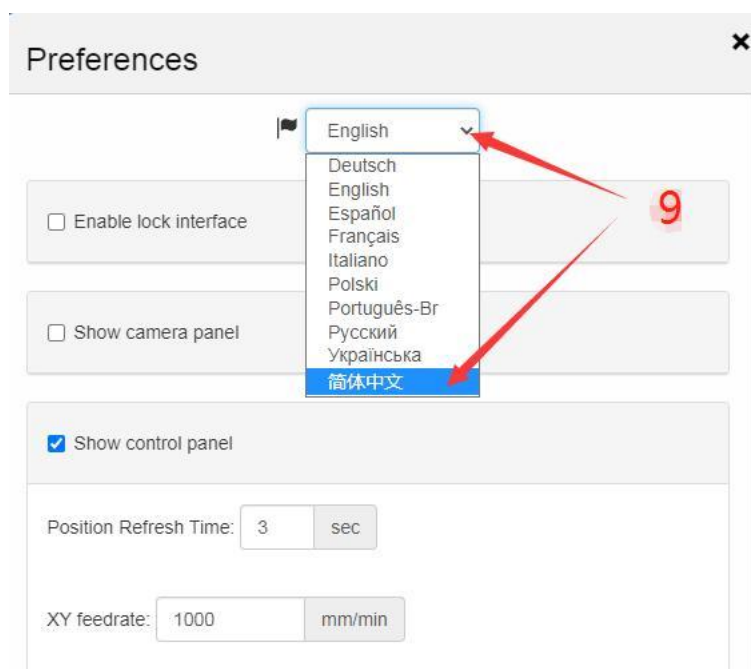
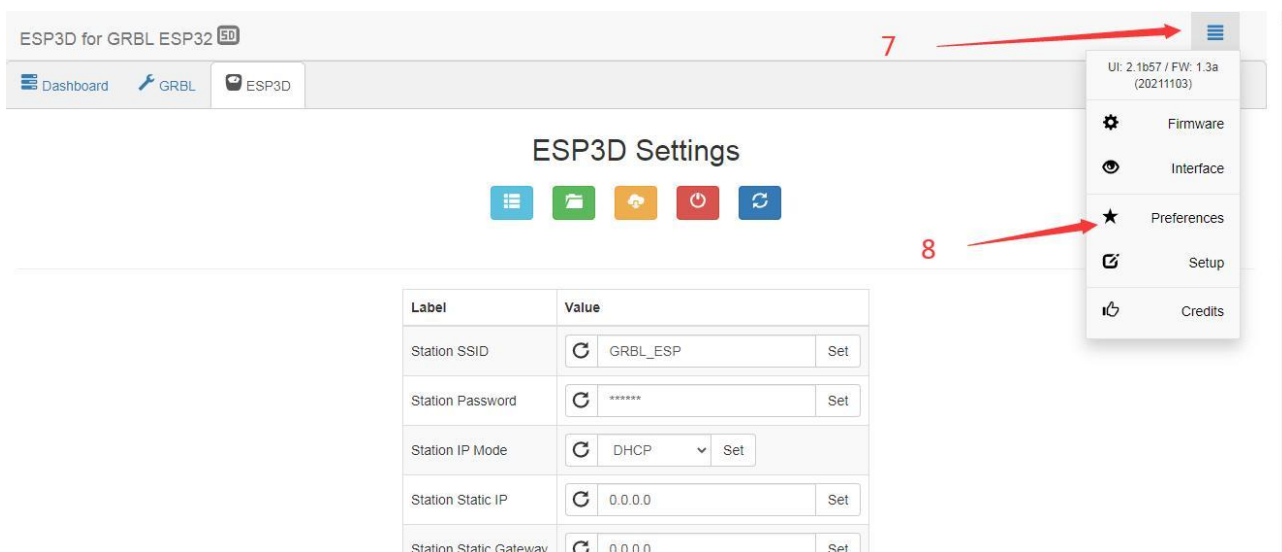
The **ESP3D Filesystem** page shows the following files:

Type	Name	Size	Action
File	index.html.gz	131.85 KB	Delete
File	preferences.json	1017 B	Delete

The filesystem status bar shows: Total: 169.38 KB | Used: 134.57 KB | 79%



6 . Right-click to refresh the page.



6. Frequently Asked Questions (Must Read)

- 1) The touch screen does not have a freezing issue. Please read the instructions carefully to see if you have clicked on a self-locking button (the button text is marked in red both in the manual and on the touch screen). To unlock it, press the coordinate display area or the **[END]** button.
- 2) If the device is slow to start up and the touchscreen keeps restarting and reconnecting, please check if the communication cable is loose or the reset port voltage level is pulled low, which may cause the controller to fail to start up. Also, check if the wifi is working in STA mode. In STA mode, the device keeps searching for the router, and if the router is turned off or the signal is weak, it will take a long time to start up. The solution is to switch to WiFi (AP) mode or NONE (off) on the **"SETTING"** page. Alternatively, try unplugging and reconnecting the communication cable after powering off the machine.
- 3) Unsupported or improperly formatted G-code will generate an error message in the format of **"Error N at line X"**. "N" refers to the error code, which can be found in the error code reference in this manual, and "X" refers to the line number where the error occurred.
- 4) Unsupported G-codes need to be removed before running the file, and they usually appear in the first few lines of the file.
- 5) The jog movement of the MPG handwheel will only take effect when the page is switched to the **"JOG"** page and the lower left **[JOG]** button is clicked twice, or when switched to the **"Probe"** page. It will not work on other pages.
- 6) The file name can only contain English letters, numbers, or English punctuation marks, and cannot contain Chinese characters or Chinese punctuation marks.
- 7) Please use a genuine TF card with a capacity of no more than 32GB (Class 4) and format it in FAT32 format. Do not use cheap counterfeit or shrunken cards, as they may not be read or may cause issues such as inability to boot or restart, or errors when reading files. Sometimes, the file list may only appear after refreshing it twice, and when running a file, an error may occur on the first attempt, requiring a second attempt to run the file.
- 8) If the card cannot be read, please check if the system is operating in Bluetooth mode. Note that the Bluetooth mode does not support SD card operations. To run SD card programs, use another mode.
- 9) Transferring files via WiFi may not be smooth. It is recommended that you format the card to FAT32 format, set the cache to 16KB, and use Google browser instead of the default browser on your computer as the latter may be blocked by the computer's security software when in AP mode (access point). This is especially true for Windows 10/11 systems, which have strict security measures. It is recommended to use Windows 7 instead. For Windows 10/11 systems, you can attempt to use STA mode (station) for improved performance.

10) Inserting the memory card into the controller provides a highly stable and smooth performance during operation. The controller can easily handle NC files that are smaller than 3GB, ensuring seamless execution.

11) The MPG handwheel operates in “**Step Motion**” mode. The x1 gear moves the machine by 0.01mm per step, the x10 gear moves by 0.1mm per step, and the x100 gear moves by 1mm per step.

- It is important not to rotate the handwheel too fast, as the machine's maximum speed and acceleration settings may not keep up with the handwheel speed. If the handwheel is rotated at a speed where it covers 100mm in one second, the machine may not be able to match that speed due to its slower settings. The machine will complete the movement equivalent to the handwheel rotation and only then come to a stop.
- If using a servo, it is recommended to increase the speed to 10000 and acceleration to 1000 for optimal responsiveness. In this case, if you haven't finished rotating to the next increment, the machine will already have completed the movement from the previous one and come to a stop. This situation occurs when your handwheel rotation speed cannot keep up with the machine's movement speed.
- If the handwheel is rotated too quickly, the system will filter the input, but there is still a risk of collision. Therefore, it is recommended to enable the soft limit function. The handwheel is only effective in the “**JOG**” page (click the [**JOG**] button twice after switching to the “**JOG**” page) and the “**Probe**” page.

12) The MPG handwheel in our online store is a regular one, which may experience missing pulses (i.e. no response for a certain increment). This is a normal situation. If other electrical leakage occurs, it may cause additional abnormal situations. For strict requirements, it is recommended to purchase genuine handwheels to ensure reliable performance. Thank you for your understanding.

13) When operating in tool change modes 4, 5, 6, or 8, it is essential to use T1 for tool alignment. Once the tool alignment is completed, proceed to the “**Probe**” page to perform the “First Tool Length Measurement” procedure. (by moving the tool above the tool setter and clicking the [**First TLM**] button. Wait patiently until it is successfully executed.)

14) To power the 10.1-inch screen, connect a 5V power supply with a minimum current rating of 2A or higher. Connect the positive (+) terminal of the power supply to the 5V terminal located near the pulse direction connection. Connect the negative (-) terminal of the power supply to any GND (ground) terminal on the controller.

15) When the MPG is not in use, switch the axis selection to the neutral position to prevent accidental movement. This ensures that the MPG remains idle and avoids unintended operation.

16) When an error or alarm message is displayed in the status bar, please click on the [**Unlock**] button on the “**Offsets**” page to resume normal operation. If an ALRM error

occurs, there is a possibility of coordinate loss, so it is recommended to re-home the machine by returning to the home position.

17) After enabling the homing function, if an ALRM occurs after restarting or resetting the machine, it indicates a coordinate loss. In this case, it is necessary to return to the home position or unlock the machine to clear the alarm and allow the machine to operate.

18) After clicking the **[Spindle On]** button, if the relay fails to engage, please check if the system is in an alarm state (if so, unlock it to return to the idle state) and verify that the "CNC mode" in the **"SETTING"** page is selected and not inadvertently switched to "Laser mode". Also, ensure that the system's power supply voltage is stable and maintained at 24V.

19) If an error or alarm occurs during tool setting, please try switching the probe signal direction to either reverse or forward in the **"SETTING"** page. Afterward, go to the **"Offsets"** page and click **[Unlock]** button before proceeding with your operations.

20) The homing function needs to be enabled in the **"SETTING"** page before performing the homing operation. Prior to homing, the travel limits should be set. After setting the travel limits, it is necessary to power off and restart the system. If the origin is not found within the defined travel range, an error will be reported.

21) When the **"JOG"** page displays an alarm message, all axes are unable to move. To resolve this, you need to either return to the home position or click the **[Unlock]** button on the **"Offsets"** page. Once done, return to the **"JOG"** page and the axes will only be able to move when the **"JOG"** page displays "IDLE".

22) Once the hard limit function is enabled, if a limit switch is triggered, the system will enter an emergency stop state and lock. To resume normal operation, press the reset button of the controller. Then, manually move the affected axis away from the triggered limit switch and perform a homing procedure. It is advisable to use the soft limit function instead of the hard limit function in this case. The soft limit function provides the same functionality but offers the advantage of automatically stopping and displaying an error when the limit is exceeded. This eliminates the need to restart the system and ensures that the coordinates are not lost, eliminating the requirement to rehome the machine.

23) If the machine abruptly halts during operation, verify if the hard limit feature is activated. Enabling the hard limit function results in an immediate emergency stop upon triggering any limit switch. Alternatively, check if the soft limit feature is enabled. Exceeding the machine's travel limits will trigger an alarm (ALRM:2). If neither the hard limit nor the soft limit features are activated and the issue persists, it may indicate a potential current leakage within the machine frame. In such cases, it is recommended to short the reset port to +24V.

24) Please refrain from accessing the USB drive without connecting the USB expansion module. Otherwise, it may result in an endless loading loop and unresponsive system. If

you have already accessed the USB drive in this state, please perform a power cycle to restart the system.

25) The presence of electrical leakage in the machine frame may cause data fluctuations on the 10.1-inch touchscreen (metal casing). Please ensure proper insulation for the touchscreen.

7. Error And Alarm Code Reference Table (Must Read)

Read the subtopics below:

7.1 Error Code Reference Table

7.2 Alarm Code Reference Table

7.1 Error Code Reference Table

CODE	ERROR DESCRIPTION
1	G-code words consist of a letter and a value. Letter was not found.
2	Missing the expected G-code word value or numeric value format is not valid.
3	System command was not recognized or supported.
4	Negative value received for an expected positive value.
5	Homing is not enabled via settings. Please enable it in the settings page.
6	Minimum step pulse time must be greater than 3usec.
7	An EEPROM read failed. Auto-restoring affected EEPROM to default values.
8	System command cannot be used unless the system is IDLE.
9	Being locked out during alarm state. Please unlock and operate.
10	Soft limits cannot be enabled without homing also enabled.
11	Max characters per line exceeded. Received command line was not executed.
12	The setting value cause the step rate to exceed the maximum supported.
13	Safety door detected as opened and door state initiated.
14	Build info or startup line exceeded EEPROM line length limit. Line not stored.
15	Jog target exceeds machine travel. Jog command has been ignored.
16	Jog command has no '=' or contains prohibited g-code.
17	Laser mode requires PWM output.
20	Unsupported or invalid g-code command found in block.
21	More than one g-code command from same modal group found in block.
22	Feed rate has not yet been set or is undefined.
23	G-code command in block requires an integer value.
24	More than one g-code command that requires X/Y/Z/A/B/C-axis words found in block.
25	Repeated g-code word found in block.
26	No X/Y/Z/A/B/C-axis words found in block for g-code command or current modal state which requires them.

27	Line number value is not in the valid range of 1 to 9,999,999.
28	G-code command has been sent, but some required P or L values are missing in that line.
29	Six workpiece coordinate systems are supported: G54 to G59. G59.1, G59.2, and G59.3 are not supported.
30	G53 only allowed with G0 and G1 motion modes.
31	Axis words found in block when no command or current modal state uses them, and the G80 motion mode cancellation is active.
32	G2 and G3 arcs require at least one in-plane axis word.
33	Motion command target is invalid. (1) Arcs can not be generated by G2/G3. (2) Invalid probe target for G38.2. (3) Depth error/tool radius error/machining radius error.
34	Arc radius value is invalid. Try to divide the arc into semicircle or quadrant circle.
35	Missing IJK or R offset words in selected plane for G2 and G3 arcs.
36	Unused value words found in block.
37	G43.1 dynamic tool length offset can not be assigned to an axis other than the Z axis.
38	Tool number greater than max supported value.
39	Timeout occurred during communication with the interface expansion module or exceeded the maximum IO expansion limit. Opening or closing operation failed.
40	Failed to perform first tool length measurement for T1 in tool changing modes 4/5/6.
42	TF card loading failed or file not found.
60	TF card not detected or format error. Format it as FAT32.
61	Failed to open file. Please try again. This error may occur if the memory card is a reduced or expanded capacity card or if the file name contains Chinese characters.
66	File name contains unsupported characters. Please rename using only English characters.

7.2 Alarm Code Reference Table

CODE	ALARM DESCRIPTION
1	Hard limit has been triggered. Machine position is likely lost due to sudden halt. Re-homing is highly recommended.
2	Soft limit alarm. G-code motion target exceeds machine travel. Machine position retained. Alarm may be safely unlocked.
3	Reset while in motion. Machine position is likely lost due to sudden halt. Re-homing is highly recommended.
4	Probe fail. Probe is not in the expected initial state before starting probe cycle. Please go to the "Settings" page and click the [PROB FWD] or [PROB REV] button to switch to the opposite state, or check the wiring. Then unlock before operating.
5	Probe fail. Probe did not contact the workpiece within the programmed travel. Please reset the probe travel and unlock before operating.
6	Homing fail. The active homing cycle was reset.
7	Homing fail. Safety door was opened during homing cycle.
8	Homing fail. Pull off travel failed to clear limit switch. Try increasing pull-off setting or check wiring. Or the origin switch is triggered ahead of time. Please check and handle the exception before performing any operation.
9	Homing fail. Could not find limit switch within search distances. Try increasing max travel, checking wiring or switch.
10	Homing fail. A second limit switch for self-alignment is not found on a two-axis machine.
11	Cutter breaks and safely ends the task.
12	E-Stop button pressed.

8. Controller Connection Instructions

Read the subtopics below:

8.1 Wiring Instructions **(Must Read)**

8.2 Bluetooth Connection Instructions

8.3 WIFI Connection Instructions

8.1 Wiring Instructions (Must Read)

Read the subtopics below:

8.1.1 Power Supply

8.1.2 Wiring Instructions for 10-inch Screen

8.1.3 Driver Wiring

8.1.4 Spindle Wiring

8.1.5 Tool Setter Wiring

8.1.6 Limit Switch Wiring

8.1.7 Buttons Wiring

8.1.8 M6 and M7

8.1.9 Power Port And Touchscreen Port

8.1.10 Reset

8.1.11 MPG Handwheel Interface Pin Wiring Instructions

8.1.12 Controller Appearance

8.1.1 Power Supply

Connect a 24V power source. Ensure that the power supply has a power rating of at least 20W.

8.1.2 Wiring Instructions for 10-inch Screen

Connect a 5V power source with a minimum current rating of 2A. Connect the positive terminal of the power supply to the 5V terminal next to the pulse direction terminal on the controller. Connect the negative terminal of the power supply to any GND (ground) terminal on the controller.

8.1.3 Driver Wiring

Choose either common anode or common cathode connection. Ensure that the pulse and direction signals have a voltage of 5V.

- 1) **Common Cathode Connection for the Driver:** (Please consult the seller of the driver to inquire about the pulse (pul/step) and direction (dir) pin locations on the driver.) Then, follow the instructions below to make the corresponding connections.
 - a. Connect the +5V terminal of the controller to the positive terminal (pul+/step+) of the driver for the pulse signal and to the positive terminal (dir+/sign+) of the driver for the direction signal.
 - b. Connect the Pul- terminal of the controller to the negative terminal (pul-/step-) of the driver for the pulse signal.
 - c. Connect the Dir- terminal of the controller to the negative terminal (dir-) of the driver for the direction signal.
 - d. Leave the enable terminal unconnected (floating).

- 2) **Common Cathode Connection for the Driver:** (Please consult the seller of the driver to inquire about the pulse (pul/step) and direction (dir) pin locations on the driver.) Then, follow the instructions below to make the corresponding connections.
 - a. Connect the GND terminal of the controller to the negative terminal (pul-/step-) of the driver for the pulse signal and to the positive terminal (dir-/sign-) of the driver for the direction signal.
 - b. Connect the Pul- terminal of the controller to the positive terminal (pul+/step+) of the driver for the pulse signal.
 - c. Connect the Dir- terminal of the controller to the positive terminal (dir+/sign+) of the driver for the direction signal.
 - d. Leave the enable terminal unconnected (floating).

If, after confirming that the wiring is correct, you find that the touchscreen coordinates are changing but the motors are not moving, please try the following steps:

- a. Go to the settings page and change the “Pulse Reverse” setting to 63. Click **[Apply]** and then power off the system. After restarting, try again.

- b. If the issue persists, go to settings page 2 and change the “Pulse Width” setting to 6. Click **[Apply]** and then power off the system. After restarting, try again.
- c. If the motors still do not move, it could indicate either incorrect wiring or a faulty motor driver.

Please double-check the wiring connections and ensure they are correctly aligned with the motor driver specifications. If the wiring is correct, it is recommended to consult a professional or the motor driver manufacturer for further assistance or to determine if the motor driver needs to be replaced.

8.1.4 Spindle Wiring

- 1) **Spindle Relay 1/2:** The controller's relay outputs function as switches with the 1st and 2nd terminals. **They cannot directly control high-power electrical devices and require an external relay/contacter for high-power applications.**

A. DC Spindle: (Please consult the inverter seller to determine the pin for the spindle's forward rotation signal and the common terminal.) Then, follow the instructions below:

There is no polarity distinction. Connect Spindle 1 to the inverter's forward rotation signal and Spindle 2 to the inverter's common terminal (com/GND). Do not load high-power devices directly; use an external relay/contacter for high-power applications.

B. Servo Spindle: (Please consult the servo seller to determine the pin for switching between speed and position modes and the corresponding common terminal.) Then, follow the instructions below:

There is no polarity distinction. Connect Spindle 1 to the servo drive's pin for switching between speed and position modes, and connect Spindle 2 to the common terminal (com/GND) of the pin for switching between speed and position modes.

For the servo drive's switching pin setting: when the switching pin and its common terminal are connected, it is in speed mode; when disconnected, it is in position mode.

- 2) **Controller AVI (0-10V):** When the voltage is not accurate, you can adjust the blue potentiometer at full speed to calibrate it to 10V.

A. DC Spindle: (Please consult the inverter seller or servo seller to determine the pins for the AVI analog 0-10V speed control voltage input and the common terminal.) Then, follow the instructions below:

Connect AVI to the AVI analog speed control voltage port of the inverter (servo drive), and connect the controller's GND to the inverter's common terminal (com/GND).

After the connections are made, ask the inverter seller to set the inverter for external terminal start and external 0-10V analog voltage frequency control mode.

B. Servo Spindle: (Please consult the servo seller to determine the pins for the AVI analog 0-10V speed control voltage input in speed mode and the common terminal.) Then, follow the instructions below:

Connect AVI to the AVI analog speed control voltage port of the servo drive, and connect the controller's GND to the servo drive's AVI common terminal (com/GND).

To use the tapping function, connect the pulse and direction to the corresponding tapping axis set in the settings page, and set the pulse equivalent of that axis to the number of pulses required for the motor to rotate the lead screw by one degree (or one revolution for firmware versions prior to June 27, 2023). When tapping, switch to position mode and turn off the spindle (M5). When milling, switch to speed mode and turn on the spindle (M3S5000).

8.1.5 Tool Setter Wiring

For the normally open type tool setter from the POCENDER online store, follow these connections:

- Connect the **green** wire to the GND (ground) terminal on the controller.
- Connect the **black** wire to the GND (ground) terminal on the controller as well.
- Connect the **yellow** wire to the tool preset signal input terminal on the controller.
- Connect the **red** wire to the reset input terminal on the controller.

Note: If you have purchased a normally closed tool setter from another store, the overtravel signal cannot be connected to the reset terminal on the controller.

Wiring Instructions for different wire configurations of tool setters:

1) Two-wire tool setter (normally open or normally closed):

- a. Connect one wire to the tool signal input terminal of the controller.
- b. Connect the other wire to the GND (ground) terminal of the controller.

2) Three-wire tool setter (normally open or normally closed):

- a. Use two wires as power supply wires and connect them to the power input terminals.
- b. Connect the remaining wire to the tool signal input terminal of the controller.

For detailed instructions on testing the functionality, please refer to the tool setting setup guide with the **Six-step Setup** chapter (4.6.1) in this manual.

8.1.6 Limit Switch Wiring

Limit switches are used as home switches during the homing process.

Wiring instructions for for different wire configurations of limit switches:

1) Two-wire configuration:

- a.** Connect one terminal of the limit switch to one wire.
- b.** Connect the other terminal of the limit switch to the GND terminal of the controller.
- c.** For a four-axis system, connect the corresponding axis- to the respective limit input interface.
- d.** For a six-axis system, connect all the limit switch signal lines together and connect them to the limit input interface.

2) Three-wire configuration (using NPN normally open switch):

- a.** Two wires are for the power supply, connect them to the power supply terminals.
- b.** The remaining wire is the signal wire.
- c.** For a four-axis system, connect the signal wire to the corresponding axis limit input terminal.
- d.** For a six-axis system, connect all the signal wires together and connect them to the limit input terminal.

8.1.7 Buttons Wiring

To connect a button, one end of the button should be connected to the input terminal, and the other end of the button should be connected to the GND (ground). It is important to use a momentary or self-resetting switch for all button inputs. This type of switch ensures that the button returns to its default position after being pressed, allowing for proper functionality and accurate input detection.

The functions of the buttons are as follows:

1) Activate/Pause/Cont.:

- When the machine is in an idle state without any alarms, this button will run the last executed NC file. It is only effective if a file has been executed before.
- During execution, it functions as a pause button.
- While paused, it functions as a resume button.
- During homing, it cancels the homing process.
- During manual jogging/incremental jogging/positioning, it cancels the current action.

2) ZXY Homing:

- Sequentially homes the Z, X, and Y axes.
- While paused, it ends the current running file.

3) ZXY Go To Zero:

- ZXY axes simultaneously return to the zero point of the current work coordinate system.

8.1.8 M6 and M7

1) M6

The M6-1 and M6-2 terminals on the controller are relay outputs for the tool change signal.

The controller's relay outputs can be seen as switches, with 1 and 2 being the terminals of the switch.

It is important to note that these outputs are not designed to handle high-power loads directly. If you need to control high-power devices, such as motors or large relays, you should use an external relay or contactor.

To enable the tool change signal output, you can use the M62 command with the parameter P0 (e.g., M62 P0) to forcefully activate the signal output. Conversely, to disable the output, you can use the M63 command with the parameter P0 (e.g., M63 P0) to forcefully deactivate the signal.

2) M7

The M7-1 and M7-2 terminals on the controller are relay outputs for the cutting fluid switch signal.

Similar to other relay outputs on the controller, these can be treated as switches, with 1 and 2 being the terminals of the switch.

It is important to note that these outputs are not designed to handle high-power loads directly. If you need to control high-power devices related to the cutting fluid, such as pumps or valves, you should use an external relay or contactor.

To turn on the cutting fluid, you can use the M7 command to activate the signal output. Conversely, to turn off the cutting fluid, you can use the M9 command to deactivate the signal. **Please note that this information is specific to the newer version of the controller, where M6 has been replaced with output 0 on the expansion module.**

8.1.9 Power Port And Touchscreen Port

1) Power Port:

- Connect the positive terminal to the 24V+ port.
- Connect the negative terminal to the 0V port or ground. Only connect the 24V power supply; the 5V connection is not required.

2) Touchscreen Port:

- The wiring sequence is already arranged. Simply plug it into the corresponding port, and it should work.
- Pay attention to the black wire, which is typically the ground (GND) connection. Make sure it is correctly aligned and inserted into the corresponding ground port on the controller. If it doesn't match, flip it over and try again.

8.1.10 Reset Wiring

The reset terminal on the controller is used as a restart button, which can also function as an emergency stop.

To connect the reset button, a normally open button should be used, where one end of the button is connected to the reset terminal and the other end is connected to GND.

It's important to note that when the machine is restarted, the coordinates will be lost, requiring a re-homing process to establish the reference position.

Starting from February 2023, this port is repurposed as the stop button. It is used to terminate all ongoing tasks. It should still be connected with a normally open button, where one end of the button is connected to the reset terminal and the other end is connected to GND. This non-restart function can be utilized as an emergency stop.

8.1.11 MPG Handwheel Pin Wiring Instructions

Pin No.	Indicator	Notes
1	Axis 6 - C-Axis	Refers to the C-Axis, which is the 6th axis
2	X1	
3	X10	
4	X100	
5	N/A	
6	X-Axis	
7	Y-Axis	
8	Z-Axis	
9	Axis 4 - A-Axis	
10	Axis 5 - B-Axis	
11	5V/VCC Indicator Positive	Typically, two wires are connected to this terminal: power positive (+) and LED positive (+).
12	0V/GND COM Negative	Typically, three wires are connected to this terminal: ground (0V), common (COM), and LED negative (-).
13	N/A	
14	Phase A Positive	Phase A negative and Phase B negative are left unconnected.
15	Phase B Positive	

8.1.12 Controller Appearance

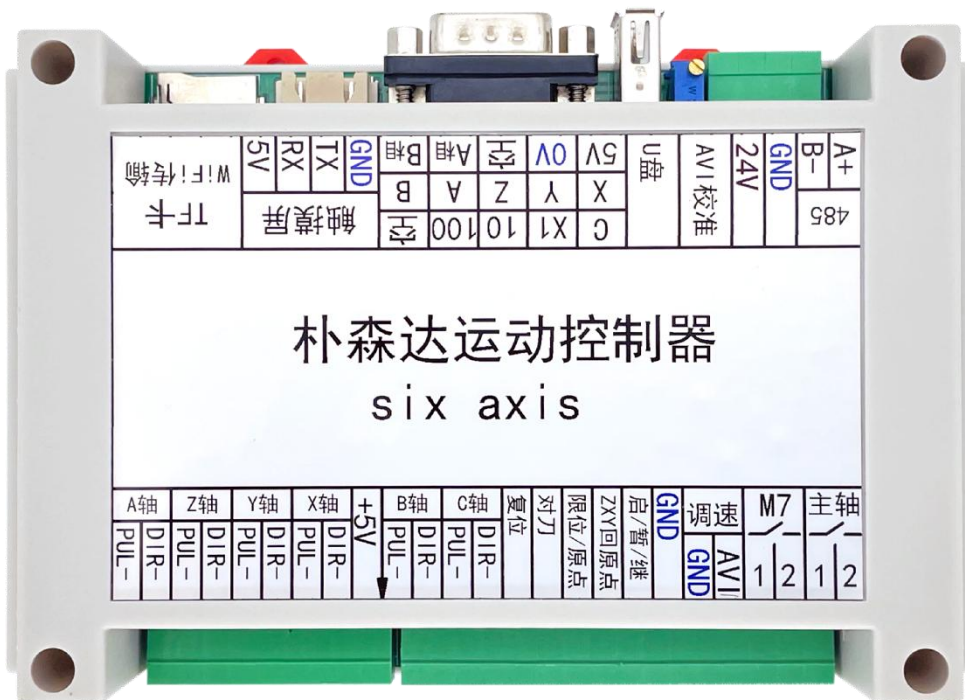
1) Four-Axis without USB Interface



2) Six-Axis without USB Interface



3) Six-Axis with USB Interface



8.2 Bluetooth Connection Instructions

Connect and control the machine using a mobile phone.

It is not recommended to use the Bluetooth function. The Bluetooth app is not developed by our company and is provided as a complimentary feature. Some functions may require additional fees. The controller's functions are continuously being upgraded, and due to the Bluetooth app not keeping up with our updates and upgrades, it may result in compatibility issues. If Bluetooth functionality is required, it is necessary to revert to the older firmware version. This statement serves as a disclaimer.

Download the app from the Q group file: 696989823.

Steps:

- a. **Enable Bluetooth:** Go to the settings page on the touchscreen and tap the **[BLUT]** button to enable Bluetooth.
- b. **Bluetooth Connection:** In the device list on your phone, locate "btgrblesp/POWSEND" and tap to connect without entering a password.
- c. Open the app and click the Bluetooth icon to select "btgrblesp" for a successful connection.
- d. Enjoy using the Bluetooth functionality for control and monitoring.

8.3 WIFI Connection Instructions

Choose between AP (Access Point) and STA (Station) mode. These modes enable you to connect and control the machine using WiFi. However, please note that simultaneous control by multiple devices is not supported.

Note:

- It is important to avoid operating other buttons during file uploads, as it may trigger actions after the upload is complete.
- It is recommended to avoid using Windows 8/10/11 systems for WiFi connection, as these systems may encounter interception issues, leading to difficulties in accessing web pages, unsuccessful file uploads, update failures, and system crashes. For optimal compatibility, we recommend using Windows 7 as the operating system and Google Chrome as the preferred browser. In case you are using Windows 10/11, you can try using the STA mode to establish the WiFi connection.

1) AP Mode:

This mode allows the controller to create a hotspot, which can be connected to using a computer or mobile device.

Follow these steps:

- Enable WiFi:** On the touchscreen interface, navigate to the settings page and tap the **[WiFi]** button. This will enable the WiFi functionality. Note that this button functions as a toggle switch, so you only need to tap it once to turn it on. After tapping the **[WiFi]** button, the system will restart, and you will need to re-home the machine.
- Connect to WiFi:** On your computer or mobile device, access the WiFi settings and select "GRBL_ESP" or "POWSEND" network.
- Click on the **[Connect]** button to establish the connection and enter the password: 12345678.
- Access the web-based control panel:** Launch a web browser on your computer or mobile device and enter the current IP address. This will take you to the web-based control panel.
- Enjoy using the WiFi functionality! You can control the machine, configure machine parameters, upgrade the controller, upload, download, and delete files.

2) STA Mode:

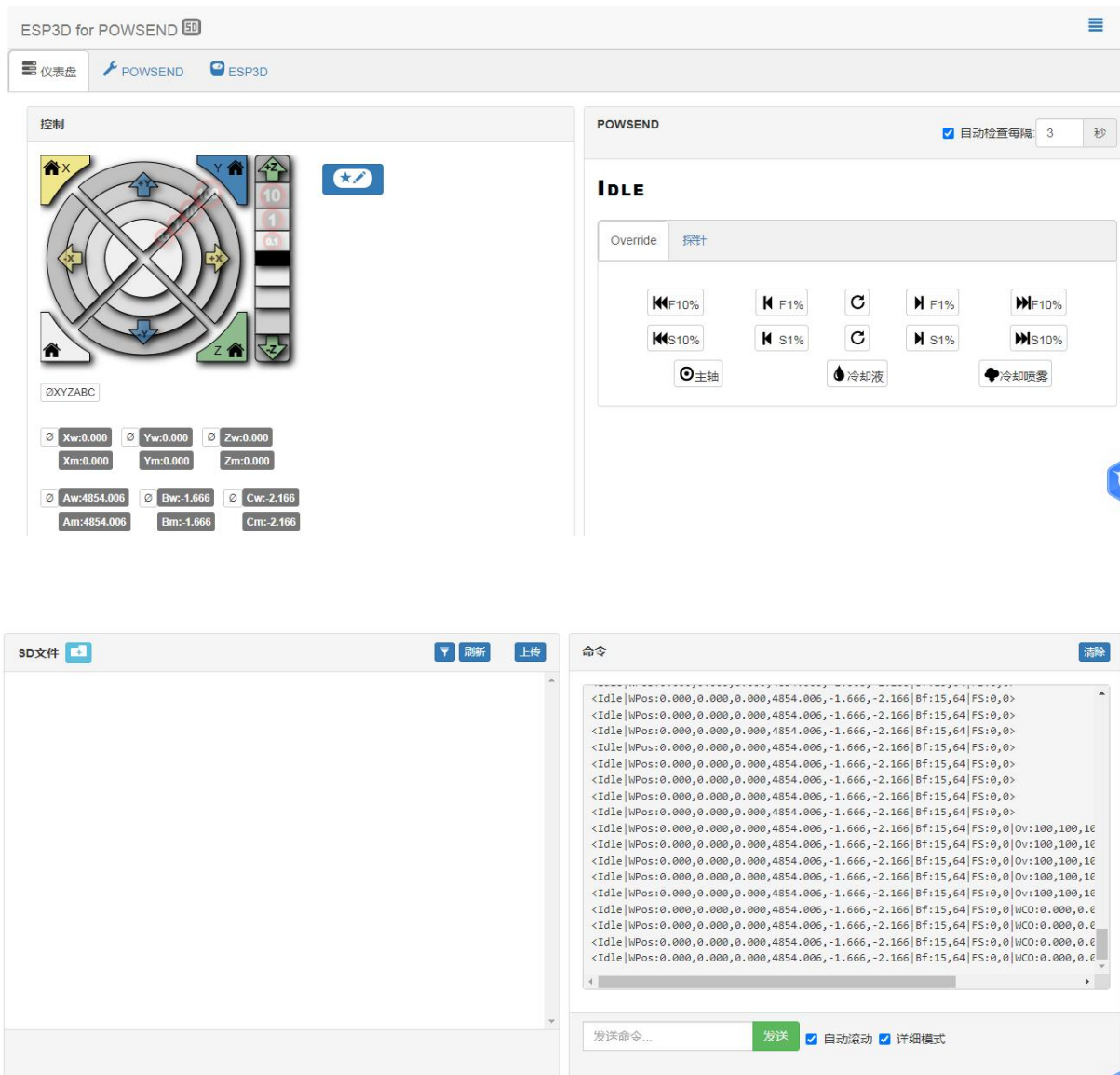
This mode allows the controller to connect to a router.

Note:

- It's important to ensure that the controller and the computer/mobile device are connected to the same router. The router name should be in English.
- The controller will automatically connect to the previously connected WiFi network on startup. If the router is turned off and the previous WiFi network is not available, it may result in a longer startup time. Switching to AP or None mode when the router is turned off will result in faster startup.

Follow these steps:

- Connect to the router:** Access the STA Settings on the second settings page and click on **[Search]** to find the router's name. (Make sure the device is in an idle state and any alarms are resolved).
- Enter the line number before the router name you want to connect to, and then enter the password for the new router. Click on **[Connect]**. **After clicking on [Connect], the system will restart, and you will need to re-home the machine. The next time you start up, it will automatically connect to the selected router.**
- Access the web-based control panel:** Launch a web browser on your computer or mobile device and enter the current connected IP address. This will give you access to the web-based control panel. Make sure that your computer or mobile device is connected to the same router.
- Now you can control the machine, configure machine parameters, upgrade the controller firmware, and perform file operations such as uploading, downloading, and deleting files using the SD card on the controller. Note that during file upload, avoid operating other buttons as it may trigger unintended actions.



The page above provides SD card file management, where you can refresh the file list, delete files, download files, and upload files.

9. RS485 Expansion Module Interface Terminals Description (Optional Feature)

Read the subtopics below:

9.1 Wiring

9.2 Functionality

9.1 Wiring

1) Power Supply:

The module can be powered by a 7-12V DC power source through the circular DC jack. Alternatively, you can connect the positive terminal of the control system's 24V power supply to the positive terminal of the IO module, and connect the control system's GND to the negative terminal of the IO module for power supply.

2) Control Signals:

Connect the control system's A+ to the A+ terminal of the IO module, and connect the control system's B- to the B- terminal of the IO module.

3) Input:

The input can be configured using either a two-wire normally open switch or a three-wire PNP normally open switch.

- **For the two-wire normally open switch:** Connect the GND/COM terminal to the power supply's negative terminal. Connect the power supply's positive terminal to one end of the switch, and connect the other end of the switch to the input port.
- **For the three-wire PNP normally open switch:** Connect the GND terminal to the power supply's negative terminal. Connect two of the switch's power supply wires to the power supply, and connect the remaining signal wire to the input port.

4) Output:

The output terminals function as relay outputs and can be used as switches. The NO (normally open) terminal represents the normally open group, while the NC (normally closed) terminal represents the normally closed group.

9.2 Functionality

1) Inputs:

- **Input 0:** Default for tool clamping cylinder safety detection.
 - ❖ **Setting it to 0:** The tool clamping detection setting on the settings page set to 1 disables the detection and activates the delay function.
 - ❖ **Setting it to 1:** The tool clamping detection setting on the settings page set to 0 enables the safety check for the tool clamping cylinder. If the check fails within 5 seconds, the system will halt.
- **Input 1:** Default for tool disk ejection cylinder signal input detection.
 - ❖ **Setting it to 2:** The tool clamping detection setting on the settings page set to 2 enables the safety check for tool clamping cylinder and signal check for the tool turret pop out cylinder. If the check fails within 5 seconds, the system will halt.
 - ❖ The tool clamping detection setting on the settings page set to 0/1 disables tool turret pop out cylinder signal input safety detection and enables delay function. (When there is no expansion module, only 0 mode delay function available)
- **Inputs 2-9:** Customized buttons. Corresponding to self-defined settings 1-7 on the setting page. Please refer to the settings page instructions for custom button input settings.
- **Inputs 7-14:** M-wait command input.

For example,

M66 P0 L3 Q10: Waits for input port 0 to become ON (HIGH), and pauses after 10 seconds timeout. L3 waits for the high level/L4 waits for the low level. (Available with expansion IO)
- **Input 15:** Emergency stop. Normally open contact.

2) Outputs:

M-code Control:

- To activate relay 0, use M62 P0.
- To activate relay 1, use M62 P1.
- To deactivate relay 0, use M63 P0.
- To deactivate relay 1, use M63 P1.

Default Definitions:

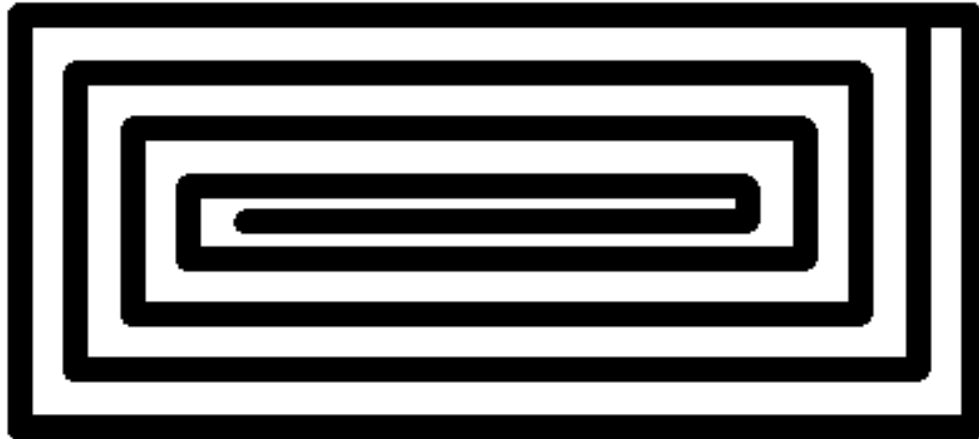
- Relay 0 is assigned to the tool engagement signal. Caution should be exercised.
- Relay 1 is assigned to the servo-driven tool magazine rack extension signal. Caution should be exercised.
- Relay 2 is assigned to the automatic switching signal between position mode and velocity mode for the servo spindle. Caution should be exercised.
- Relay 3 is assigned to the output signal for the chip door cylinder in the tool magazine. Caution should be exercised.
- **Relay 9 is assigned to the air blow output signal for the tool setter. Caution should be exercised.**
- Relay 10 is assigned to the alarm display signal for the tricolor lamp **(red)**.
- Relay 11 is assigned to the pause/idle/manual output signal for the tricolor lamp **(yellow)**.
- Relay 12 is assigned to the running signal output for the tricolor lamp **(green)**.

10. Specific G-code Format Instructions for POCENDER CNC Control System

Read the subtopics below:

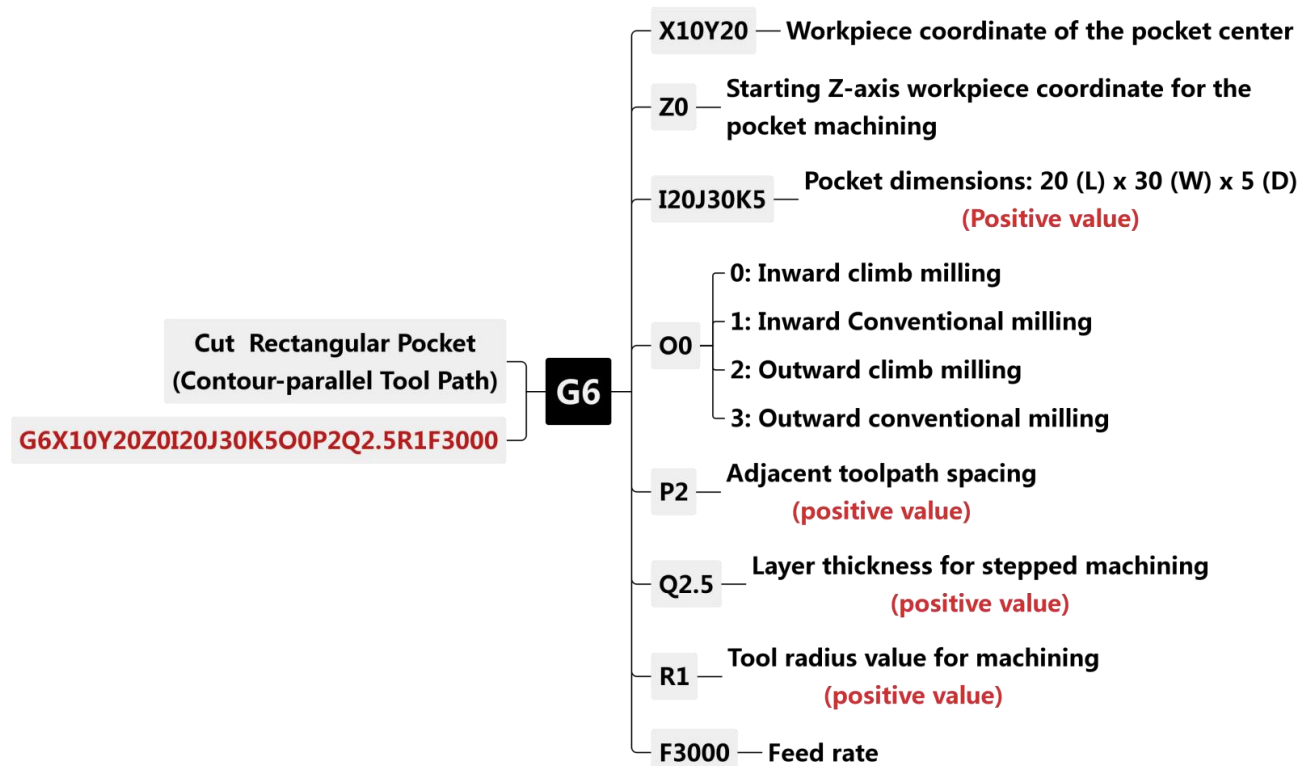
- 10.1 G6/G14 - Cut Rectangular/Rounded Rectangular Pocket
- 10.2 G7- Milling Hole/Thread With Helical Interpolation (Operation Mode G90)
- 10.3 G8 - Vortex Milling Circular Pocket (Operation Mode G90)
- 10.4 G9 - Zigzag Surfacing a Rectangular Plane
- 10.5 G11 - Cut Rectangle/RoundRect. (Operation Mode G90)
- 10.6 G12 - Matrix Drilling (Operation Mode G90)
- 10.7 G13 - Index Drilling (Operation Mode G90)
- 10.8 G73/G81/G82/G83 - Drilling/Tapping

10.1 G6/G14 - Cut Rectangular/Rounded Rectangular Pocket

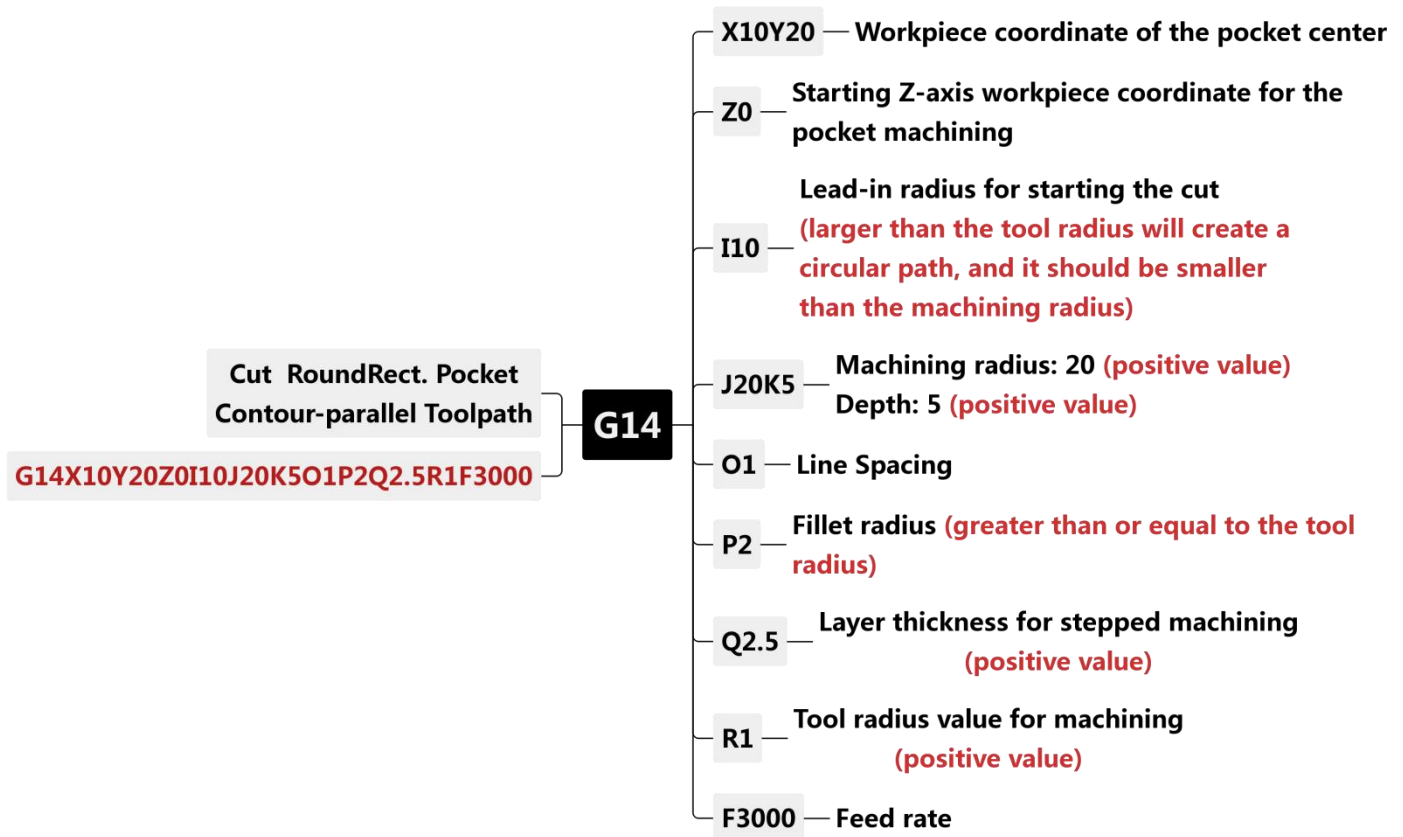


Contour-parallel Tool Path

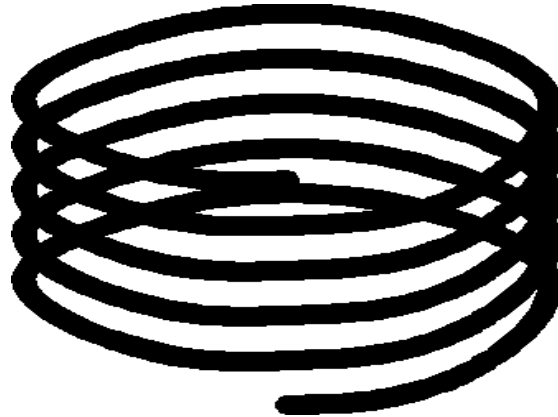
1) G6: Cut Rectangular Pocket



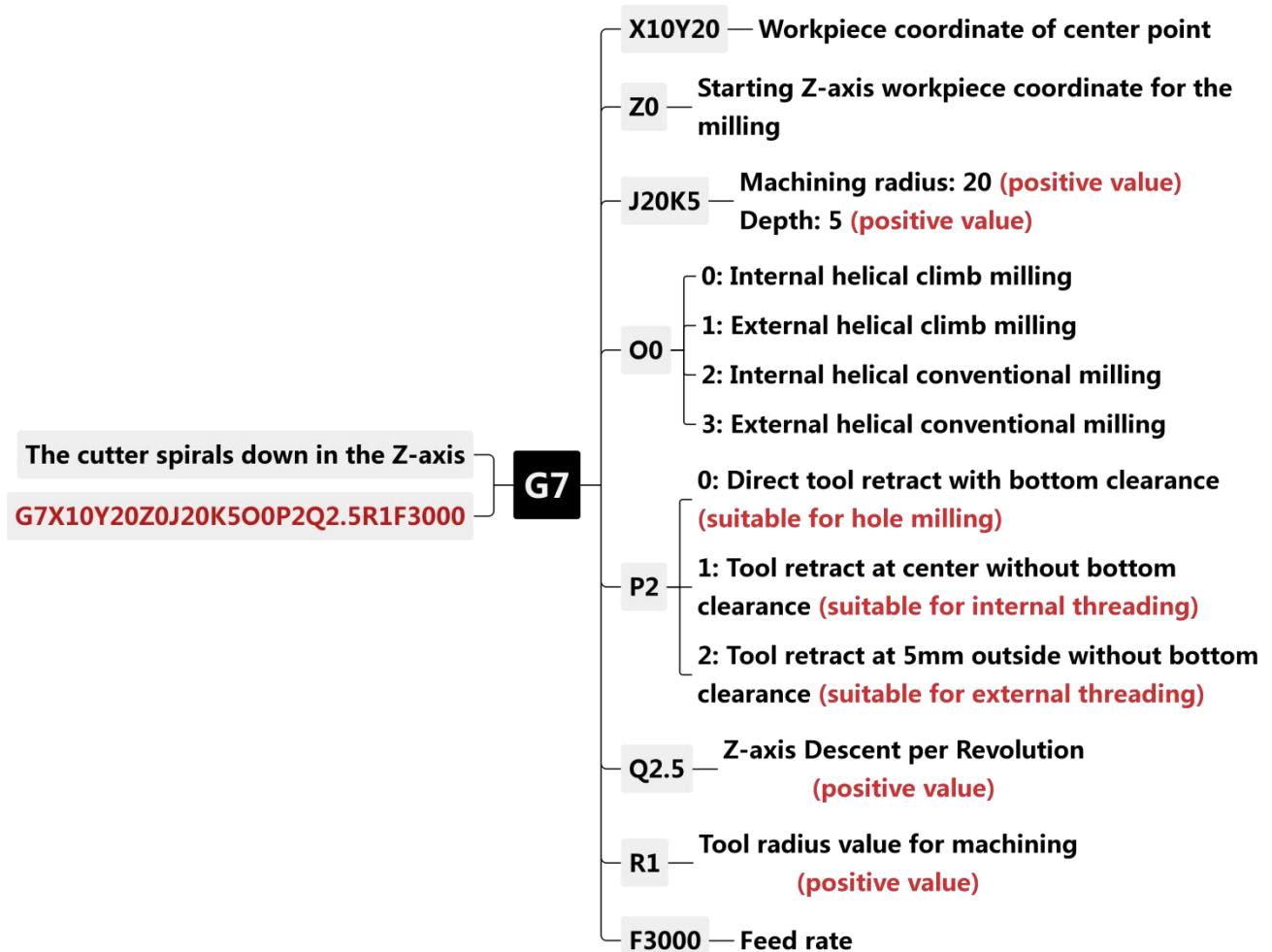
2) G14: Cut Rounded Rectangular Pocket



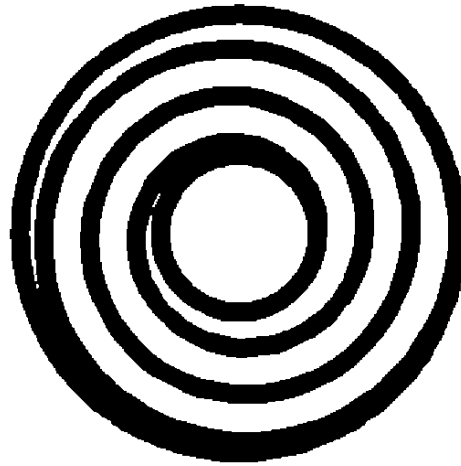
10.2 G7 - Milling Hole/Thread With Helical Interpolation (Operation Mode G90)



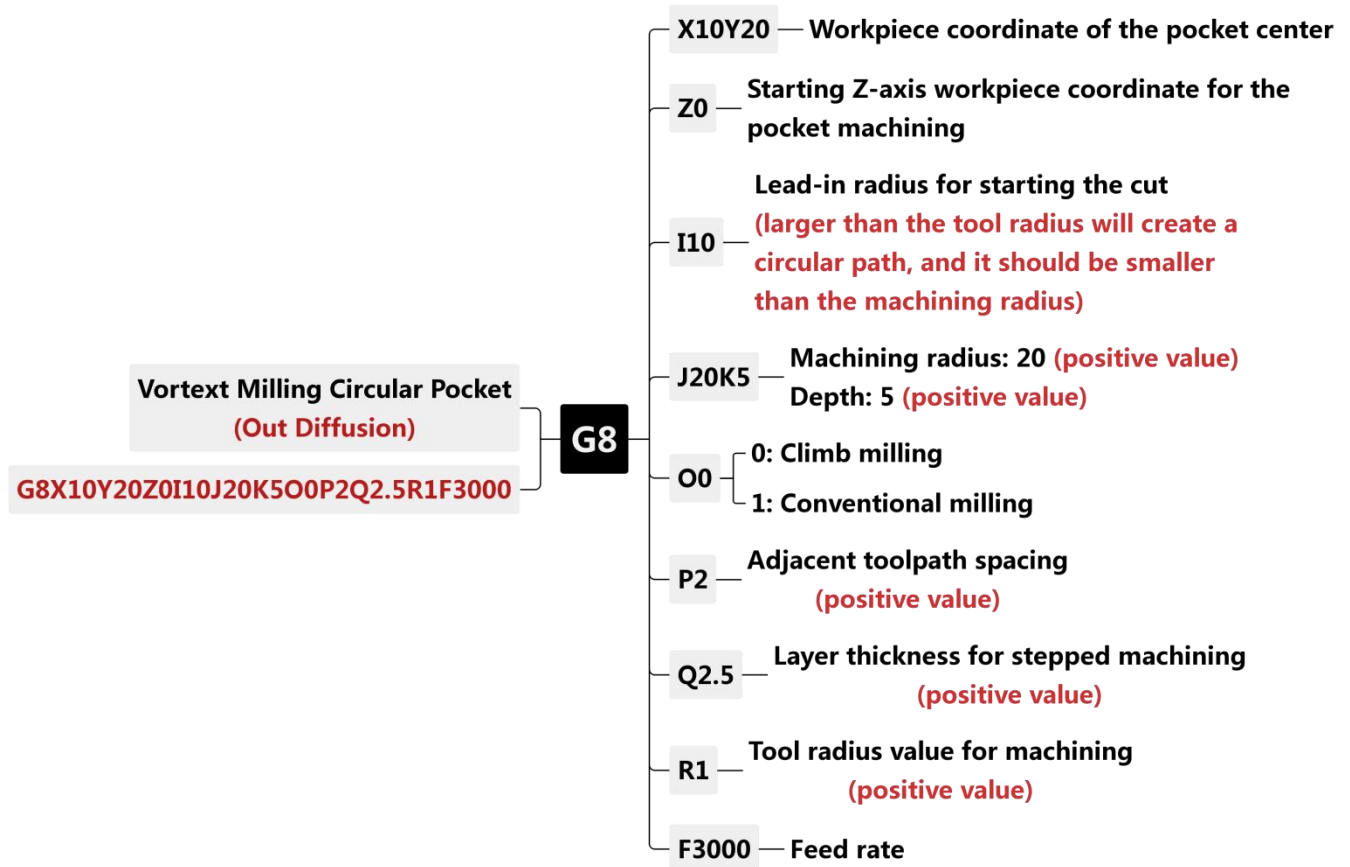
Helical Interpolation



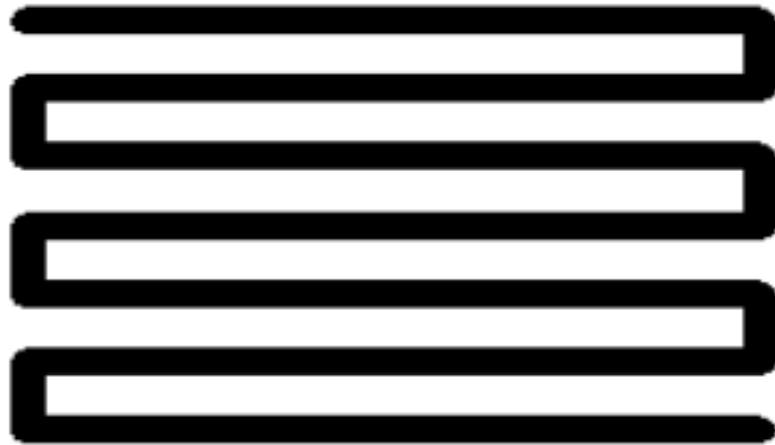
10.3 G8 - Vortex Milling Circular Pocket (Operation Mode G90)



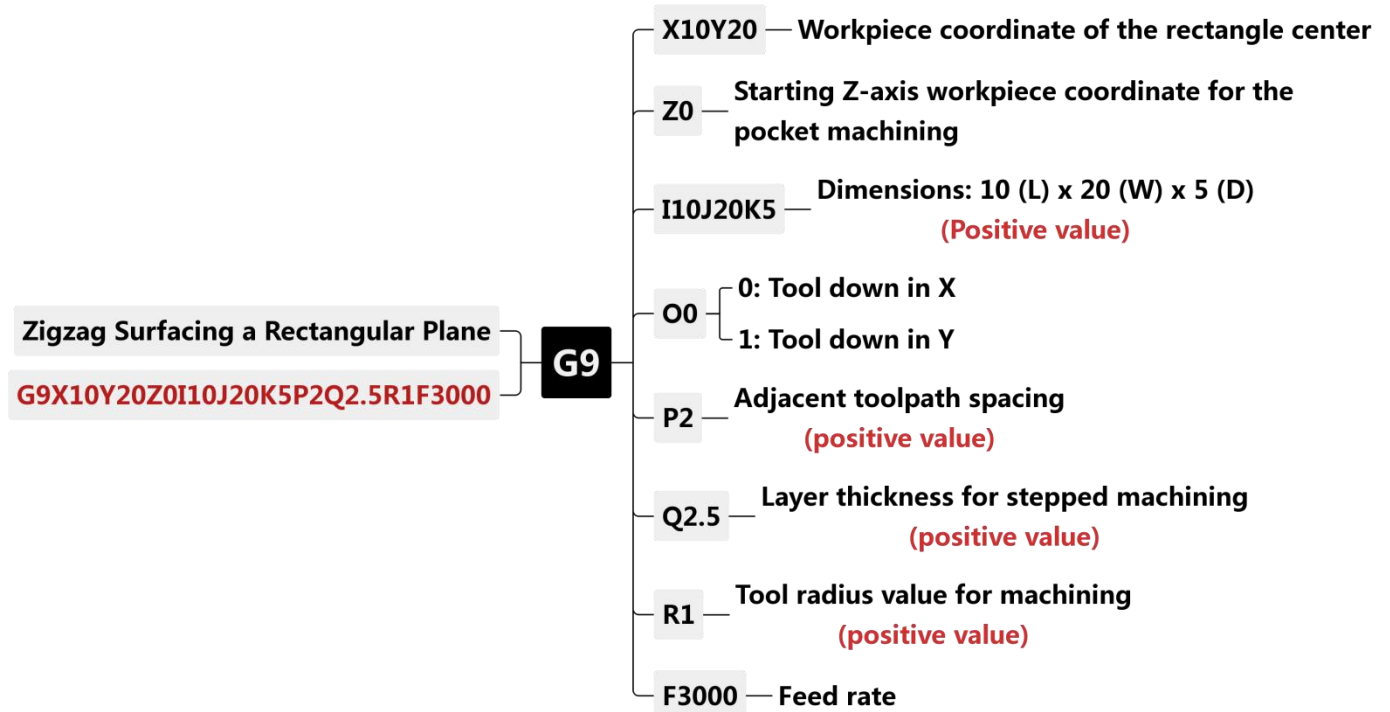
Vortex Milling



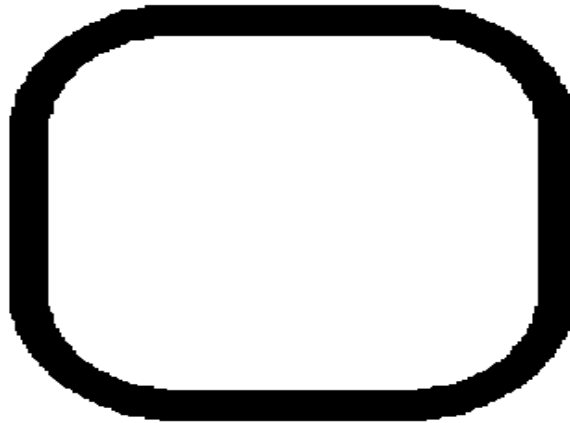
10.4 G9 - Zigzag Surfacing a Rectangular Plane



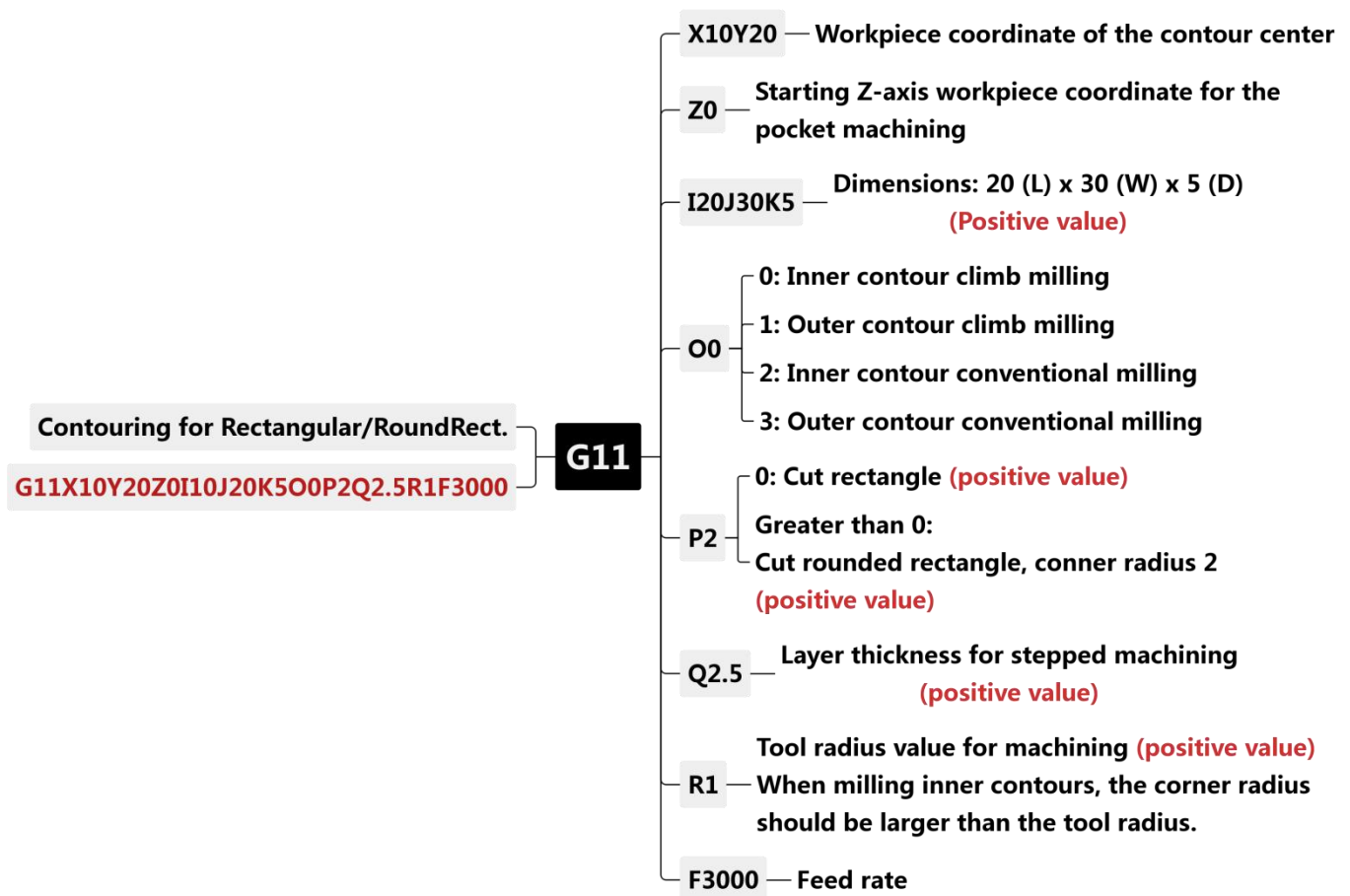
Zigzag Surfacing



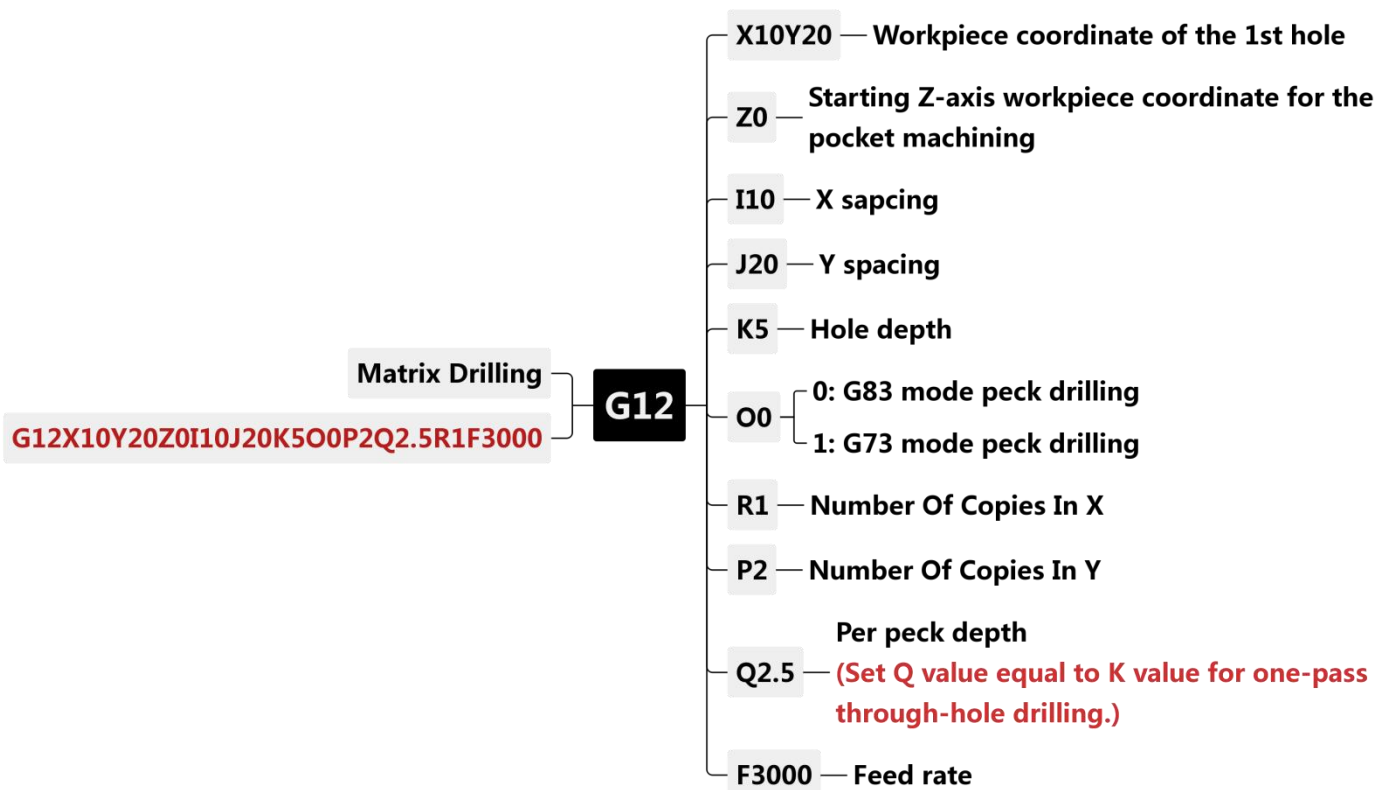
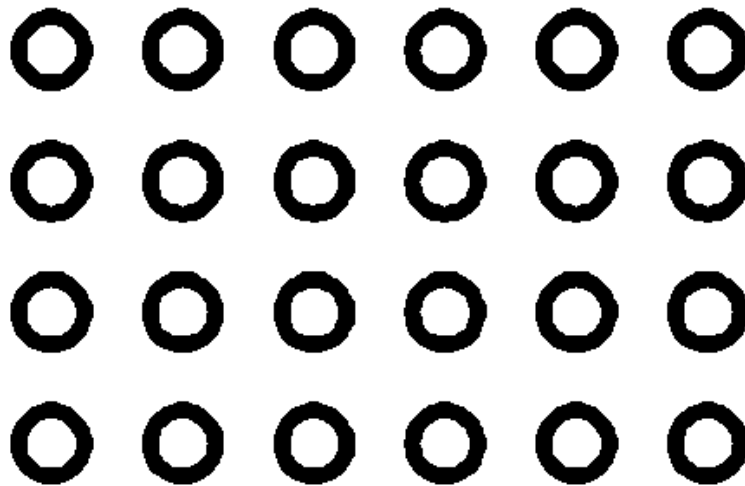
10.5 G11 - Cut Rectangle/RoundRect. (Operation Mode G90)



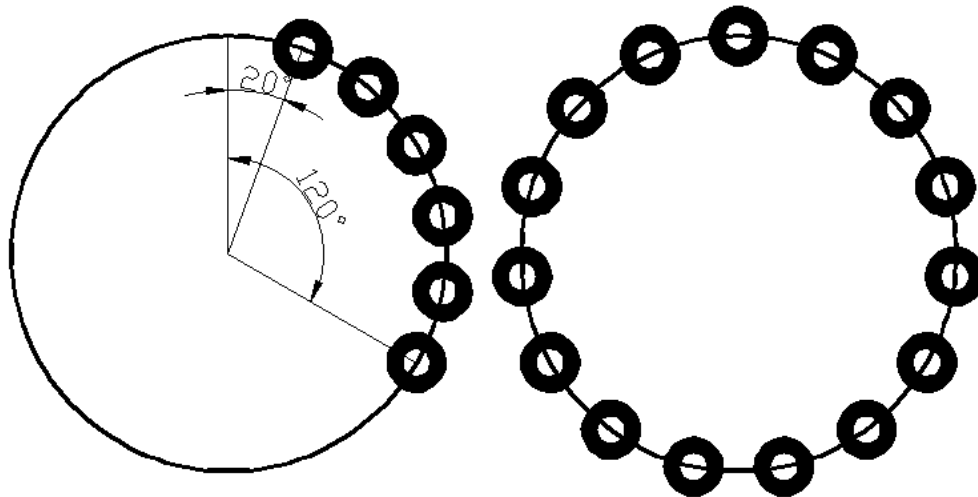
Profile Milling



10.6 G12 - Matrix Drilling (Operation Mode G90)



10.7 G13 - Index Drilling (Operation Mode G90)



Index Drilling G13X10Y20Z0I10J20K5O0P2Q2.5R1F3000	G13	X10Y20 — Workpiece coordinate for indexing center
		Z0 — Starting Z-axis workpiece coordinate for the pocket machining
		I10 — Start angle
		J20 — When O is 0 or 1: End angle When O is 2 or 3: Increment angle
		K5 — Hole depth
		O0 — 0 or 1: Uniform spacing index drilling from start angle to end angle 0: G73 mode peck drilling 1: G83 mode peck drilling 2 or 3: Index drilling with incremental angle 2: G73 mode peck drilling 3: G83 mode peck drilling
		P2 — When O is 0 or 1: Number of equally spaced holes within the Start Angle and End Angle When O is 2 or 3: Number of holes with incremental angle spacing
		Q2.5 — Per peck depth (Set Q value equal to K value for one-pass through-hole drilling.)
		R1 — Index radius
		F3000 — Feed rate

10.8 G73/G81/G82/G83 - Drilling/Tapping

NOTE: When using the tapping command, remember to set the tapping axis as axis letter A, B, or C, which correspond to the axis numbers 4, 5, and 6 respectively. Then, set the pulse equivalent of the tapping axis to the number of pulses required for one degree (or one rotation for firmware version prior 27 June, 2023) of the tap, and set the servo spindle to the position mode.

If you have an IO expansion module installed, the 2nd relay can be used to output an automatic switching signal for the servo spindle between position mode and speed mode. During tapping operations, the signal will be automatically activated to switch the servo spindle to position mode. During drilling operations, the signal will be automatically



INFORMATION: You can set the return plane using G98 to return to the initial plane after completing the operation, or use G99 to return to the R-plane after completing the operation.

deactivated to switch the servo spindle to speed mode.

1) G73

This command is canned cycle used for the peck drilling/tapping. The retract distance is determined by the hole depth (D value) set in the settings page.

Example program

G73X0Y0Z-5R0O1.3Q2.3F500

Syntax & Parameters

- **X0Y0**: Workpiece coordinate of the center
- **Z-5**: Workpiece coordinate when machining reaches Z-5.
- **R0**: Starting Z-axis workpiece coordinate for the machining
- **O1.3**: Specify the pitch as 1.3mm. When set to 0 or omitted, it indicates drilling mode.
- **Q2.3**: Set the peck depth to 2.3mm per peck.

A drilling/tapping cycle start command used for peck drilling.
The retract distance is determined by the hole depth (D value) set in the configuration.

G73X0Y0Z-5R0O1.3Q2.3F500

G73

- X0Y0** — Workpiece coordinate of the center
- Z-5** — Workpiece coordinate position when machining reaches Z-5
- R0** — Starting Z-axis workpiece coordinate for the machining
- O1.3** — Specify the pitch as 1.3mm. When set to 0 or omitted, it indicates drilling mode.
- Q2.3** — Set the peck depth to 2.3mm per peck.

2) G81

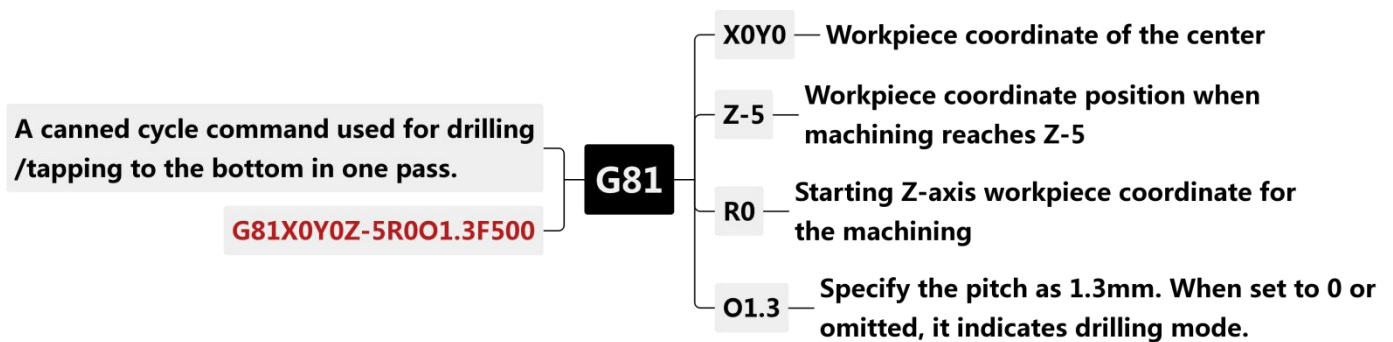
This command is canned cycle used for drilling/tapping to the bottom in one pass.

Example program

```
G81X0Y0Z-5R0O1.3F500
```

Syntax & Parameters

- **X0Y0**: Workpiece coordinate of the center
- **Z-5**: Workpiece coordinate when machining reaches Z-5.
- **R0**: Starting Z-axis workpiece coordinate for the machining
- **Q1.3**: Specify the pitch as 1.3mm. When set to 0 or omitted, it indicates drilling mode.



3) G82

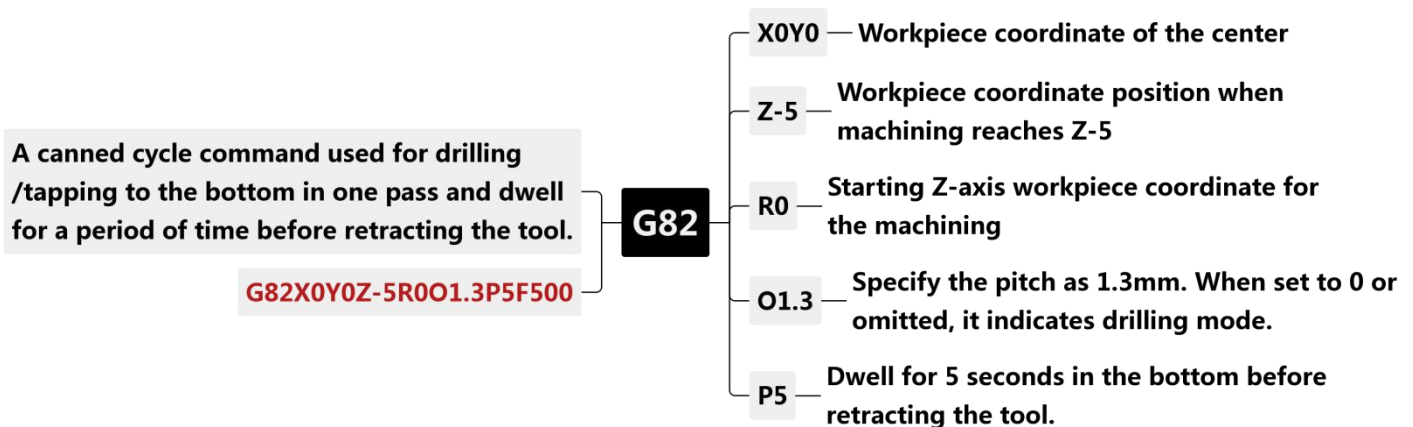
This command is canned cycle used for drilling/tapping to the bottom in one pass and dwell for a period of time before retracting the tool.

Example program

G82X0Y0Z-5R0O1.3P5F500

Syntax & Parameters

- **X0Y0**: Workpiece coordinate of the center
- **Z-5**: Workpiece coordinate when machining reaches Z-5.
- **R0**: Starting Z-axis workpiece coordinate for the machining
- **Q1.3**: Specify the pitch as 1.3mm. When set to 0 or omitted, it indicates drilling mode.
- **P5**: Dwell for 5 seconds in the bottom before retracting the tool.



4) G83

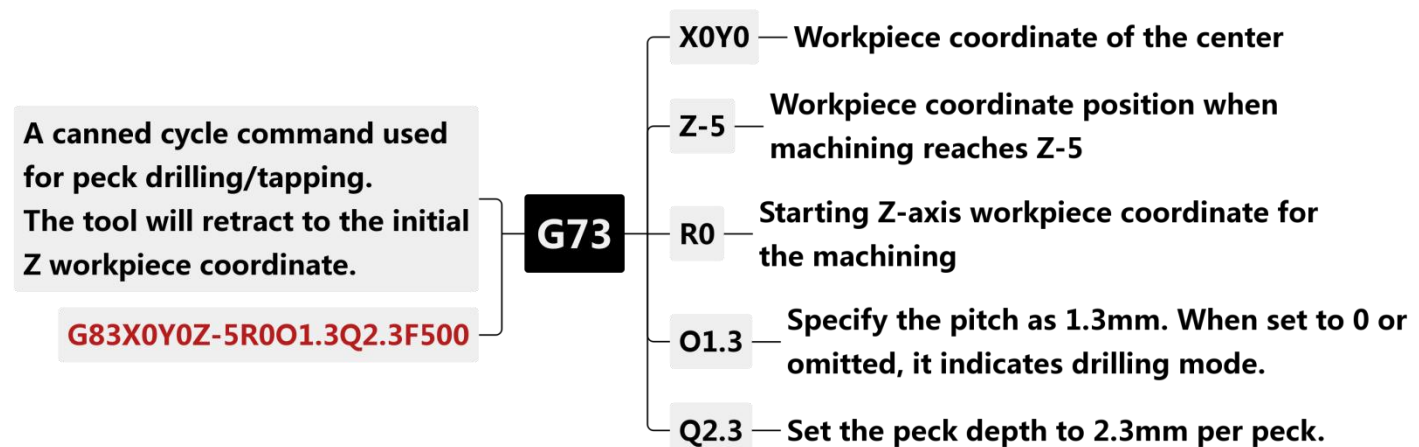
This command is canned cycle used for peck drilling/tapping. The tool will retract to the initial Z workpiece coordinate.

Example program

G83X0Y0Z-5R0O1.3Q2.3F500

Syntax & Parameters

- **X0Y0**: Workpiece coordinate of the center
- **Z-5**: Workpiece coordinate when machining reaches Z-5.
- **R0**: Starting Z-axis workpiece coordinate for the machining
- **Q1.3**: Specify the pitch as 1.3mm. When set to 0 or omitted, it indicates drilling mode.
- **Q2.3**: Set the peck depth to 2.3mm per peck.



11. Detailed Explanation for Supported G Codes & M Codes

Read other subtopics below:

- 11.1 G0 - Rapid Motion Positioning
- 11.2 G1 - Linear Interpolation Motion
- 11.3 G2, G3 - Circular Interpolation Motion
- 11.4 G4 - Dwell
- 11.5 G10 L2 - Set Coordinate System
- 11.6 G10 L20 - Set Coordinate System
- 11.7 G17,G18,G19 - Plane Select
- 11.8 G20,G21 - Units
- 11.9 G28G28.1 - Go/Set Predefined Position
- 11.10 G30 G301 - Go/Set Predefined Position
- 11.11 G38.n - Straight Probe
- 11.12 G40 - Cutter Compensation Cancel
- 11.13 G43 - Tool Length Offset
- 11.14 G43.1 - Dvnamic Tool Length Offset
- 11.15 G43.4 - Rotary Tool Center Point (Similar to FUNAC)
- 11.16 G49 - Cancel Tool Length Offset and RTCP (Similar to FUNAC)
- 11.17 G53 - Move in Machine Coordinates
- 11.18 G54. G55, G56, G57, G58, G59 - Select Coordinate System
- 11.19 G60 - Program Loop
- 11.20 G61 Exact Path Mode
- 11.21 G68.2G69 -Enable/Cancel 3+2 Function
- 11.22 G73 - Drilling Cycle with Chip Breaking
- 11.23 G80 - Cancel Canned Cycle
- 11.24 G81 - Drilling Canned Cycle
- 11.25 G82 - Drilling Canned Cycle With Dwell

11.26 G83 - Peck Drilling Canned Cycle for Deeper Holes

11.27 G90.G91 - Distance Mode

11.28 G911 - Arc IK Incremental Mode

11.29 G92 - Coordinate System Offset

11.30 G921 - Reset G92 Offsets

11.31 G93.G94 - Feed Rate Mode

11.32 G98G99 - Canned Cycle Return Level

11.33 M0 - Program Pause

11.34 M2M30 - Program End

11.35 M3M5 - Spindle Control

11.36 M6 - Auto Tool Change

11.37 M7.M8.M9 - Coolant Control

11.38 M56 - Parking Override and Cancel

11.39 M62M63 - Digital Output Control

11.40 M66 Wait on Input

11.41 \$HX(\$HZ\$HY...) - Homing

11.1 G0 - Rapid Motion Positioning

G0 axes

For rapid motion, program G0 axes, where all the axis words are optional. The G0 is optional if the current motion mode is G0. This will produce coordinated motion to the destination point at the maximum rapid rate (or slower). G0 is typically used as a positioning move.

G0 Example

G90 (set absolute distance mode)

G0 X1 Y-2.3 (Rapid linear move from current location to X1 Y-2.3)

M2 (end program)

If G53 is programmed on the same line, the motion will also differ.

The path of a G0 rapid motion can be rounded at direction changes and depends on the trajectory control settings and maximum acceleration of the axes.

It is an error if:

- An axis letter is without a real value.
- An axis letter is used that is not configured

11.2 G1 - Linear Interpolation Motion

G1 axes

For linear (straight line) motion at programmed feed rate (for cutting or not), program G1 'axes', where all the axis words are optional. The G1 is optional if the current motion mode is G1. This will produce coordinated motion to the destination point at the current feed rate (or slower).

G1 Example

G90 (set absolute distance mode)

G1 X1.2 Y-3 F10 (linear move at a feed rate of 10 from current position to X1.2 Y-3)

Z-2.3 (linear move at same feed rate from current position to Z-2.3)

Z1 F25 (linear move at a feed rate of 25 from current position to Z1)

M2 (end program)

If G53 is programmed on the same line, the motion will also differ.

It is an error if:

- No feed rate has been set.
- An axis letter is without a real value.
- An axis letter is used that is not configured

11.3 G2, G3 - Circular Interpolation Motion

G2 or G3 axes offsets (center format)

G2 or G3 axes R- (radius format)

G2 or G3 offsets|R- <P> (full circles)

A circular or helical arc is specified using either G2 (clockwise arc) or G3 (counterclockwise arc) at the current feed rate. The direction (CW, CCW) is as viewed from the positive end of the axis about which the circular motion occurs.

The axis of the circle or helix must be parallel to the X, Y, or Z axis of the machine coordinate system. The axis (or, equivalently, the plane perpendicular to the axis) is selected with G17 (Z-axis, XY-plane), G18 (Y-axis, XZ-plane), or G19 (X-axis, YZ-plane). Planes 17.1, 18.1, and 19.1 are not currently supported. If the arc is circular, it lies in a plane parallel to the selected plane.

To program a helix, include the axis word perpendicular to the arc plane, for example, if in the G17 plane, include a Z word. This will cause the Z axis to move to the programmed value during the circular XY motion.

To program an arc that gives more than one full turn, use the P word specifying the number of full turns plus the programmed arc. The P word must be an integer. If P is unspecified, the behavior is as if P1 was given that is, only one full or partial turn will result. For example, if a 180 degree arc is programmed with a P2, the resulting motion will be 1 1/2 rotations. For each P increment above 1 an extra full circle is added to the programmed arc. Multi turn helical arcs are supported and give motion useful for milling holes or threads.

Warning

If the pitch of the helix is very small (less than the naive CAM tolerance then the helix might be converted into a straight line.

If a line of code makes an arc and includes rotary axis motion, the rotary axes turn at a constant rate so that the rotary motion starts and finishes when the XYZ motion starts and finishes. Lines of this sort are hardly ever programmed.

The arc center can be relative as set by G91.1.

Two formats are allowed for specifying an arc: Center Format and Radius Format.

It is an error if:

- No feed rate has been set.
- The P word is not an integer.

1) Center Format Arcs

Center format arcs are more accurate than radius format arcs and are the preferred format to use.

The end point of the arc along with the offset to the center of the arc from the current location are used to program arcs that are less than a full circle. It is OK if the end point of the arc is the same as the current location.

The offset to the center of the arc from the current location and optionally the number of turns are used to program full circles.

When programming arcs an error due to rounding can result from using a precision of less than 4 decimal places (0.0000) for inch and less than 3 decimal places (0.000) for millimeters.

Incremental Arc Distance Mode

Arc center offsets are a relative distance from the start location of the arc. Incremental Arc Distance Mode is default.

One or more axis words and one or more offsets must be programmed for an arc that is less than 360 degrees.

No axis words and one or more offsets must be programmed for full circles. The P word defaults to 1 and is optional.

XY-plane (G17)

G2 or G3 <X- Y- Z- I- J- P->

- Z - helix
- I - X offset
- J - Y offset
- P - number of turns

XZ-plane (G18)

G2 or G3 <X- Z- Y- I- K- P->

- Y - helix
- I - X offset
- K - Z offset
- P - number of turns

YZ-plane (G19)

G2 or G3 <Y- Z- X- J- K- P->

- X - helix
- J - Y offset
- K - Z offset
- P - number of turns

It is an error if:

- No feed rate is set with the F word.
- No offsets are programmed.
- When the arc is projected on the selected plane, the distance from the current point to the center differs from the distance from the end point to the center by more than (.05 inch/.5 mm) OR ((.0005 inch/.005mm) AND .1% of radius).

Deciphering the Error message Radius to end of arc differs from radius to start:

- start - the current position
- center - the center position as calculated using the i, j, or k words
- end - the programmed end point
- r1 - radius from the start position to the center
- r2 - radius from the end position to the center

2) Center Format Examples

Calculating arcs by hand can be difficult at times. One option is to draw the arc with a cad program to get the coordinates and offsets. Keep in mind the tolerance mentioned above, you may have to change the precision of your cad program to get the desired results. Another option is to calculate the coordinates and offset using formulas. As you can see in the following figures a triangle can be formed from the current position the end position and the arc center.

In the following figure you can see the start position is X0 Y0, the end position is X1 Y1. The arc center position is at X1 Y0. This gives us an offset from the start position of 1 in the X axis and 0 in the Y axis. In this case only an I offset is needed.

G2 Example Line

G0 X0 Y0

G2 X1 Y1 I1 F10 (clockwise arc in the XY plane)

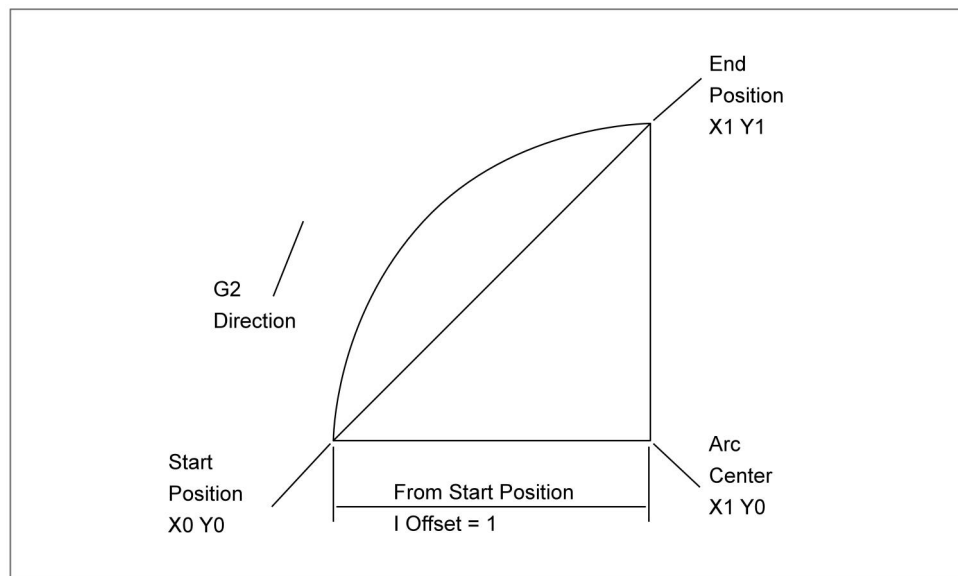


Figure 1. G2 Example

In the next example we see the difference between the offsets for Y if we are doing a G2 or a G3 move. For the G2 move the start position is X0 Y0, for the G3 move it is X0 Y1. The arc center is at X1 Y0.5 for both moves. The G2 move the J offset is 0.5 and the G3 move the J offset is -0.5.

G2-G3 Example Line

G0 X0 Y0

G2 X0 Y1 I1 J0.5 F25 (clockwise arc in the XY plane)

G3 X0 Y0 I1 J-0.5 F25 (counterclockwise arc in the XY plane)

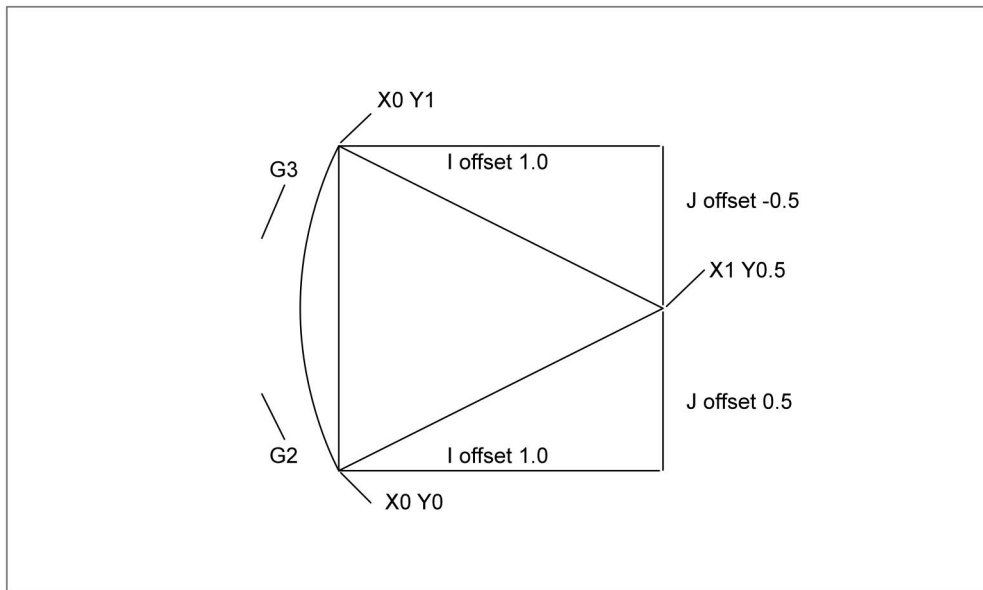


Figure 2. G2-G3 Example

In the next example we show how the arc can make a helix in the Z axis by adding the Z word.

G2 Example Helix

G0 X0 Y0 Z0

G17 G2 X10 Y16 I3 J4 Z-1 (helix arc with Z added)

In the next example we show how to make more than one turn using the P word.

P word Example

G0 X0 Y0 Z0

G2 X0 Y1 Z-1 I1 J0.5 P2 F25

In the center format, the radius of the arc is not specified, but it may be found easily as the distance from the center of the circle to either the current point or the end point of the arc.

3) Radius Format Arcs

G2 or G3 axes R- <P>

- R - radius from current position

It is not good practice to program radius format arcs that are nearly full circles or nearly

semicircles because a small change in the location of the end point will produce a much larger change in the location of the center of the circle (and, hence, the middle of the arc). The magnification effect is large enough that rounding error in a number can produce out-of-tolerance cuts. For instance, a 1% displacement of the endpoint of a 180 degree arc produced a 7% displacement of the point 90 degrees along the arc. Nearly full circles are even worse. Other size arcs (in the range tiny to 165 degrees or 195 to 345 degrees) are OK.

In the radius format, the coordinates of the end point of the arc in the selected plane are specified along with the radius of the arc. Program G2 axes R- (or use G3 instead of G2). R is the radius. The axis words are all optional except that at least one of the two words for the axes in the selected plane must be used. The R number is the radius. A positive radius indicates that the arc turns through less than 180 degrees, while a negative radius indicates a turn of more than 180 degrees. If the arc is helical, the value of the end point of the arc on the coordinate axis parallel to the axis of the helix is also specified.

It is an error if:

- both of the axis words for the axes of the selected plane are omitted
- the end point of the arc is the same as the current point.

G2 Example Line

G17 G2 X10 Y15 R20 Z5 (radius format with arc)

The above example makes a clockwise (as viewed from the positive Z-axis) circular or helical arc whose axis is parallel to the Z-axis, ending where X=10, Y=15, and Z=5, with a radius of 20. If the starting value of Z is 5, this is an arc of a circle parallel to the XY-plane; otherwise it is a helical arc.

11.4 G4 - Dwell

G4 P-

- P - seconds to dwell (floating point)

The P number is the time in seconds that all axes will remain unmoving. The P number is a floating point number so fractions of a second may be used. G4 does not affect spindle, coolant and any I/O.

G4 Example Line

G4 P0.5 (wait for 0.5 seconds before proceeding)

It is an error if:

- the P number is negative or not specified.

11.5 G10 L2 - Set Coordinate System

G10 L2 P- <axes R->

- P - coordinate system (0-6)
- R - rotation about the Z axis

G10 L2 offsets the origin of the axes in the coordinate system specified to the value of the axis word. The offset is from the machine origin established during homing. The offset value will replace any current offsets in effect for the coordinate system specified. Axis words not used will not be changed.

Program P0 to P6 to specify which coordinate system to change.

Table 1. Coordinate System

P Value	Coordinate System	G code
0	Active	n/a
1	1	G54
2	2	G55
3	3	G56
4	4	G57
5	5	G58
6	6	G59

Optionally program R to indicate the rotation of the XY axis around the Z axis. The direction of rotation is CCW as viewed from the positive end of the Z axis.

All axis words are optional.

Being in incremental distance mode (G91) has no effect on G10 L2.

Important Concepts:

- G10 L2 Pn does not change from the current coordinate system to the one specified by P, you have to use G54-59 to select a coordinate system.

- When a rotation is in effect jogging an axis will only move that axis in a positive or negative direction and not along the rotated axis.
- If a G52 local offset or G92 origin offset was in effect before G10 L2, it will continue to be in effect afterwards.
- When programming a coordinate system with R, any G52 or G92 will be applied after the rotation.
- The coordinate system whose origin is set by a G10 command may be active or inactive at the time the G10 is executed. If it is currently active, the new coordinates take effect immediately.

It is an error if:

- The P number does not evaluate to an integer in the range 0 to 6.
- An axis is programmed that is not defined in the configuration.

G10 L2 Example Line

G10 L2 P1 X3.5 Y17.2

In the above example the origin of the first coordinate system (the one selected by G54) is set to be X=3.5 and Y=17.2. Because only X and Y are specified, the origin point is only moved in X and Y; the other coordinates are not changed.

G10 L2 Example Line

G10 L2 P1 X0 Y0 Z0 (clear offsets for X,Y & Z axes in coordinate system 1)

The above example sets the XYZ coordinates of the coordinate system 1 to the machine origin.

11.6 G10 L20 - Set Coordinate System

G10 L20 P- axes

- P - coordinate system (0-6)

G10 L20 is similar to G10 L2 except that instead of setting the offset/entry to the given value, it is set to a calculated value that makes the current coordinates become the given value.

G10 L20 Example Line

G10 L20 P1 X1.5 (set the X axis current location in coordinate system 1 to 1.5)

It is an error if:

- The P number does not evaluate to an integer in the range 0 to 6.
- An axis is programmed that is not defined in the configuration.

11.7 G17, G18, G19 - Plane Select

These codes set the current plane as follows:

- G17 - XY (default)
- G18 - ZX
- G19 - YZ

It is a good idea to include a plane selection in the preamble of each G code file.

11.8 G20, G21 - Units

- G20 - to use inches for length units.
- G21 - to use millimeters for length units.

It is a good idea to include units in the preamble of each G code file.

11.9 G28, G28.1 - Go/Set Predefined Position

Warning

Only use G28 when your machine is homed to a repeatable position and the desired G28 position has been stored with G28.1.

G28 uses the machine coordinates stored in G28.1 as the final point to move to. If no positions are stored with G28.1 then all axes will go to the machine origin.

- G28 - makes a rapid move from the current position to the absolute position set by G28.1.
- G28 axes - makes a rapid move to the position specified by axes including any offsets, then will make a rapid move to the absolute position of the values with G28.1 for all axes specified. Any axis not specified will not move.
- G28.1 - stores the current absolute position as a predefined position.

G28 Example Line

G28 Z2.5 (rapid to Z2.5 then to Z location specified in G28.1)

11.10 G30, G30.1 - Go/Set Predefined Position

Warning

Only use G30 when your machine is homed to a repeatable position and the desired G30 position has been stored with G30.1.

G30 uses the machine coordinates stored in G30.1 as the final point to move to. If no positions are stored with G30.1 then all axes will go to the machine origin.

- G30 - makes a rapid move from the current position to the absolute position set by G30.1.
- G30 axes - makes a rapid move to the position specified by axes including any offsets, then will make a rapid move to the absolute position of the values with G30.1 for all axes specified. Any axis not specified will not move.
- G30.1 - stores the current absolute position as a predefined position.

G30 Example Line

G30 Z2.5 (rapid to Z2.5 then to Z location specified in G30.1)

11.11 G38.n - Straight Probe

G38.n axes

- G38.2 - probe toward workpiece, stop on contact, signal error if failure
- G38.3 - probe toward workpiece, stop on contact
- G38.4 - probe away from workpiece, stop on loss of contact, signal error if failure
- G38.5 - probe away from workpiece, stop on loss of contact

Important

You will not be able to use a probe move until your machine has been set up to provide a probe input signal.

Program G38.n axes to perform a straight probe operation. The axis words are optional, except that at least one of them must be used. The axis words together define the destination point that the probe will move towards, starting from the current location. If the probe is not tripped before the destination is reached G38.2 and G38.4 will signal an error.

The tool in the spindle must be a probe or contact a probe switch.

In response to this command, the machine moves the controlled point (which should be at the center of the probe ball) in a straight line at the current feed rate toward the programmed point. In inverse time feed mode, the feed rate is such that the whole motion from the current point to the programmed point would take the specified time. The move stops (within machine acceleration limits) when the programmed point is reached, or when the requested change in the probe input takes place, whichever occurs first.

It is an error if:

- the current point is the same as the programmed point.
- no axis word is used
- the feed rate is zero
- the probe is already in the target state

11.12 G40 - Cutter Compensation Cancel

G40 - turn cutter compensation off. If tool compensation was on the next move must be a linear move and longer than the tool diameter. It is OK to turn compensation off when it is already off.

G40 Example

; current location is X1 after finishing cutter compensated move

G40 (turn compensation off)

G0 X1.6 (linear move longer than current cutter diameter)

M2 (end program)

It is an error if:

- A G2/G3 arc move is programmed next after a G40.
- The linear move after turning compensation off is less than the tool diameter.

11.13 G43 - Tool Length Offset

G43 <H>

- H - tool number (optional)

G43 enables tool length compensation. G43 changes subsequent motions by offsetting the axis coordinates by the length of the offset. G43 does not cause any motion. The next time a compensated axis is moved, that axis's endpoint is the compensated location.

G43 without an H word uses the currently loaded tool from the last Tn M6.

G43 Hn uses the offset for tool n.

Note

G43 H0 is a little special. Its behavior is different on random tool changer machines and nonrandom tool changer machines. On nonrandom tool changer machines, G43 H0 applies the Tool Length Offset of the tool currently in the spindle, or a TLO of 0 if no tool is in the spindle. On random tool changer machines, G43 H0 applies the TLO of the tool T0 defined in the tool table file (or causes an error if T0 is not defined in the tool table).

G43 H- Example Line

G43 H1 (set tool offsets using the values from tool 1 in the tool table)

It is an error if:

- the H number is not an integer, or
- the H number is negative, or
- the H number is not a valid tool number (though note that 0 is a valid tool number on nonrandom tool changer machines, it means "the tool currently in the spindle")

11.14 G43.1 - Dynamic Tool Length Offset

G43.1 axes

- G43.1 axes - change subsequent motions by replacing the current offset(s) of axes. G43.1 does not cause any motion. The next time a compensated axis is moved, that axis's endpoint is the compensated location.

G43.1 Example

G90 (set absolute mode)

T1 M6 G43 (load tool 1 and tool length offsets, Z is at machine 0 and DRO shows Z1.500)

G43.1 Z0.250 (offset current tool offset by 0.250, DRO now shows Z1.250)

M2 (end program)

- motion is commanded on the same line as G43.1

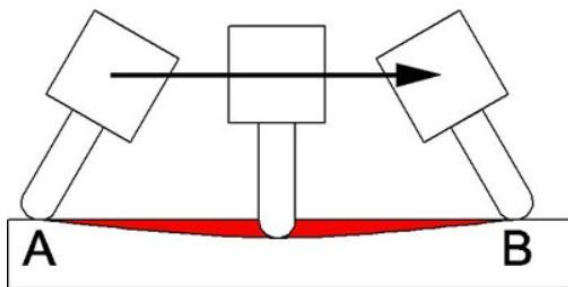
Note

G43.1 does not write to the tool table.

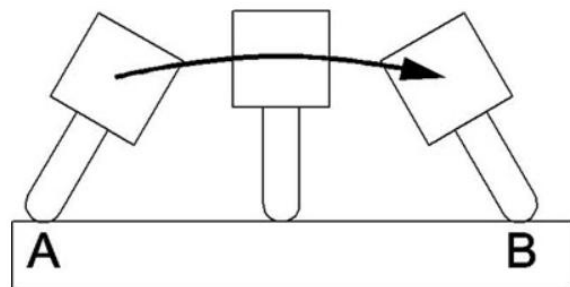
11.15 G43.4 - Rotary Tool Center Point (Similar to FUNAC)

The primary function of G43.4 (Rotary Tool Center Point) (RTCP) is to translate the tool-tip path defined in the part program into the position and orientations for the machine's linear and rotary axes. Tool center point control solves the problem of local tool gouging. Since the CNC knows the profile of the workpiece surface, it can coordinate all of the axes to keep the tool tip in precise contact with the surface when moving between points. So when the program includes an X-axis move between A and B and a B-axis orientation change from -30° and $+30^\circ$, the CNC automatically coordinates the Z-axis to keep the tool tip on the surface path.

RTCP works well on head/head, table/table and head/table machines. The most common practice is for the CAM system to output coordinates relative to the part in top plane. If the part turns, the coordinate system turns with it. The control tracks the part as it turns keeping the zero point pinned to where you set it on the part. RTCP is mostly used for 5 axis paths but can also be used for 3+2 paths. When a machine is in RTCP mode, it will move the tool along a theoretical straight line between points. If you are not used to using RTCP this can cause unexpected moves.



Tool Path Without RTCP



Tool Path Without RTCP

11.16 G49 - Cancel Tool Length Offset and RTCP (Similar to FUNAC)

- G49 - cancels tool length compensation when being used with G43.1
- G49 - cancels RTCP when being used with G43.1

It is OK to program using the same offset already in use. It is also OK to program using no tool length offset if none is currently being used.

11.17 G53 - Move in Machine Coordinates

G53 axes

To move in the machine coordinate system, program G53 on the same line as a linear move. G53 is not modal and must be programmed on each line. G0 or G1 does not have to be programmed on the same line if one is currently active.

For example G53 G0 X0 Y0 Z0 will move the axes to the home position even if the currently selected coordinate system has offsets in effect.

G53 Example

G53 G0 X0 Y0 Z0 (rapid linear move to the machine origin)

G53 X2 (rapid linear move to absolute coordinate X2)

- See G0 section for more information.

It is an error if:

- G53 is used without G0 or G1 being active,
- or G53 is used while cutter compensation is on.

11.18 G54, G55, G56, G57, G58, G59 - Select Coordinate System

- G54 - select coordinate system 1
- G55 - select coordinate system 2
- G56 - select coordinate system 3
- G57 - select coordinate system 4
- G58 - select coordinate system 5
- G59 - select coordinate system 6

11.19 G60 - Program Loop

The G60 code is used to enable the looped execution of a program on a CNC machine. By specifying the number of repetitions with the P value, the program will be executed repeatedly for the specified number of times.

- G60: Command for looped program execution
- P10: Number of repetitions (in this case, 10 times)

G60 Example Line

G60P10 (execute the program file 10 times)

11.20 G61 - Exact Path Mode

- G61 - Exact path mode, movement exactly as programmed. Moves will slow or stop as needed to reach every programmed point. If two sequential moves are exactly co-linear movement will not stop.

11.21 G68.2, G69 - Enable/Cancel 3+2 Function

The G68.2 enables 3+2 function.

AC-axis tilting table and BC-axis tilting table both support 3+2.

The coordinate system adopts ZXZ/ZYZ, Euler angle and IJK calculation method (similar to FUNAC).

▼ Workplane Definition	
Calculation Method	Euler
Workplane Source	Original Workplane
Ignore Toolpath Workplane Shift	No
Euler Convention	ZXZ Rotating
Euler Angles Style	Winding and Unwinding
Zero Euler angles correspond to	Initial Tool Vector

G69 cancels G68.2.

11.22 G73 - Drilling Cycle with Chip Breaking

G73 X- Y- Z- R- Q- <L->

- R - retract position along the Z axis.
- Q - delta increment along the Z axis.
- L - repeat

The G73 cycle is drilling or milling with chip breaking. This cycle takes a Q number which represents a delta increment along the Z axis.

1. Preliminary motion.

- If the current Z position is below the R position, The Z axis does a rapid move to the R position.
- Move to the X Y coordinates

2. Move the Z-axis only at the current feed rate downward by delta or to the Z position, whichever is less deep.

3. Rapid up a bit.

4. Repeat steps 2 and 3 until the Z position is reached at step 2.

5. The Z axis does a rapid move to the R position.

It is an error if:

- the Q number is negative or zero.
- the R number is not specified

11.23 G80 - Cancel Canned Cycle

- G80 - cancel canned cycle modal motion. G80 is part of modal group 1, so programming any other G code from modal group 1 will also cancel the canned cycle.

It is an error if:

- Axis words are programmed when G80 is active.

G80 Example

G90 G81 X1 Y1 Z1.5 R2.8 (absolute distance canned cycle)

G80 (turn off canned cycle motion)

G0 X0 Y0 Z0 (rapid move to coordinate home)

The following code produces the same final position and machine state as the previous code.

G0 Example

G90 G81 X1 Y1 Z1.5 R2.8 (absolute distance canned cycle)

G0 X0 Y0 Z0 (rapid move to coordinate home)

The advantage of the first set is that, the G80 line clearly turns off the G81 canned cycle. With the first set of blocks, the programmer must turn motion back on with G0, as is done in the next line, or any other motion mode G word.

If a canned cycle is not turned off with G80 or another motion word, the canned cycle will attempt to repeat itself using the next block of code that contains an X, Y, or Z word. The following file drills (G81) a set of eight holes as shown in the following caption.

G80 Example 1

N100 G90 G0 X0 Y0 Z0 (coordinate home)

N110 G1 X0 G4 P0.1

N120 G81 X1 Y0 Z0 R1 (canned drill cycle)

N130 X2

N140 X3

N150 X4

N160 Y1 Z0.5

N170 X3

N180 X2

N190 X1

N200 G80 (turn off canned cycle)

N210 G0 X0 (rapid move home)

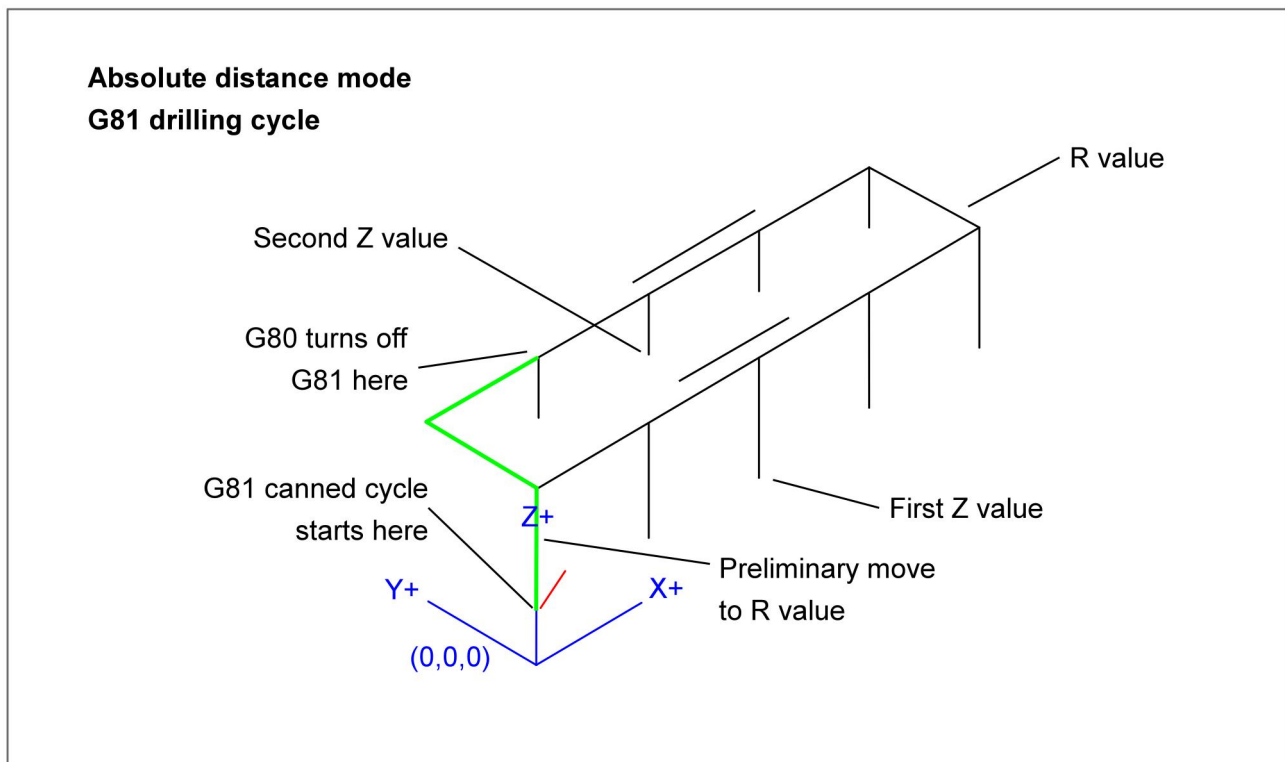
N220 Y0

N230 Z0

N240 M2 (program end)

Note

Notice the z position change after the first four holes. Also, this is one of the few places where line numbers have some value, being able to point a reader to a specific line of code.



G80 Cycle

The use of G80 in line N200 is optional because the G0 on the next line will turn off the G81

cycle. But using the G80 as shown in Example 1, will provide for easier to read canned cycle. Without it, it is not so obvious that all of the blocks between N120 and N200 belong to the canned cycle.

11.24 G81 - Drilling Canned Cycle

G81 (X- Y- Z- R- L-

The G81 cycle is intended for drilling.

The cycle functions as follows:

1. Preliminary motion, as described in the Preliminary and In-Between Motion section.
2. Move the Z-axis at the current feed rate to the Z position.
3. The Z-axis does a rapid move to clear Z.

Example 1 - Absolute Position G81

Suppose the current position is (X1, Y2, Z3) and the following line of NC code is interpreted.

G90 G98 G81 X4 Y5 Z1.5 R2.8

This calls for absolute distance mode (G90) and OLD_Z retract mode (G98) and calls for the G81 drilling cycle to be performed once.

The X value and X position are 4.

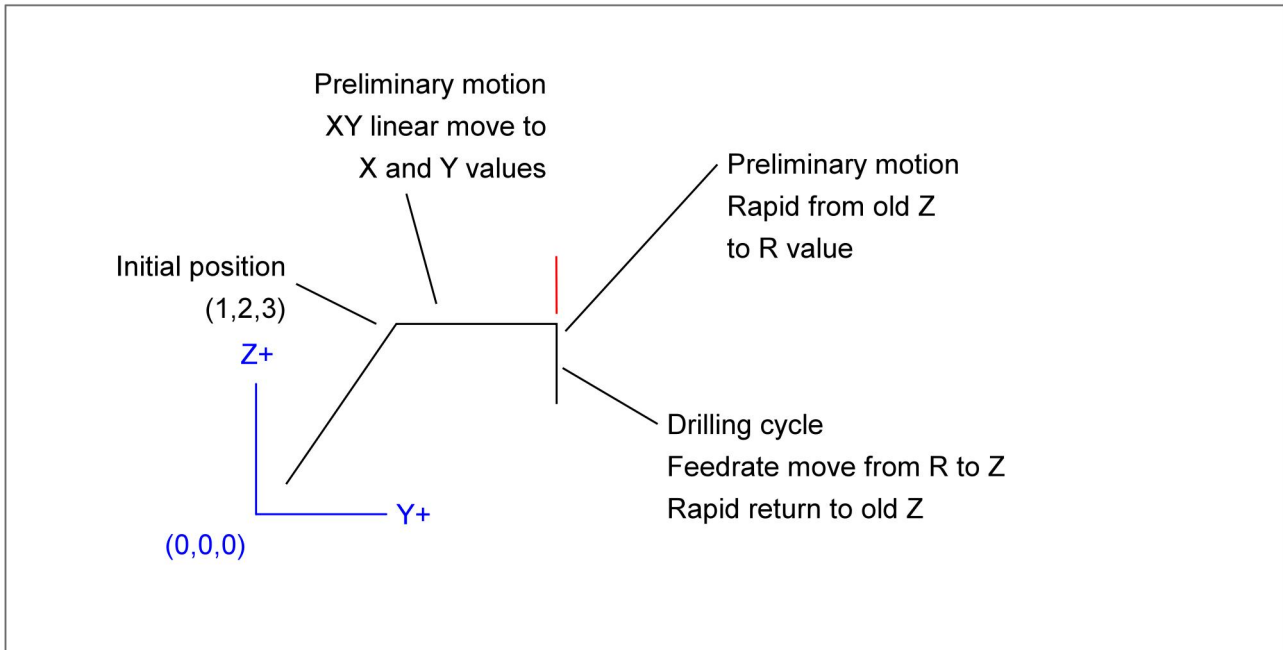
The Y value and Y position are 5.

The Z value and Z position are 1.5.

The R value and clear Z are 2.8. OLD_Z is 3.

The following moves take place:

1. a rapid move parallel to the XY plane to (X4, Y5)
2. a rapid move move parallel to the Z-axis to (Z2.8).
3. move parallel to the Z-axis at the feed rate to (Z1.5)
4. a rapid move parallel to the Z-axis to (Z3)



Example 2 - Relative Position G81

Suppose the current position is (X1, Y2, Z3) and the following line of NC code is interpreted.

G91 G98 G81 X4 Y5 Z-0.6 R1.8 L3

This calls for incremental distance mode (G91) and OLD_Z retract mode (G98). It also calls for the G81 drilling cycle to be repeated three times. The X value is 4, the Y value is 5, the Z value is -0.6 and the R value is 1.8. The initial X position is 5 (=1+4), the initial Y position is 7 (=2+5), the clear Z position is 4.8 (=1.8+3), and the Z position is 4.2 (=4.8-0.6). OLD_Z is 3.

The first preliminary move is a maximum rapid move along the Z axis to (X1,Y2,Z4.8), since $OLD_Z < clear\ Z$.

The first repeat consists of 3 moves.

1. a rapid move parallel to the XY-plane to (X5, Y7)
2. move parallel to the Z-axis at the feed rate to (Z4.2)
3. a rapid move parallel to the Z-axis to (X5, Y7, Z4.8)

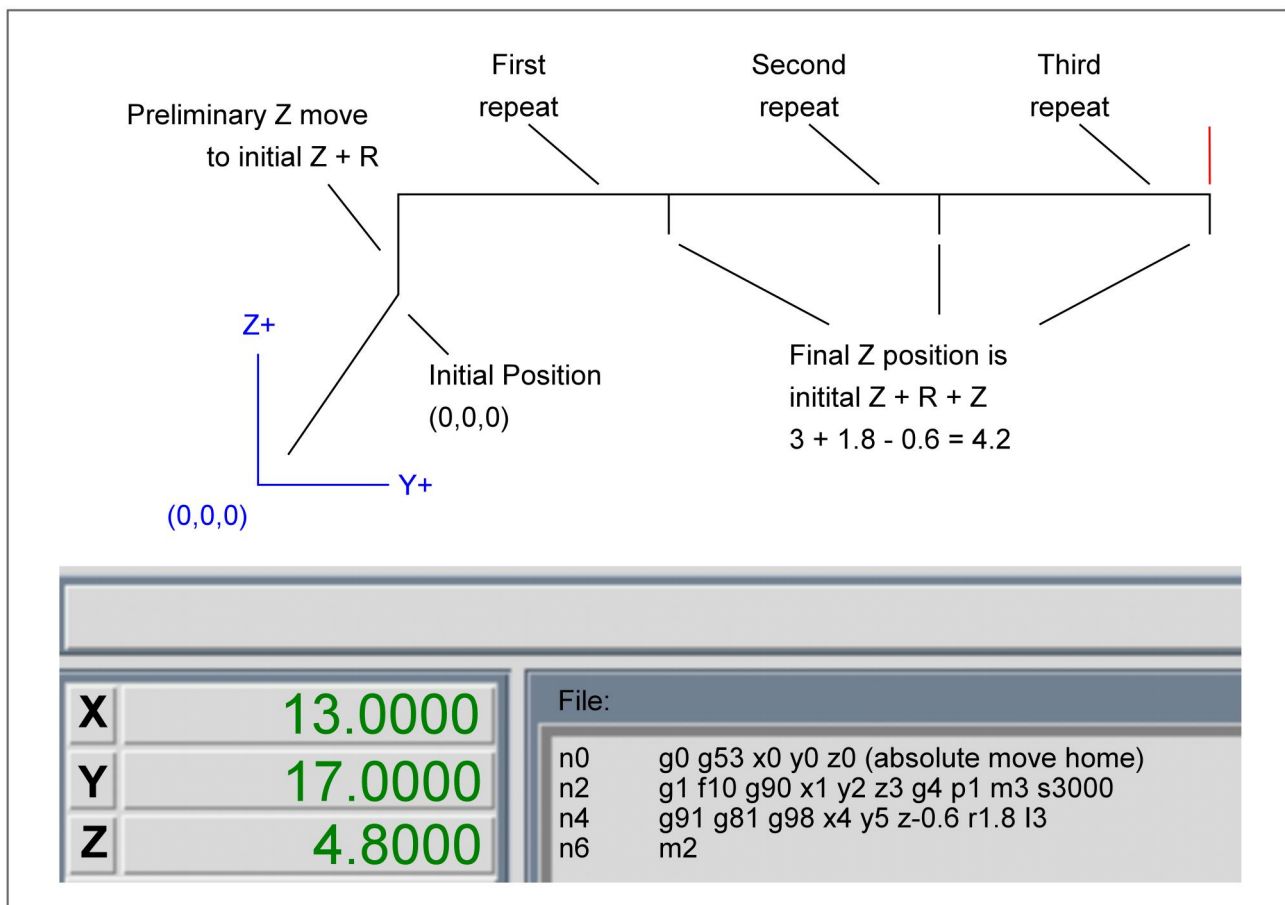
The second repeat consists of 3 moves. The X position is reset to 9 (=5+4) and the Y position to 12 (=7+5).

1. a rapid move parallel to the XY-plane to (X9, Y12, Z4.8)
2. move parallel to the Z-axis at the feed rate to (X9, Y12, Z4.2)

3. a rapid move parallel to the Z-axis to (X9, Y12, Z4.8)

The third repeat consists of 3 moves. The X position is reset to 13 (=9+4) and the Y position to 17 (=12+5).

1. a rapid move parallel to the XY-plane to (X13, Y17, Z4.8)
2. move parallel to the Z-axis at the feed rate to (X13, Y17, Z4.2)
3. a rapid move parallel to the Z-axis to (X13, Y17, Z4.8)

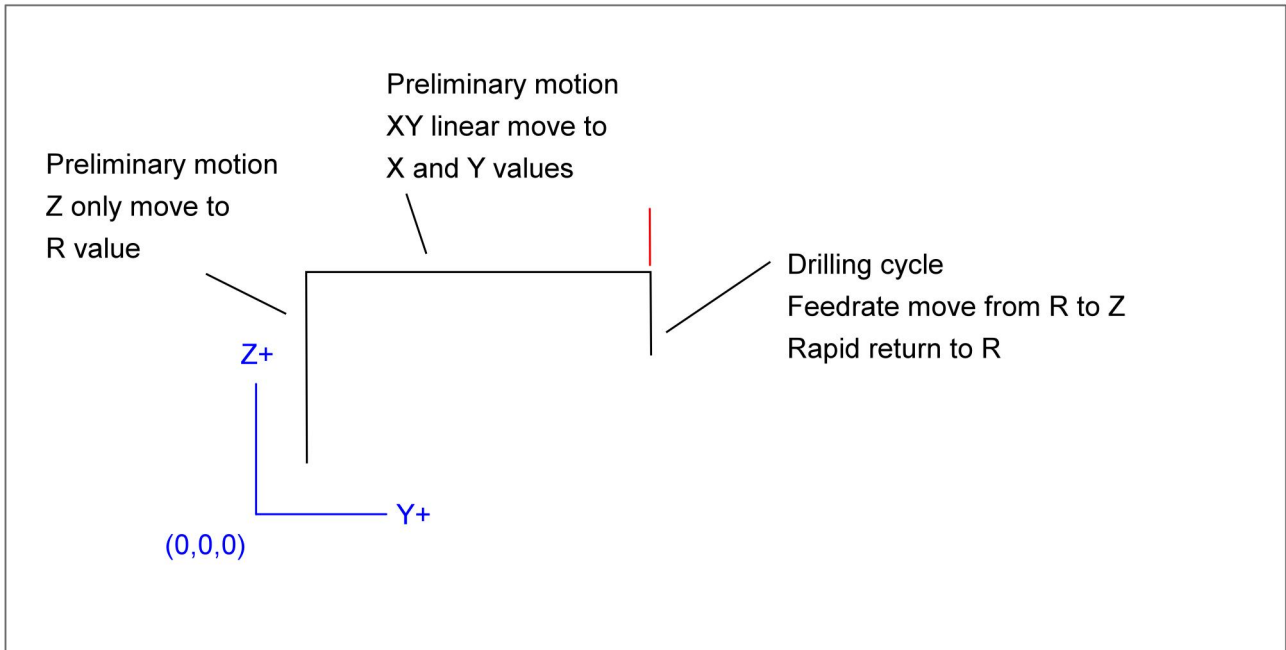


Example 3 - Relative Position G81

Now suppose that you execute the first G81 block of code but from (X0, Y0, Z0) rather than from (X1, Y2, Z3).

G90 G98 G81 X4 Y5 Z1.5 R2.8

Since OLD_Z is below the R value, it adds nothing for the motion but since the initial value of Z is less than the value specified in R, there will be an initial Z move during the preliminary moves.

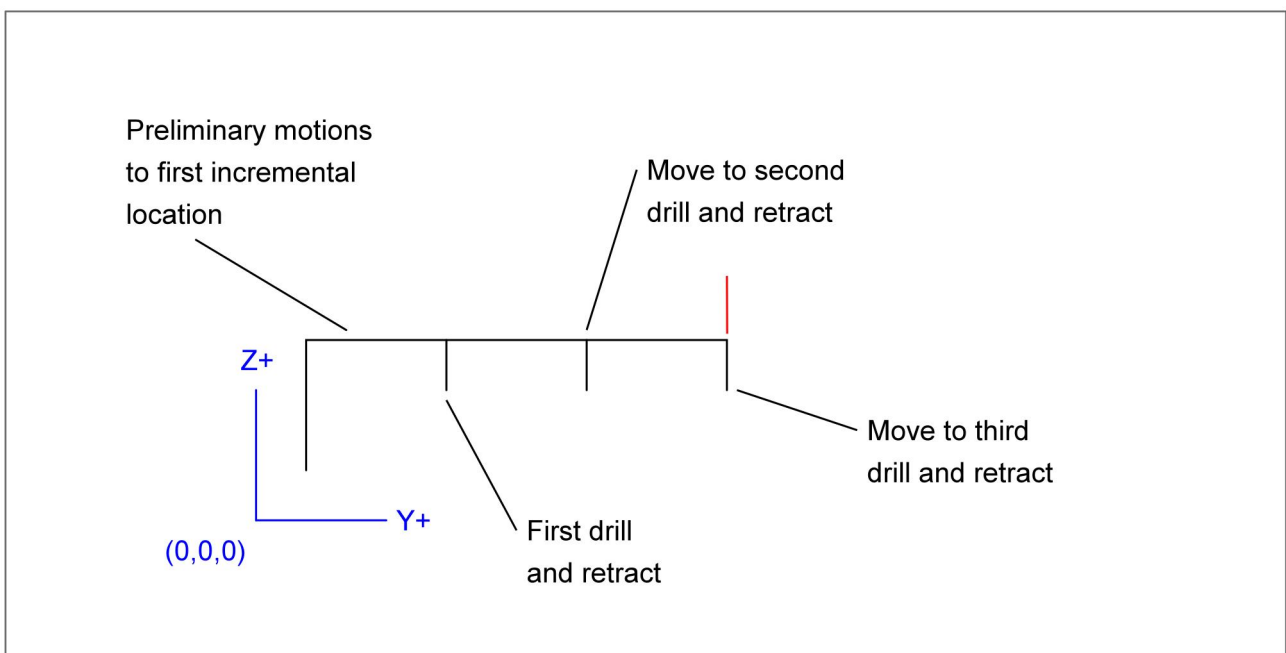


Example 4 - Absolute G81 R > Z

This is a plot of the path of motion for the second g81 block of code.

G91 G98 G81 X4 Y5 Z-0.6 R1.8 L3

Since this plot starts with (X0, Y0, Z0), the interpreter adds the initial Z0 and R1.8 and rapid moves to that location. After that initial Z move, the repeat feature works the same as it did in example 3 with the final Z depth being 0.6 below the R value.



Example 5 - Relative position R > Z

G90 G98 G81 X4 Y5 Z-0.6 R1.8

Since this plot starts with (X0, Y0, Z0), the interpreter adds the initial Z0 and R1.8 and rapid moves to that location as in Example 4. After that initial Z move, the rapid move to X4 Y5 is done. Then the final Z depth being 0.6 below the R value. The repeat function would make the Z move in the same location again.

11.25 G82 - Drilling Canned Cycle With Dwell

G82 X- Y- Z-R- L- P-

The G82 cycle is intended for drilling with a dwell at the bottom of the hole.

1. Preliminary motion, as described in the Preliminary and In-Between Motion section.
2. Move the Z-axis at the current feed rate to the Z position.
3. Dwell for the P number of seconds.
4. The Z-axis does a rapid move to clear Z.

The motion of a G82 canned cycle looks just like G81 with the addition of a dwell at the bottom of the Z move. The length of the dwell is specified by a P- word in the G82 block.

11.26 G83 - Peck Drilling Canned Cycle for Deeper Holes

G83 X-Y-Z-R-L-Q-

The G83 cycle (often called peck drilling) is intended for deep drilling or milling with chip breaking. The retracts in this cycle clear the hole of chips and cut off any long stringers (which are common when drilling in aluminum). This cycle takes a Q number which represents a delta increment along the Z-axis. The retract before final depth will always be to the retract plane even if G98 is in effect. The final retract will honor the G98/99 in effect. G83 functions the same as G81 with the addition of retracts during the drilling operation.

1. Preliminary motion, as described in the Preliminary and In-Between Motion section.
2. Move the Z-axis at the current feed rate downward by delta or to the Z position, whichever is less deep.
3. Rapid move back out to the retract plane specified by the R word.
4. Rapid move back down to the current hole bottom, backed off a bit.
5. Repeat steps 2, 3, and 4 until the Z position is reached at step 2.
6. The Z-axis does a rapid move to clear Z.

It is an error if:

- the Q number is negative or zero.

11.27 G90, G91 - Distance Mode

- G90 - absolute distance mode In absolute distance mode, axis numbers (X, Y, Z, A, B, C) usually represent positions in terms of the currently active coordinate system.
- G91 - incremental distance mode In incremental distance mode, axis numbers usually represent increments from the current coordinate.

G90 Example

G90 (set absolute distance mode)

G0 X2.5 (rapid move to coordinate X2.5 including any offsets in effect)

G91 Example

G91 (set incremental distance mode)

G0 X2.5 (rapid move 2.5 from current position along the X axis)

- See G0 section for more information.

11.28 G91.1 - Arc IJK Incremental Mode

G91.1 - incremental distance mode for I, J & K offsets. G91.1 Returns I, J & K to their default behavior.

11.29 G92 - Coordinate System Offset

G92 axes

Warning

Only use G92 after your machine has been positioned to the desired point.

G92 makes the current point have the coordinates you want (without motion), where the axis words contain the axis numbers you want. All axis words are optional, except that at least one must be used. If an axis word is not used for a given axis, the offset for that axis will be zero.

When G92 is executed, the origins of all coordinate systems move. They move such that the value of the current controlled point, in the currently active coordinate system, becomes the specified value. All of the coordinate system's origins (G53-G59) are offset this same distance.

For example, suppose the current point is at X=4 and there is currently no G92 offset active. Then G92 X7 is programmed. This moves all origins -3 in X, which causes the current point to become X=7. This -3 is saved in system.

Being in incremental distance mode (G91 instead of G90) has no effect on the action of G92.

G92 offsets may be already be in effect when the G92 is called. If this is the case, the offset is replaced with a new offset that makes the current point become the specified value.

It is an error if:

- all axis words are omitted.

The system stores the G92 offsets and reuses them on the next run of a program. To prevent this, one can program a G92.1 (to erase them).

11.30 G92.1 - Reset G92 Offsets

G92.1 - turn off G92 offsets and reset the predefined parameters to zero.

Note

G92.1 only clears G92 offsets, to change G53-G59 coordinate system offsets in G code use either G10 L2 or G10 L20.

11.31 G93, G94 - Feed Rate Mode

- G93 - is Inverse Time Mode. In inverse time feed rate mode, an F word means the move should be completed in [one divided by the F number] minutes. For example, if the F number is 2.0, the move should be completed in half a minute.
- When the inverse time feed rate mode is active, an F word must appear on every line which has a G1, G2, or G3 motion, and an F word on a line that does not have G1, G2, or G3 is ignored. Being in inverse time feed rate mode does not affect G0 (rapid move) motions.
- G94 - is Units per Minute Mode. In units per minute feed mode, an F word is interpreted to mean the controlled point should move at a certain number of inches per minute, millimeters per minute, or degrees per minute, depending upon what length units are being used and which axis or axes are moving.

It is an error if:

- Inverse time feed mode is active and a line with G1, G2, or G3 (explicitly or implicitly) does not have an F word.
- A new feed rate is not specified after switching to G94

11.32 G98, G99 - Canned Cycle Return Level

- G98 - retract to the position that axis was in just before this series of one or more contiguous canned cycles was started.
- G99 - retract to the position specified by the R word of the canned cycle.

Program a G98 and the canned cycle will use the Z position prior to the canned cycle as the Z return position if it is higher than the R value specified in the cycle. If it is lower, the R value will be used. The R word has different meanings in absolute distance mode and incremental distance mode.

G98 Retract to Origin

G0 X1 Y2 Z3

G90 G98 G81 X4 Y5 Z-0.6 R1.8 F10

The G98 to the second line above means that the return move will be to the value of Z in the first line since it is higher than the R value specified.

The initial (G98) plane is reset any time cycle motion mode is abandoned, whether explicitly (G80) or implicitly (any motion code that is not a cycle). Switching among cycle modes (say G81 to G83) does NOT reset the initial plane. It is possible to switch between G98 and G99 during a series of cycles.

11.33 M0 - Program Pause

M0 - pause a running program temporarily. Pressing the **[Cont.]** button will restart the program at the following line.

11.34 M2, M30 - Program End

- M2 - end the program. Pressing **[Run]** will restart the program at the beginning of the file.
- M30 - exchange pallet shuttles and end the program. Pressing **[Run]** will start the program at the beginning of the file.

Both of these commands have the following effects:

1. The spindle is stopped (like M5).
2. Coolant is turned off (like M9).

Note

Lines of code after M2/M30 will not be executed. Pressing **[Run]** will start the program at the beginning of the file.

11.35 M3, M5 - Spindle Control

- M3 - start the selected spindle clockwise at the S speed.
- M5 - stop the selected spindle.

This example will start the spindle clockwise at the speed of 1000:

M3S1000

It is OK to use M3 if the S spindle speed is set to zero. If this is done (or if the speed override switch is enabled and set to zero), the spindle will not start turning. If, later, the spindle speed is set above zero (or the override switch is turned up), the spindle will start turning. It is OK to use M3 when the spindle is already turning or to use M5 when the spindle is already stopped.

11.36 M6 - Auto Tool Change

To change a tool in the spindle from the tool currently in the spindle to the tool most recently selected (using a T word), program M6. When the tool change is complete:

- The spindle will be stopped.
- The tool that was selected (by a T word on the same line or on any line after the previous tool change) will be in the spindle.
- If the selected tool was not in the spindle before the tool change, the tool that was in the spindle (if there was one) will be placed back into the tool changer magazine.
- No other changes will be made. For example, coolant will continue to flow during the tool change unless it has been turned off by an M9.

Warning

The tool length offset is not changed by M6, use G43 after the M6 to change the tool length offset.

The tool change may include axis motion. It is OK (but not useful) to program a change to the tool already in the spindle. It is OK if there is no tool in the selected slot; in that case, the spindle will be empty after the tool change. If slot zero was last selected, there will definitely be no tool in the spindle after a tool change.

11.37 M7, M8, M9 - Coolant Control

- M7 - turn mist coolant on.
- M8 - turn flood coolant on.
- M9 - turn both M7 and M8 off.

Connect one or both of the coolant control pins before M7 or M8 will control an output. M7 and M8 can be used to turn on any output via G code.

It is OK to use any of these commands, regardless of the current coolant state.

11.38 M56 - Parking Override and Cancel

- M56 - Parking override
- M56P0 - Cancel parking override

11.39 M62, M63 - Digital Output Control

- M62 P- - turn on digital output synchronized with motion. The P- word specifies the digital output number.
- M63 P- - turn off digital output synchronized with motion. The P- word specifies the digital output number.

The P-word ranges from 0 to a default value of 3. If needed the the number of I/O can be increased by using an IO expansion module.

The M62 & M63 commands will be queued. Subsequent commands referring to the same output number will overwrite the older settings. More than one output change can be specified by issuing more than one M62/M63 command.

The actual change of the specified outputs will happen at the beginning of the next motion command. If there is no subsequent motion command, the queued output changes won't happen. It's best to always program a motion G code (G0, G1, etc) right after the M62/63.

Note

M62-65 will not function unless the appropriate pins are connected.

11.40 M66 - Wait on Input

M66 P- | <L->

- P- - specifies the digital input number from 0 to 3.
- L- - specifies the wait mode type.
 - Mode 3: HIGH - waits for the selected input to go to the HIGH state.
 - Mode 4: LOW - waits for the selected input to go to the LOW state.
- Q- - specifies the timeout in seconds for waiting. If the timeout is exceeded, the wait is interrupt, and the variable #5399 will be holding the value -1. The Q value is ignored if the L-word is zero (IMMEDIATE). A Q value of zero is an error if the L-word is non-zero.
- Mode 0 is the only one permitted for an analog input.

M66 Example Lines

M66 P0 L3 Q5 (wait up to 5 seconds for digital input 0 to turn on)

M66 wait on an input stops further execution of the program, until the selected event (or the programmed timeout) occurs.

11.41 \$HX(\$HZ,\$HY...) - Homing

X home (Z home, Y home...), only one axis can be homed in a single line.